
Chapter 4

Optical Fiber Measurement

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4.0 INTRODUCTION

Optical fiber is an indispensable part of fiber-optic communication systems; it provides a low-loss and wideband transmission medium. The performance of an optical fiber system depends, to a large extent, on the characteristics of optical fibers. The realization of low-loss optical fibers in the early 1970s made optical fiber communication a viable technology, and the research and development of optical fibers have since become among the central focuses of telecommunications industry.

In addition to standard multimode fiber and standard single-mode fiber, many different types of optical fibers have been developed to provide modified chromatic dispersion properties, engineered nonlinear properties, and enlarged low-loss windows. Effects such as chromatic dispersion, polarization mode dispersion (PMD), polarization-dependent loss (PDL), and nonlinearities, which may be negligible in low-speed optical systems, have become extremely critical in modern optical communications using high-speed time-division multiplexing (TDM) and wavelength-division multiplexing (WDM). The improved fiber parameters and the introduction of new types of optical fibers enabled high-speed and long-distance optical transmission systems as well as various applications of fiber optics such as optical sensing and imaging. Precise measurement and characterization of optical fiber properties are important in the research, development, and fabrication of fibers as well as in the performance evaluation of optical systems.

This chapter reviews various techniques to characterize the properties of optical fibers, their operating principles, and the comparison among techniques. [Section 4.1](#) briefly reviews various types of optical fibers, including standardized fibers for telecommunications and specialty fibers that are developed for various other applications ranging from optical signal processing to optical sensing. [Section 4.2](#) discusses mode field distributions, the definition of mode field diameter, and techniques to measure near-field and far-field profiles. [Section 4.3](#) introduces fiber attenuation measurement techniques. The primary focus of this section is to discuss the operating principle, accuracy considerations, and application of optical time-domain reflectometers (OTDRs).

In [Section 4.4](#) we discuss the measurement of fiber dispersions, including modal dispersion in multimode fibers and chromatic dispersion in single-mode fibers. Both time-domain and frequency-domain measurement techniques are discussed and compared. [Section 4.5](#) reviews a number of techniques for the characterization of PMD in optical fibers, including pulse delay, interferometric methods, Poincare-arc length, fixed analyzer, Jones Matrix, and Mueller Matrix methods. Understanding these measurement techniques, their pros and cons, and the comparison between them is critical in experimental system design

and accuracy assessment. Section 4.6 discusses the measurement of PDL in optical fibers based on the Mueller Matrix technique. Since both the PMD and the PDL in optical systems are randomly varying, the measured results have to be presented to reflect the statistic nature of the process. Section 4.7 discusses implementations of PMD sources and emulators, which are useful instruments in optical transmission equipment testing and qualification. Section 4.8, the last section of this chapter, reviews various fiber nonlinear effects and techniques to characterization them.

4.1 CLASSIFICATION OF FIBER TYPES

As we all know, optical fiber is a cylindrical waveguide that supports low-loss propagation of optical signals. The general properties of optical fibers have been discussed in Chapter 1. In recent years, numerous fiber types have been developed and optimized to meet the demand of various applications. Some popular fiber types that are often used in optical communication systems have been standardized by the International Telecommunication Union (ITU-T). The list includes graded index multimode fiber (G.651), nondispersion-shifted single-mode fiber (G.652), dispersion-shifted fiber (G.653), and nonzero dispersion shifted fiber (G.655). In addition to fibers designed for optical transmission, there are also various specialty fibers for optical signal processing, such as dispersion compensating fibers (DCF), polarization maintaining (PM) fibers, photonic crystal fibers (PCF), and rare-earth doped active fibers for optical amplification. Unlike transmission fibers, these specialty fibers are less standardized.

4.1.1 Standard Optical Fibers for Transmission

The ITU-T G.651 multimode fiber (MMF) has a 50- μm core diameter and a 125- μm cladding diameter. The attenuation parameter for this fiber is on the order of 0.8 dB/km at 1310 nm wavelength. Because of its large core size, MMF is relatively easy to handle with large misalignment tolerance for optical coupling and connection. However, due to its large modal dispersion, MMF is often used for short-reach and low data-rate optical communication systems. Although this fiber is optimized for use in the 1300-nm band, it can also operate in the 850-nm and 1550-nm wavelength bands [1, 2].

The ITU-T G.652 fiber, also known as *standard single-mode fiber*, is the most commonly deployed fiber in long distance optical communication systems. This fiber has a simple step-index structure with a 9- μm core diameter and a 125- μm cladding diameter. It is single-mode with a zero-dispersion wavelength around $\lambda_0 = 1310$ nm. The typical chromatic dispersion value at 1550 nm is about

17 ps/nm-km. The attenuation parameter for G.652 fiber is typically 0.5 dB/km at 1310 nm and 0.2 dB/km at 1550 nm. An example of this type of fiber is Corning SMF-28 [1].

Although standard SMF has low loss in the 1550 nm wavelength window, which makes it suitable for long-distance optical communications, it shows relatively high chromatic dispersion values in this wavelength window. In high-speed optical transmission systems, chromatic dispersion introduces significant waveform distortion, which may significantly degrade system performance. The trend of shifting the transmission wavelength window from 1310 nm to 1550 nm in the early 1990s initiated the development of dispersion-shifted fiber (DSF). Through proper cross-section design, DSF shifts the zero-dispersion wavelength λ_0 from 1310 nm to approximately 1550 nm, making the 1550 nm wavelength window have both the lowest loss and the lowest dispersion. The core diameter of the DSF is about 7 μm , which is slightly smaller than the standard SMF. DSF is able to significantly extend the dispersion-limited transmission distance if there is only a single optical channel propagating in the fiber, but people soon realized that DSF is not suitable for multiwavelength WDM systems due to the high-level nonlinear crosstalk between optical channels through four-wave mixing and cross-phase modulation. For this reason, the deployment of DSF did not last very long.

To reduce chromatic dispersion while maintaining reasonably low nonlinear crosstalk in WDM systems, nonzero dispersion-shifted fibers (NZDSF) were developed. NZDSF moves the zero-dispersion wavelength near to but outside the 1550 nm window so that the chromatic dispersion for optical signals in the 1550 nm wavelength is less than that in standard SMF but higher than that of DSF. The basic idea of this approach is to keep a reasonable level of chromatic dispersion at 1550 nm, which prevents high nonlinear crosstalk from happening but without the need of dispersion compensation in the system. There are several types of NZDSF depending on the selected value of zero-dispersion wavelength λ_0 . In addition, since λ_0 can be either longer or shorter than 1550 nm, the dispersion at 1550 nm can be either negative (normal) or positive (anomalous). The typical chromatic dispersion for G.655 fiber at 1550 nm is 4.5 ps/nm-km. Although NZDSF usually has core sizes smaller than standard SMF, which increases the nonlinear effect, some designs such as Corning LEAF (large effective area fiber) have the same effective core area as standard SMF, which is approximately 80 μm^2 .

Figure 4.1.1 shows typical dispersion vs. wavelength characteristics for several major fiber types that have been offered for long-distance links [3]. The detailed specifications of these fibers are listed in Table 4.1.1, where M is the product of the chromatic dispersion and the effective cross section area. In optical communication system applications, the debate on which fiber has the best performance can never settle down, because data rate, optical modulation formats, channel spacing, the number of WDM channels, and the optical powers used in each channel may affect the conclusion. In general, standard SMF has

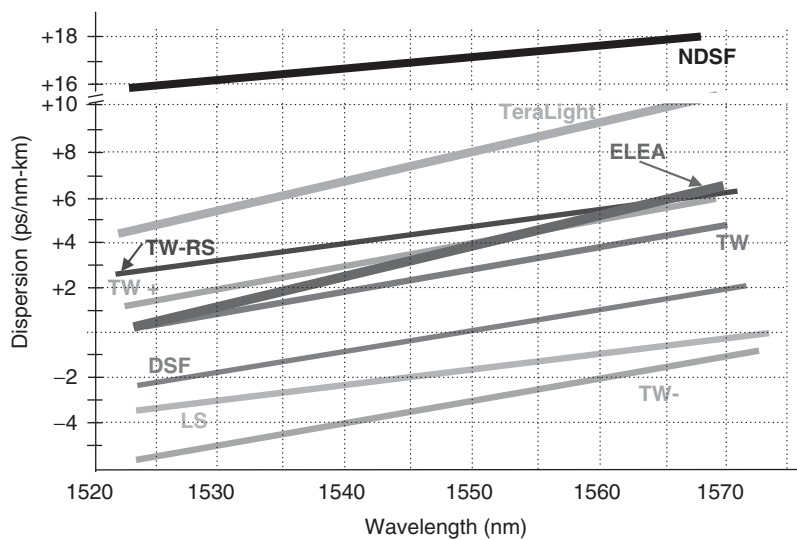


Figure 4.1.1 Chromatic dispersions of non-dispersion-shifted fiber (NDSF) and various different non-zero-dispersion-shifted fibers (NZDSF)

Table 4.1.1
Important Parameters for Standard SMF (NDSF), Long-Span (LS) Fiber, Truewave (TW) Fiber, Truewave Reduced-Slope (TW-RS) Fiber, Large Effective Area Fiber (LEAF), and Teralight Fiber [3]

Fiber Type	ITU	Dispersion @ 1550nm (ps/nm/km)	Dispersion Slope (ps/km/nm ²)	Effective Area (A _{eff}) (μm ²)	M
NDSF(SMF-28)	NDSF/G.652	16.70	0.06	86.6	1446
LS	NZNDSF/G.655	−1.60	0.075	50	80
Truewave (TW)	NZNDSF/G.655	2.90	0.07	55.42	161
TW – RS	NZNDSF/G.655	4.40	0.042	55.42	244
LEAF	NZNDSF/G.655	3.67	0.105	72.36	266
TERALIGHT	NZNDSF/G.655	8.0	0.058	63	504

high chromatic dispersion so that the effect of nonlinear crosstalk between channels is generally small; however, a large amount of dispersion compensation must be used, introducing excess loss and requiring higher optical gain in the amplifiers. This in turn will degrade OSNR at the receiver. On the other hand, low dispersion fibers may reduce the requirement of dispersion compensation but at the risk of increasing nonlinear crosstalk.

4.1.2 Specialty Optical Fibers

In addition to the fibers designed for optical transmission, a number of specialty fibers have also been developed for various purposes ranging from linear and nonlinear optical signal processing to interconnection between equipment. The following are a few examples of specialty optical fibers that have been widely used.

Dispersion compensating fiber (DCF) is a widely used specialty fiber that usually provides a large value of negative (normal) dispersion in a 1550 nm wavelength window. It is developed to compensate for chromatic dispersion in optical transmission systems that are based primarily on standard SMF. The dispersion coefficient of DCF is typically on the order of $D = -95$ ps/nm-km at a 1550 nm wavelength window. Therefore approximately 14 km DCF is required to compensate for the chromatic dispersion of 80 km standard SMF in an amplified optical span. For practical system applications, DCFs can be packaged into modules, which are commonly referred to as dispersion compensating module (DCM).

Compared to other types of dispersion compensation techniques such as fiber Bragg gratings, the distinct advantage of DCF is its wide wavelength window, which is critical for WDM applications, and its high reliability and negligibly small dispersion ripple over the operating wavelength. In addition, DCF can be designed to compensate the slope of chromatic dispersion, thus making it an ideal candidate for WDM applications involving wide wavelength windows. However, due to the limited dispersion value per unit length, DCF usually has relatively higher attenuation compared to fiber gratings, especially when the required total dispersion is high. In addition, because a large value of waveguide dispersion is needed to achieve normal dispersion in 1550 nm wavelength window, the effective core area of DCF can be as small as $A_{eff} \approx 15 \mu\text{m}^2$, which is less than $1/5^{\text{th}}$ that of a standard SMF. Therefore the nonlinear effect in DCF is expected to be significant, which has to be taken into account in designing a measurement setup involving DCF.

Polarization maintaining (PM) fiber is another important category of specialty fibers. It is well known that in an ideal single-mode fiber with circular cross-section geometry, two degenerate modes coexist, with mutually orthogonal polarization states and identical propagation constants. The effect of external stress may cause the fiber to become birefringent and the propagation constants of these two degenerate modes will become different. The partitioning of the propagating optical signal into the two polarization modes not only depends on the coupling condition from the source to the fiber but also on the energy coupling between the two modes while propagating in the fiber, which is usually random. As a consequence, the polarization state of the output

optical signal is usually random, even after only a few meters of propagation length in the fiber; the mode coupling and the output polarization state are very sensitive to external perturbations such as temperature variation, mechanical stress change, and micro and macro bending.

It is also known that energy coupling between the two orthogonal polarization modes can be minimized if the difference between the propagation constants of these two modes is large enough. This can be accomplished by incorporating extra elements in the fiber cladding that apply asymmetric stress to the fiber core. Due to the difference in the thermal expansion coefficients of different materials, asymmetrical stress in the fiber core can be achieved from the manufacturing process when the fiber is drawn from a preform. Depending on the shape of the stress-applying parts (SAP), PM fibers can be classified as “Panda” and “Bowtie,” as illustrated in Fig. 4.1.2. The bowtie structure is depicted in which the SAPs are arranged in an arced manner around the fiber core; the Panda structure is named based on its similarity to the face of a panda bear. In the direction of the SAPs, which is parallel to the field of tension (horizontal in Figure 4.1.2), the fiber core has a slightly higher refractive index so that this axis is also referred to as the *slow axis* because the horizontally polarized mode propagates slower than the vertically polarized mode. The principle axis perpendicular to the slow axis is thus called the *fast axis*.

It is important to note that a PM fiber is simply a highly birefringent fiber, in which the coupling between the two orthogonally polarized propagation modes is minimized. However, for a PM fiber to maintain the polarization state of an optical signal, the polarization state of input optical signal has to be aligned to either the slow or the fast axis of the PM fiber. Otherwise, both of the two degenerate modes will be excited, although there is minimum energy coupling between them; their relative optical phases will still be affected by the fiber perturbation, and the output polarization state will not be maintained because of the vectorial summation of these two mode fields. To further explain, assume

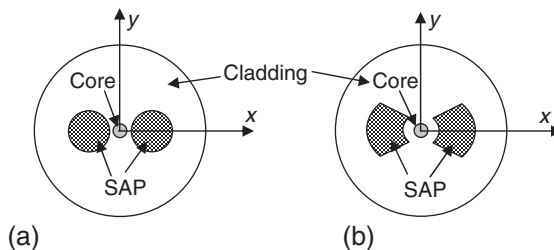


Figure 4.1.2 Cross-sections of polarization fibers: (a) Panda fiber and (b) Bowtie fiber. SAP: stress-applying parts.

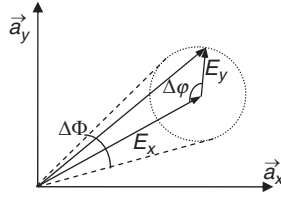


Figure 4.1.3 Illustration of the output field vector of a PM fiber when the input field polarization state is not aligned to the principle axis of the fiber.

that the input optical field is linearly polarized with an orientation angle θ with respect to the birefringence axis of the PM fiber. The optical field E_0 will be split into the two polarization modes such that $E_x = E_0 \cos \theta$ and $E_y = E_0 \sin \theta$. At the output of the fiber, the composite optical field vector $\vec{E} = E_0 e^{-\alpha L} (\vec{a}_x \cos \theta + \vec{a}_y \sin \theta \cdot e^{j\Delta\varphi})$, where $\vec{a}_x$ and $\vec{a}_y$ are unit vectors, L is the fiber length, and α is the attenuation parameter. $\Delta\varphi = (\beta_x - \beta_y)L$ with β_x and β_y the propagation constants of the two modes. It is important to note that the differential phase $\Delta\varphi$ is very sensitive to external perturbations of the fiber. When $\Delta\varphi$ varies, the polarization state of the output optical field also varies if $\theta \neq 0$ (or 90°), as illustrated in [Figure 4.1.3](#). In this case, since the phase difference $\Delta\varphi$ between the two mode fields E_x and E_y are considered random, the variation of the output polarization orientation can be as high as $\Delta\Phi$, as shown in [Figure 4.1.3](#).

Therefore, in an optical system, if a PM fiber is used, one has to be very careful in the alignment of the signal polarization state at the fiber input. Otherwise the PM fiber could be even worse than a standard single-mode fiber in terms of the output polarization stability. Another issue regarding the use of PM fibers is the difficulty of connecting and splicing. When connecting between two PM fibers, we must make sure that their birefringence axes are perfectly aligned. Misalignment between the axes would cause the same problem as the misalignment of input polarization state, as previously discussed. To provide the functionality of precisely controlled axes rotation and alignment, a PM fiber splicer can be five times more expensive than a conventional fiber splicer due to its complexity.

Photonic crystal fiber (PCF), also known as photonic bandgap fiber, is an entirely new category of optical fibers because of its different wave-guiding mechanisms. As shown in [Figure 4.1.4](#), a PCF usually has large number of air holes periodically distributed in its cross-section; for that reason it is also known as the “holey” fiber. The guiding mechanism of PCF is based on the Bragg resonance effect in the transversal direction of the fiber; therefore the low-loss transmission window heavily depends on the bandgap structure design.

[Figure 4.1.4\(a\)](#) shows the cross-section view of a hollow core PCF in which the optical signal is guided by the air core. In contrast to the conventional optical wave-guiding mechanism where high refractive index solid dielectric material

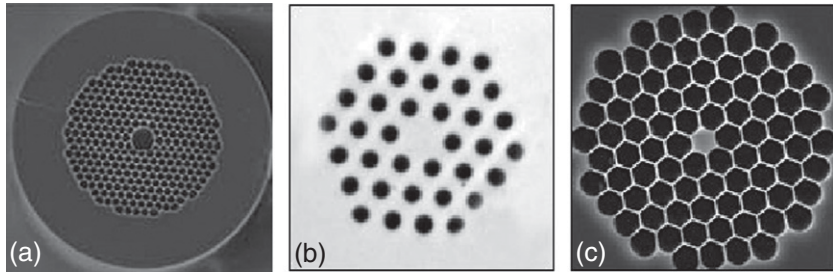


Figure 4.1.4 Cross-sectional view of photonic crystal fibers: (a) hollow-core PCF, (b) large core area PCF, and (c) highly nonlinear PCF [4].

is required for the core, the photonic bandgap structure in the cladding of PCF acts as a virtual mirror that confines the propagating lightwave to the hollow core. In general, the periodical photonic bandgaps structure of PCF is formed by the index contrast between silica and air by incorporating the holes into a silica matrix. In most of the hollow core PCFs, more than 95 percent of the optical power exists in the air; therefore the interaction between signal optical power and the glass material is very small. Because the nonlinearity of the air is approximately three orders or magnitude lower than in the silica, hollow core PCF can have extremely low nonlinearity and can be used to deliver optical signals with very high optical power. In addition, because of the design flexibility of the photonic bandgap structures, by varying the size and the pitch of the holes, zero dispersion can be achieved at virtually any wavelength. One drawback, though, for the hollow core PCF is its relatively narrow transmission window, which is typically on the order of 200 nm. This is the consequence of the very strong resonance effect of the periodic structure that confines the signal energy in the hollow core.

Another category of PCF as shown in Figure 4.1.4(b) is referred to as *large core area PCF*. This type of fiber has a solid silica core to guide the optical signal, whereas the periodic holes in the cladding are used to facilitate optical wave guiding and help confine signal optical power in the core area. Generally, standard single-mode fiber with a 10 μm core diameter has a cutoff wavelength at approximately 1100 nm, below which should be micro meter (μm) the fiber will have multiple propagation modes. Large core PCF, on the other hand, allows for single-mode operation in a very wide wavelength window—for example, from 750 to 1700 nm—while maintaining a very large core area. Compared to hollow core PCF, large core area PCF has much wider low-loss window. Though a large core area PCF has a lower nonlinear parameter than standard single-mode fiber, its nonlinear parameter is typically much higher than a hollow core PCF.

Highly nonlinear PCF, as shown in Figure 4.1.4(c), is also a very useful fiber for optical signal processing which has very small solid core cross-section; therefore the power density in the core is extremely high. For example, for a highly

nonlinear PCF with zero-dispersion wavelength at $\lambda_0 = 710$ nm, the core diameter is as small as $1.8 \mu\text{m}$ and the nonlinear parameter is $\gamma > 100\text{W}^{-1} \text{km}^{-1}$, which is 40 times higher than that of a standard single-mode fiber. Highly nonlinear PCFs are usually used in nonlinear optical signal processing, such as parametric amplification and supercontinuum generation.

Photonic crystal fibers are considered high-end fiber types that are usually expensive due to their complex fabrication processes. They are generally sold in meters instead of kilometers because relatively short PCF fibers are enough for most of the applications for which PCFs are designed. They are also delicate and difficult to handle; for example, end surface treatment, termination, connection, and fusion splicing are not straightforward because of the existence of air holes in the fiber.

Plastic optical fiber (POF) is a low-cost type of optical fiber that is easy to handle. The core material of POF is typically made of PMMA (polymethyl methacrylate), which is a general-purpose resin, whereas the cladding is usually made of fluorinated polymer, which has a lower refractive index than the core. The cross-section design of POF is more flexible than silica fibers, and various core sizes and core/cladding ratio can be easily obtained. For example, in a large-diameter POF, 95 percent of the cross-section is the core that allows the transmission of light. The fabrication of POF does not need an expensive MOCVD process as usually required for the fabrication of silica-based optical fibers; this is one of the reasons for the low cost of the POFs. Although silica-based fiber is widely used for telecommunications, POF has found more and more applications due to its low cost and high flexibility. The costs associated with POF connection and installation are also low, which are especially suitable for fiber to the home application. On the other hand, POFs have transmission losses on the order of 0.25 dB/m , which is almost three orders of magnitude higher than silica fibers. This excluded POFs from being used in long-distance optical transmission. In addition, the majority of POFs are multi-mode, and therefore they are often used in low-speed, short-distance applications such as fiber-to-the-home networks, optical interconnections, networks inside automobile, and flexible illumination and instrumentation.

4.2 MEASUREMENT OF FIBER MODE-FIELD DISTRIBUTION

Mode-field distribution is an important parameter in the specification of an optical fiber. Many practical characteristics of the optical fiber, such as the mode-field diameter, the coupling efficiency between fibers, and the effective cross-section area, are all determined by mode-field distribution. Although the core diameter can be easily determined by the geometry of the fiber, the determination of its mode-field

distribution is more complex, depending on the refractive index profile as well as the wavelength of the optical signal propagating in the fiber. Mode-field distribution can be described by near-field, far-field, or a specially defined mode-field diameter [5, 6, 7]. In this section, we discuss various mode-field definitions and how these parameters can be measured.

4.2.1 Near-Field, Far-Field, and Mode-Field Diameter

In practice, because of the circular geometry of an optical fiber, the field distribution of the fundamental mode in a single-mode optical fiber is circularly symmetrical. This simplifies the problem, and the fiber mode field can be specified by a single parameter known as the *mode-field diameter* (MFD), and the electrical field distribution can often be assumed as Gaussian:

$$E(r) = E_0 \exp\left(-\frac{r^2}{W_0^2}\right) \quad (4.2.1)$$

where r is the radius, E_0 is the optical field at $r = 0$, and W_0 is the width of the field distribution. Specifically, the MFD is defined as $2W_0$, which is

$$2W_0 = 2 \left(\frac{2 \int_0^\infty r^3 E^2(r) dr}{\int_0^\infty r E^2(r) dr} \right)^{1/2} \quad (4.2.2)$$

Figure 4.2.1 illustrates the mode-field distribution of a single mode fiber in which Gaussian approximation is used. The physical meaning of the MFD definition given by Equation 4.2.2 can be explained as follows: The denominator in

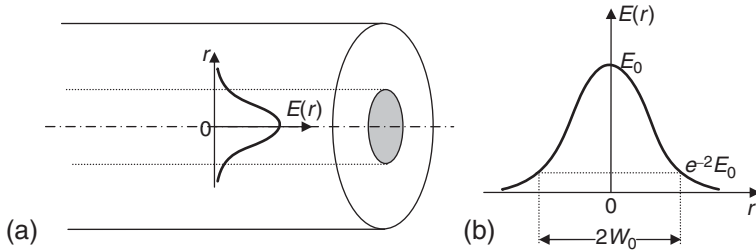


Figure 4.2.1 (a) Illustration of mode distribution in a single-mode fiber and (b) definition of mode-field diameter under Gaussian approximation.

Equation 4.2.2 is proportional to the integration of the power density across the entire fiber cross-section, which is the total power of the fundamental mode, whereas the numerator is the integration of the square of the radial distance (r^2) weighted by the power density over the fiber cross-section. Therefore MFD defined by Equation 4.2.2 represents a root mean square (*rms*) value of the mode distribution of the optical field on the fiber cross-section.

It is important to point out that mode-field distribution $E(r)$ in Equation 4.2.1 represents the field distribution inside the fiber; thus it is equivalent to the optical field distribution exactly on the output end surface of the fiber. It is commonly referred to as the *near-field* (NF) distribution. Near-field distribution is a very important parameter of the fiber that determines the effective cross-section area of the fiber as

$$A_{eff} = \frac{2\pi \left[\int_0^\infty r |E(r)|^2 dr \right]^2}{\int_0^\infty r |E(r)|^4 dr} \quad (4.2.3)$$

As introduced in Equation 1.3.95, if the total optical power P carried by the fiber is known, the power density in the fiber core can be determined using the effective cross-section area as $I_{density} = P/A_{eff}$.

On the other hand, the field radiation pattern at a large distance from the fiber output facet is usually defined as the *far-field* (FF) distribution, which is in general different from the NF distribution; their relationship is [Artiglia 1989]

$$\Psi(R, \theta) = \frac{jk}{R} e^{ikR} E_H(\theta) \quad (4.2.4)$$

where

$$E_H(\theta) = \int_0^\infty E(r) J_0(r \cdot k \sin \theta) r dr \quad (4.2.5)$$

represents the angular distribution of the field, which is the 0th-order Hankel transform of the near-field distribution $E(r)$ on the fiber exit facet. As illustrated in Figure 4.2.2, R is the distance between the observation point in the far-field region and the center of the fiber exit facet, θ is the angle of the observation point with respect to the fiber axis, and r is the distance between the emission point on the fiber facet and the center of the fiber core.

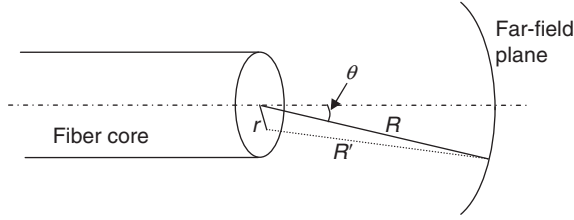


Figure 4.2.2 Coordinates of fiber end surface and far-field plane.

$k = 2\pi/\lambda$ is the wave number and λ is the free-space wavelength. J_0 is the 0th-order Bessel function. As defined previously, the far-field plane has to be far away from the fiber exit facet, and this in general requires $R \approx R'$ and $R \gg W_0^2/\lambda$. In general, if the far-field angular radiation pattern $E_H(\theta)$ can be measured, according to the property of the Hankel transform, the near-field distribution can be easily obtained by an inverse Hankel transform,

$$E(r) = \int_0^\infty E_H(q) J_0(2\pi q r) q dq \quad (4.2.6)$$

where $q = (\sin \theta)/\lambda$ is commonly referred to as the *spatial frequency*.

Similar to the mode-field diameter for the near field as defined by Equation 4.2.2, the mode-field diameter D_0 for the far field can also be defined as

$$2D_0 = 2 \left(\frac{2 \int_0^\infty \rho^3 E_H^2(\rho) d\rho}{\int_0^\infty \rho E_H^2(\rho) d\rho} \right)^{1/2} \quad (4.2.7)$$

where $\rho = R \sin \theta$ is the radial distance between the observation point and the center on the observation plane. Both the near-field and the far-field mode-field diameters are determined by the distribution of the optical power density on the fiber cross-section; thus the most important task is the precise measurement of the actual distribution of the mode field. There are a number of techniques to measure mode-field distribution, such as the far-field scanning technique and the near-field scanning technique. Again, since the relationship between the near-field and the far-field diameters is deterministic through Equations 4.2.2 to 4.2.7, the measurement of either one of them is considered to be sufficient and the other one can be derived accordingly.

4.2.2 Far-Field Measurement Techniques

Figure 4.2.3 shows the block diagram of the far-field measurement setup. The lightwave signal from a laser diode is coupled into the input end of the optical fiber through a focusing lens. A cladding mode filter is usually necessary to remove the optical power carried in the cladding of the fiber and to accelerate the process of achieving equilibrium mode-field distribution. The aperture of the optical detection head determines the spatial resolution of the measurement, which can be a pinhole in front of a photodiode, or simply using an optical fiber as the probe.

The normalized far-field power distribution $P_{angle}(\theta) = u|E_H(\theta)|^2$ defined by Equation 4.2.5 can be obtained through scanning the angular position of the detector head, where u is a constant representing the detector efficiency and the aperture size. Figure 4.2.4 shows examples of measured angular distributions of far-field power density for three different types of fibers. To make

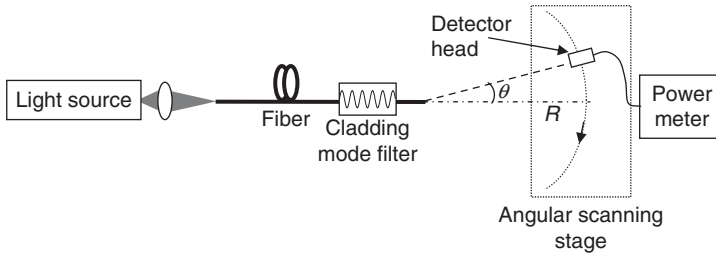


Figure 4.2.3 Far-field measurement setup using angular scanning.

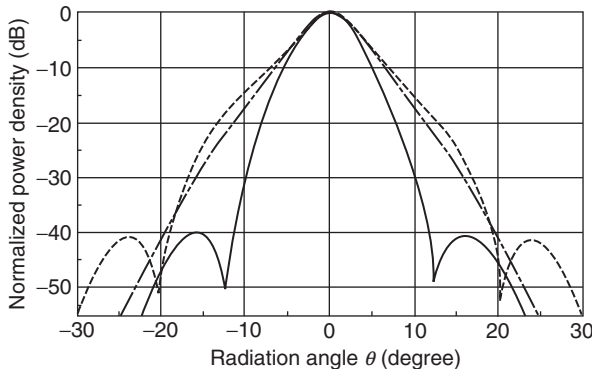


Figure 4.2.4 Examples of measured angular distribution of far-field power density of conventional single-mode step-index fiber at $\lambda = 1300$ nm (solid line), dispersion shifted fiber at $\lambda = 1300$ nm (dot-dashed line), and dispersion flattened fiber at $\lambda = 1550$ nm (dashed line) [8]. Used with permission.

precise estimation of the mode diameter, the far-field power distribution pattern has to be measured over a wide angular range to cover at least the entire main lobe. One practical problem is that the absolute power density $|\Psi(R, \theta)|^2$ of the far field is inversely proportional to the distance R , as indicated by Equation 4.2.4, and thus the measured signal power levels will be very weak if the detector is placed far away from the fiber facet. But the definition of “far field” does require the detector to be far away from the fiber facet such that $R \gg W_0^2/\lambda$. Meanwhile, the acceptable spatial resolution of the measurement requires the small size of the pinhole in front of the detector, which further reduces the optical power at the detector. To make things worse, the power level is even weaker when the detector is placed at large angles with respect to the fiber axis. Therefore very sensitive detectors have to be used in these measurements, with extremely low noise levels. Lock-in amplifiers are usually used in most of the practical implementations. The detection and amplification circuitry also must have reasonably large dynamic ranges to ensure the measurement accuracy. As shown in Figure 4.2.4, the dynamic range has to be higher than 55 dB to clearly show the side lobes of mode distribution.

Fortunately, the mode-field distribution is stationary and the measurement does not require high-speed detection. Therefore, the far-field detection can also be accomplished using a two-dimensional charge-coupled detection (CCD) array, which is basically a video camera. The mode-field distribution at various emission angles can be simultaneously detected by sensing pixels in the camera and can be processed in parallel. This avoids the requirement of mechanically scanning the position of the detector head, which could easily introduce uncertainties. As shown in Figure 4.2.5, since the CCD is a planar array, the far-field power density $P_{plane}(z, \rho)$ measured on the plane is slightly different from the one measured by the angular scanning, and their relationship is

$$P_{plane}(z, \rho) \propto P_{angle}(\theta) \cos^3 \theta \quad (4.2.8)$$

where $\rho = z \tan \theta$ is the radial distance from the center of the array plan. The factor $\cos^2 \theta$ in Equation 4.2.8 comes from the fact that $R = z/\cos \theta$ and the far-field power density is proportional to R^2 , as shown in Equation 4.2.4. The other $\cos \theta$ factor comes from the angle of the detector surface normal, which has an angle θ in respect to the incident light ray.

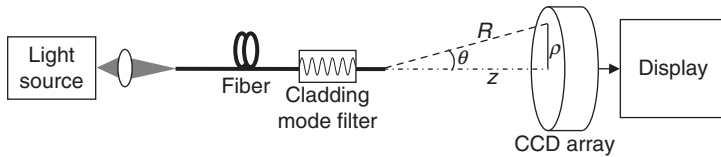


Figure 4.2.5 Far-field measurement setup using a planar detection array.

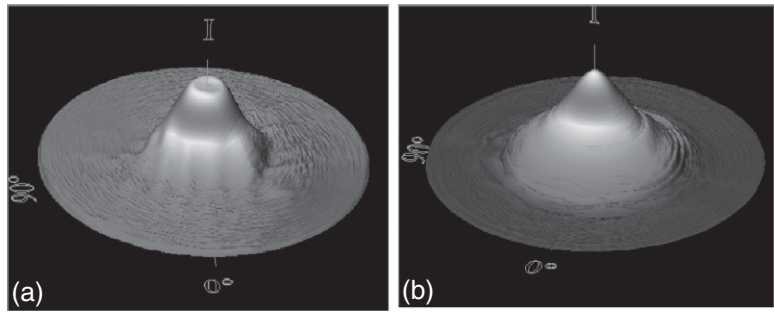


Figure 4.2.6 Examples of the measured far-field profiles of two different fibers [9]. Used with permission.

Instrumentations are commercially available now for the measurement of mode-field distributions [10]. Figure 4.2.6 shows examples of the measured far-field profiles of two different fibers using the far-field imaging technique. Obviously the two mode-field profiles are different, determined by the difference of the index profiles of the two fibers.

4.2.3 Near-Field Measurement Techniques

As signified by its definition, near field is the radial distribution of the electrical field on the fiber end facet. Since the transversal dimension of a fiber facet is very small, it is not feasible to directly measure the field distribution using a probe. It is usually necessary to magnify the image on the fiber exit facet with a microscope objective lens. A block diagram of a near-field measurement setup is shown in Figure 4.2.7, where an objective lens magnifies the image of the fiber exit facet and an optical probe mounted on a translation stage scans across the image plane. A photodiode converts the optical signal sampled by the optical probe into a voltage signal. In this measurement, the scanning probe can simply be a piece of single-mode fiber with the core diameter of approximately 9 μm .

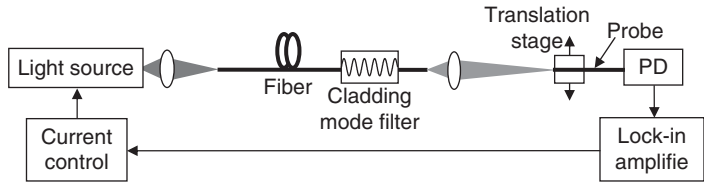


Figure 4.2.7 Near-field measurement setup using a transversal translation stage and a lock-in amplifier.

To make accurate measurements with enough sampling points, the magnification of the optical system has to be large enough, say $100\times$, with good linearity. A careful calibration has to be made because the absolute value of image magnification is critical in determining the mode-field diameter of the fiber under test. Similar power budget concerns as in the case of far-field measurement also exist in the near-field measurement: For a $100\times$ magnification of the optical system, the power density on the image plane is 10^4 times (or equivalently 40 dB) lower than that on the fiber exit facet. In addition to having a large dynamic range, the noise level in the detection system has to be extremely low; therefore a lock-in amplifier is usually required.

Another consideration of the near-field measurement setup is the numerical aperture of the magnification system. Due to diffraction, the detailed feature of the near-field distribution on the imaging plane may be smoothed out. Based on the Rayleigh criterion, to be able to resolve two adjacent points separated by d on the object plane, the numerical aperture of the optical system has to be $NA_{opt} \geq 0.61\lambda/d$. As an example, if the signal wavelength carried in the fiber is 1550 nm, to resolve 10 sampling points along a fiber core diameter of $9\text{ }\mu\text{m}$ ($d = 1\text{ }\mu\text{m}$), the numerical aperture of the optical system has to be $NA_{opt} \approx 0.95$.

Again, similar to the far-field measurement setup shown in Figure 4.2.5, the scanning optical probe, the translation stage, and the photodiode can be replaced by a two-dimensional CCD array. This eliminates the repeatability concern and position uncertainties due to translation stage, and the measurement is also much faster due to parallel sampling and signal processing. As an example, Figure 4.2.8 shows the measured near-field image on the fiber facet together with the near-field intensity profiles in the horizontal and vertical directions. In recent years, the

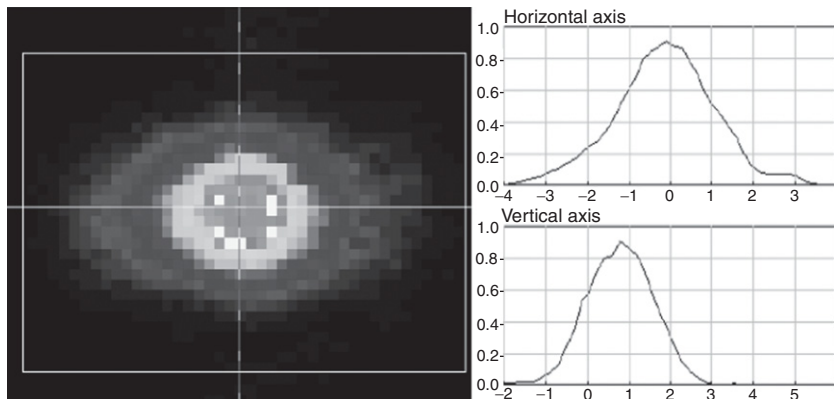


Figure 4.2.8 Examples of the measured near-field profile with the numerical values in the horizontal and vertical directions (scale: $1\text{ }\mu\text{m}/\text{div}$) [9]. Used with permission.

advances in digital signal processing and high-sensitivity CCD technology greatly simplified the processing and analysis of digital images, allowing real-time observation of the far-field and near-field images [10].

In principle, the far-field and the near-field are related and they are all determined by the optical field distribution $E(r)$ on the fiber cross section. In practice, near-field measurement requires a lens with large magnification, which may introduce image distortion for the measurement. In comparison, far-field measurement is more straightforward which only requires a scanning probe. If mechanical scanning stage is stable enough far-field measurement can potentially be more accurate with better signal to noise ratio and dynamic range, which can be observed by comparing Figure 4.2.6 with Figure 4.2.8 [9].

There are a number of alternative techniques to measure fiber mode-field diameter, such as the transverse offset method, the variable aperture method, and the mask method. The details of these methods can be found in [11].

4.3 FIBER ATTENUATION MEASUREMENT AND OTDR

Optical attenuation in an optical fiber is one of the most important issues affecting all applications that use optical fibers. A number of factors may contribute to fiber attenuation, such as material absorption, optical scattering, micro or macro bending, and interface reflection and connection. Some of these factors are uniform, whereas some of them vary along the fiber, especially if different spools of fibers are fusion-spliced together. Characterization of fiber attenuation is fundamental to optical system design, implementation, and performance estimation.

4.3.1 Cutback Technique

In early days, the cutback technique was often used to measure fiber attenuation. As illustrated in Figure 4.3.1, the cutback technique measures fiber transmission losses at different lengths. Suppose the attenuation coefficient α is uniform; the power distribution along the fiber is $P(z) = P_0 e^{-\alpha z}$. With the same amount of optical power coupled into the fiber, if the output power measured as P_1 , P_2 , and P_3 at the fiber lengths of L_1 , L_2 , and L_3 , respectively, the fiber attenuation coefficient can be calculated as $\alpha = [\ln(P_2/P_1)]/(L_1 - L_2)$ or $\alpha = [\ln(P_3/P_2)]/(L_2 - L_3)$.

For a single-mode fiber, there are only two orthogonal fundamental modes and the difference in their attenuations is generally negligible. For a multimode fiber, on the other hand, there are literally hundreds of propagation modes and

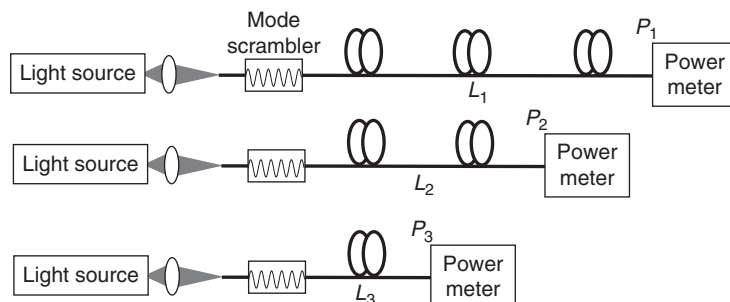


Figure 4.3.1 Illustration of the cutback technique to measure fiber loss.

different modes may have different attenuation coefficients. Therefore, launching condition from the source to the fiber is an important consideration in the loss measurement, especially in comparing results measured by different laboratories. A mode scrambler can be used to stabilize the power distribution of the guided modes and stripe out the cladding mode.

It is well known that a single-mode fiber will turn into multimode if the signal wavelength is shorter than the cutoff wavelength. Immediately below the cutoff wavelength, the second-order LP_{11} mode starts to exist in addition to the fundamental mode. Cutback technique can be used to evaluate the single-mode condition in a fiber by measuring the attenuation coefficient of the second-order LP_{11} mode. Figure 4.3.2 illustrates a technique of measuring the attenuation coefficients of both the fundamental LP_{01} mode and the second-order LP_{11} mode based on the fact that the bending loss of LP_{11} mode is much higher than LP_{01} mode [12]. The measurement procedure is as follows: First, bend the fiber with small radius ($\sim 1\text{--}2\text{cm}$) near the input end of the fiber and measure the power P_1 and P_2 at fiber lengths of $L_1 + L_2$, and L_1 , respectively.

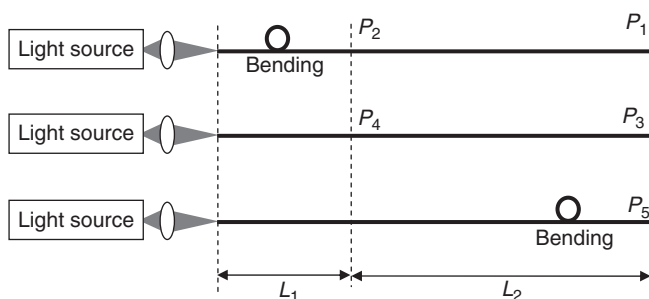


Figure 4.3.2 Technique to selectively measure the attenuation coefficient of the fundamental mode LP_{01} and the second-order mode LP_{11} .

In this case, the bending is equivalent to a mode filter that removes the power carried by the LP_{11} mode while introducing negligible loss for the fundamental LP_{01} mode. The attenuation coefficient of LP_{01} mode can be obtained as

$$\alpha_{01} = \frac{1}{L_2} \ln \left(\frac{P_2}{P_1} \right) \quad (4.3.1)$$

Next, release the bending and measure the powers for the straight fiber at the same locations. In this case, the measured powers P_3 and P_4 include both the LP_{01} mode and the LP_{11} mode. Then bend the fiber near the output terminal so that the measured power P_5 only includes the LP_{01} mode. Through these measurements, the attenuation coefficient of LP_{11} mode can be found as

$$\alpha_{11} = \frac{1}{L_2} \ln \left(\frac{P_1 P_4 - P_2 P_5}{P_1 P_3 - P_1 P_5} \right) \quad (4.3.2)$$

The cutback measurement technique is simple, direct, and it can be very accurate, however, it has several disadvantages. First, cutting the fiber step by step will certainly make the fiber less useable. Especially if the fiber loss is low; a very long fiber has to be used to have measurable losses. For example, if the attenuation coefficient of the fiber is $\alpha = 0.25$ dB/km, the output optical power only increases by 0.025 dB by cutting off 100 m of fiber, and obviously the accuracy of the measurement is limited by the accuracy and the resolvable digits of the power meter. In addition, it would be difficult to use the cutback technique to evaluate fibers after cabling; it is even impossible to measure in-service fibers that are already buried under ground. Therefore nondestructive techniques, such as optical time-domain reflectometer, would be more practical.

4.3.2 Optical Time-Domain Reflectometers

Rayleigh backscattering is one of the most important linear effects in a single-mode optical fiber; it sets a fundamental limit of fiber loss and is responsible for the major part of the attenuation in modern optical fibers in which other losses are already minimized. Meanwhile, Rayleigh backscattering can also be utilized to measure the fiber attenuation distribution along the fiber [13]. In this backscattering measurement technique, a short and high-peak power optical pulse train is launched into the fiber and the waveform of the backscattered optical signal from the fiber is detected, providing the detailed local loss information throughout the fiber. An important advantage of this technique is that it only needs to access a single end of the fiber because both the source and the detector are colocated, so the method is nondestructive. Since Rayleigh

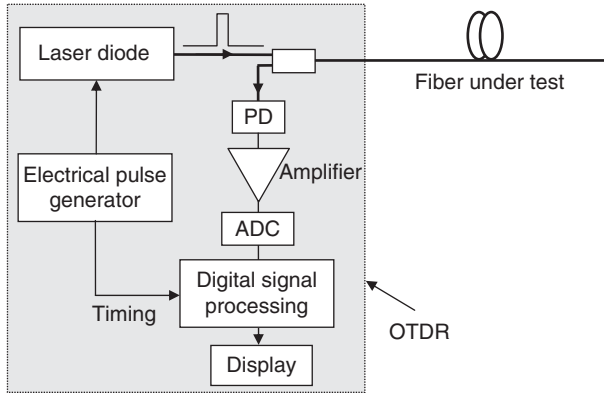


Figure 4.3.3 Block diagram of an OTDR. *PD*: photodetector, *ADC*: analog-to-digital converter.

backscattering is a linear process, at any point along the fiber, the magnitude of the backscattered optical power is linearly proportional to the optical power at that location. Due to the effect of fiber loss, both the transmitted and the backscattered powers are gradually attenuated along the fiber. The measurement of the time-dependent waveform of the backscattered power at the fiber input terminal provides the information about the loss distribution along the fiber; this information can be used to precisely calculate the attenuation coefficient.

Figure 4.3.3 shows the block diagram of an optical time-domain reflectometer (OTDR). A laser diode is driven by an electrical pulse generator that produces a train of short optical pulses. A photodiode is used to detect the backscattered optical power from the fiber through a directional coupler. The detected waveform of the optical signal is amplified and digitized through an ADC and analyzed by a digital-signal processing (DSP) unit. The timing of the DSP is synchronized with the source optical pulses so that the propagation delay of each backscattered pulse can be precisely calculated.

In general, the loss in an optical fiber may be caused by absorption, scattering, bending, and connecting. Most of these effects can be nonuniform along the fiber, especially if different fiber spools are connected together. Therefore, the fiber attenuation coefficient is a function of the location along the fiber. The power distribution along the fiber can be expressed as

$$P(z) = P(0) \exp \left\{ - \int_0^z [\alpha_0(x) + \alpha_{SC}(x)] dx \right\} \quad (4.3.3)$$

where $P(0)$ is the input optical power, $\alpha_{SC}(x)$ is the attenuation coefficient due to Rayleigh scattering, and $\alpha_0(x)$ is that caused by attenuation effects other than

scattering along the fiber. Suppose an optical pulse is injected into the fiber at time t_0 with the pulse width τ ; neglecting the effect of chromatic dispersion and fiber nonlinearity, the locations of the pulse leading edge and the trailing edge along the fiber at time t are $z_{le} = v_g(t - t_0)$ and $z_{tr} = v_g(t - t_0 - \tau)$, respectively, where v_g is the group velocity of the optical pulse.

Then we consider the Rayleigh backscattering caused by a short fiber section of length dz . According to Equation 4.3.3, the optical power loss due to Rayleigh scattering within this short fiber section is

$$dP_{SC}(z) = P(z)\alpha_{SC}(z)dz \quad (4.3.4)$$

Only a small fraction of this scattered energy is coupled to the guided mode, which can propagate back to the input side of the fiber. Therefore the reflected optical power that is originated from the location z and reaches to the input end of the fiber can be calculated as

$$dP_{BS}(z) = P(0) \cdot \eta \cdot \alpha_{SC}(z) \cdot dz \cdot \exp\left\{-2 \int_0^z \alpha(x)dx\right\} \quad (4.3.5)$$

where $\alpha(x) = \alpha_0(x) + \alpha_{SC}(x)$ is the composite attenuation coefficient of the fiber, which includes both the scattering and other attenuation effects. $\eta = (1 - \cos \theta_1)/2$ is the conversion efficiency from the scattered light to that captured by the fiber, as explained by Equation 1.3.59, and θ_1 is the maximum trace angle of the guided mode in the fiber, which is proportional to the numerical aperture, as illustrated in Figure 1.3.12. More precisely, if the normalized frequency, V , is between $1.5 < V < 2.4$, this conversion efficiency can be estimated as [Nakazawa, 1983]

$$\eta \approx \frac{1}{4.55} \left(\frac{NA}{n_1} \right) \quad (4.3.6)$$

where n_1 is the refractive index of the fiber core. As an example, in a silica-based single-mode fiber with $NA = 0.11$ and $n_1 = 1.45$, the conversion efficiency is approximately $\eta \approx 0.13\%$.

As illustrated in Figure 4.3.4, at time t_1 , the backscattered optical signal that reaches the input terminal of the fiber is originated from a short fiber section of length $\Delta z/2 = (z_{le} - z_{tr})/2$, and the amplitude is therefore

$$P_{BS}(z) = \frac{v_g \tau}{2} P(0) \cdot \eta \cdot \alpha_{SC}(z) \cdot \exp\left\{-2 \int_0^z \alpha(x)dx\right\} \quad (4.3.7)$$

where $\Delta z = v_g \tau$ has been used. Assume that both the scattering loss α_{sc} and the capturing coefficient η are constants along the fiber; the attenuation

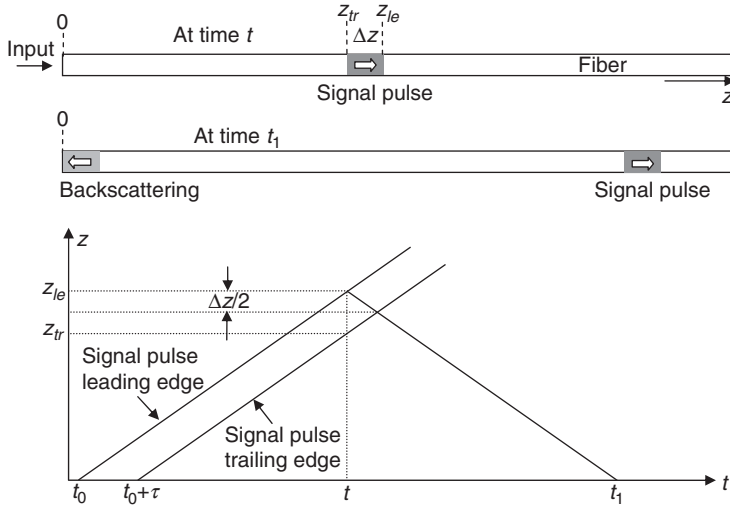


Figure 4.3.4 Illustration of the locations of the signal optical pulse, the scattering section, and the backscattered pulse in the fiber.

coefficient at location z can be evaluated from the differential return loss from Equation 4.3.7:

$$\alpha_{dB}(z) = -\frac{4.343}{2} \cdot \frac{d}{dz} \ln \left(\frac{P_{BS}(z)}{P(0)} \right) = -\frac{1}{2} \frac{d}{dz} \left(\frac{P_{BS}(z)}{P(0)} \right)_{dB} \quad (4.3.8)$$

where the unit of the attenuation coefficient α_{dB} is in dB and $\left(P_{BS}(z)/P(0) \right)_{dB}$ is the normalized return loss, also measured in dB. It is worthwhile to note that the slope of the return loss as the function of z alone is sufficient to determine the fiber attenuation coefficient. However, the source power $P(0)$, the pulse duration τ , and the efficiency of capturing the backscattered power into guided mode $\eta_{\alpha_{SC}}$ are important to determine the actual scattering optical power that reaches the receiver, and thus to evaluate the signal-to-noise ratio (SNR). Although the Rayleigh backscattering power P_{BS} is usually measured as the function of time, it is straightforward to convert the time scale into the fiber length scale z based on the well-known group velocity of the optical pulse. The following is an example of OTDR measurement.

A standard single-mode fiber has the attenuation coefficient $\alpha_{dB} = 0.25 \text{ dB/km}$ in a 1550 nm wavelength window. Assume that the attenuation is uniform along the fiber and a large part of this attenuation is caused by Rayleigh backscattering such that $\alpha_{SC,dB} = 0.2 \text{ dB/km}$ and $\alpha_{0,dB} = 0.05 \text{ dB/km}$. We also assume that the efficiency of capturing the backscattered power into guided mode is

$\eta = 1.19 \times 10^{-3}$ and the signal optical pulse launched into the optical fiber has the peak power of $P(0) = 10 \text{ mW}$. The backscattered optical power that reaches the receiver at the fiber input terminal is

$$\begin{aligned}
 P_{BS}(z) &= \frac{v_g \tau}{2} P(0) \cdot \eta \cdot \frac{\alpha_{SC,dB}(z)}{4.343} \cdot \exp \left\{ -2 \int_0^z \alpha(x) dx \right\} \\
 &= 5.59 \times 10^{-3} \tau \cdot \exp \left\{ -2 \times \frac{0.25}{4.343} z \right\}
 \end{aligned} \tag{4.3.9}$$

where τ is the pulse width and $v_g = 2.04 \times 10^{-8} \text{ m/s}$ has been used considering that the refractive index of silica is approximately 1.47.

Figure 4.3.5 shows the backscattered optical power that reaches the receiver at the fiber input terminal as the function of the pulse location z along the fiber or as the function of time t . Due to the low capturing efficiency of the backscattered optical power by the fiber propagating mode, the level of the received signal power is at least 60 dB lower than the original optical signal peak power when the pulse width is $1 \mu\text{s}$. This power level is linearly reduced with further decreasing the pulse width. Therefore, in practical implementations, there are very stringent requirements on the receiver sensitivity, the dynamic range, and the linearity.

If two fibers are connected together either by a connector or by fusion splicing, there will be a loss at the location of connection. In addition, because of the discontinuity possibly caused by the air gap if a connector is used, there is usually a discrete reflection peak in the OTDR trace. The connection loss can be

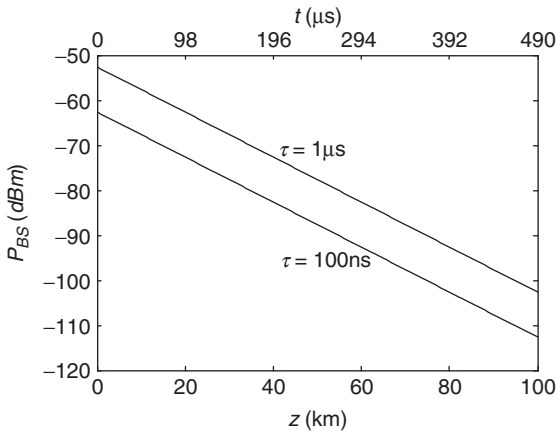


Figure 4.3.5 The backscattered power that reaches the receiver as the function of the location of the signal optical pulse. The input pulse peak power is 10 dBm.

evaluated straightforwardly by OTDR measurements. Assuming that the two fibers are identical with a uniform attenuation coefficient, the backscattered optical power measured by the OTDR is

$$P_{BS}(z) = \begin{cases} \frac{v_g \tau}{2} P(0) \cdot \eta \cdot \frac{\alpha_{SC,dB}}{4.343} \cdot \exp^{-2\alpha z} & (z < L_s) \\ A_s^2 \frac{v_g \tau}{2} P(0) \cdot \eta \cdot \frac{\alpha_{SC,dB}}{4.343} \cdot \exp^{-2\alpha z} & (z > L_s) \end{cases} \quad (4.3.10)$$

where L_s is the location and A_s is the fractional loss of the connection. A_s is squared in Equation 4.3.10 because the roundtrip pass of the optical signal across the connection point. Therefore the connection loss can be obtained from the measured OTDR trace; simply compare the backscattered optical power immediately before and immediately after the connection point, as illustrated in Figure 4.3.6:

$$A_s = \sqrt{\frac{P_{BS}(L_s + \delta)}{P_{BS}(L_s - \delta)}} \bigg|_{\delta \rightarrow 0} \quad (4.3.11)$$

where δ is a very short fiber length.

However, this measurement may not be accurate if the two fibers are not of the same type. In fact, even if they are of the same type, different fiber spools may still have slightly different numerical apertures or different attenuation coefficients. In this more practical case, the backscattered optical powers measured immediately before and after the splicing point are

$$P_{BS}(L_s - \delta) = \frac{v_g \tau}{2} P(0) \cdot \eta_1 \cdot \frac{\alpha_{SC1,dB}}{4.343} \cdot \exp^{-2\alpha_1 L_s} \quad (4.3.12a)$$

$$P_{BS}(L_s + \delta) = A_s^2 \frac{v_g \tau}{2} P(0) \cdot \eta_2 \cdot \frac{\alpha_{SC2,dB}}{4.343} \cdot \exp^{-2\alpha_1 L_s} \quad (4.3.12b)$$

where η_1 and η_2 are the conversion efficiencies defined by Equation 4.3.6 for the first and second fiber sections and α_{SC1} and α_{SC2} are the scattering

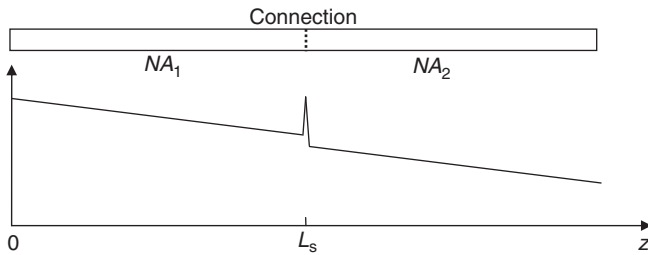


Figure 4.3.6 Illustration of an OTDR trace for a fiber with splicing.

coefficients for the two fiber sections, respectively. Since $\delta \rightarrow 0$, the fiber attenuation in the second fiber section is negligible; therefore only α_1 is used in Equation 4.3.12, which is the fiber loss coefficient in the first fiber section. Obviously,

$$A_{s,L} = \sqrt{\frac{P_{BS}(L_s + \delta)}{P_{BS}(L_s - \delta)}} \bigg|_{\delta \rightarrow 0} = A_s \sqrt{\frac{\eta_2 \cdot \alpha_{SC2,dB}}{\eta_1 \cdot \alpha_{SC1,dB}}} \quad (4.3.13)$$

may not accurately predict the splicing loss. The measurement error can be caused by the difference in the scattering and capturing efficiencies between the two fiber sections.

To overcome this measurement error, a useful technique often used in fiber testing is to perform OTDR measurement from both sides of the fiber. First, if the OTDR is used from the left side of the fiber shown in Figure 4.3.6, the measured OTDR trace in the vicinity of the splicing point, $z = L_s$, is given by Equation 4.3.12. As the second step, the OTDR measurement is launched from the right side of the fiber. The measured backscattering values immediately after and before the splicing point are,

$$P'_{BS}(L_s - \delta) = A_s^2 \frac{v_g \tau}{2} P(0) \cdot \eta_1 \cdot \frac{\alpha_{SC1,dB}}{4.343} \cdot \exp^{-2\alpha_2 L'_s} \quad (4.3.14a)$$

$$P'_{BS}(L_s + \delta) = \frac{v_g \tau}{2} P(0) \cdot \eta_2 \cdot \frac{\alpha_{SC2,dB}}{4.343} \cdot \exp^{-2\alpha_2 L'_s} \quad (4.3.14b)$$

where α_2 is the loss coefficient and L'_s is the length of the second fiber section, respectively (from the splicing point to the right end). From this second measurement, one can obtain

$$A_{s,R} = \sqrt{\frac{P'_{BS}(L_s - \delta)}{P'_{BS}(L_s + \delta)}} \bigg|_{\delta \rightarrow 0} = A_s \sqrt{\frac{\eta_1 \cdot \alpha_{SC1,dB}}{\eta_2 \cdot \alpha_{SC2,dB}}} \quad (4.3.15)$$

Because the actual connection loss is reciprocal and should be independent of the direction in which the OTDR signal is launched, the value of A_s in Equations 4.3.13 and 4.3.15 should be identical. Combining the results of these two measurements, the correct connection loss can be obtained as

$$A_s = \sqrt{A_{s,L} A_{s,R}} \quad (4.3.16)$$

This automatically removes the measurement error introduced by fiber type mismatch [15].

In general, an optical fiber may consist of many sections that are concatenated together by splicing and connecting. Figure 4.3.7 illustrates a typical OTDR trace for a fiber which is composed of multiple sections. In this case each fiber section may have different attenuation coefficient which can be seen from the slope of the section on the OTDR trace. The sharp peaks in the OTDR trace

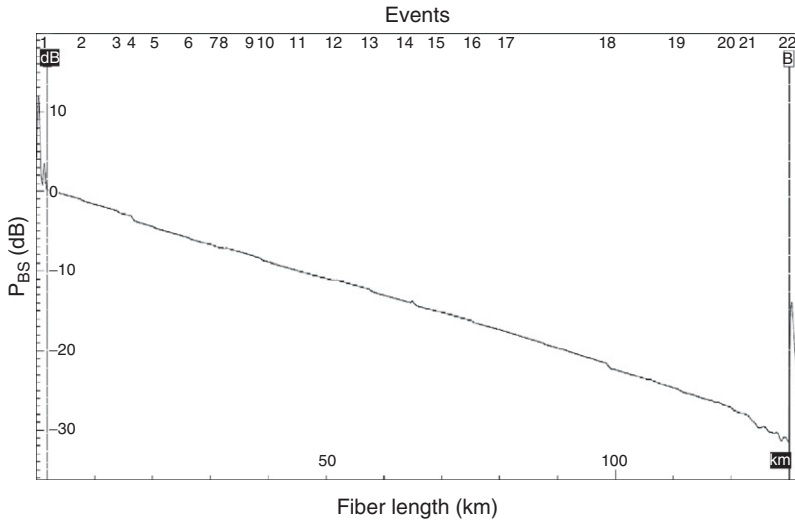


Figure 4.3.7 Illustration of an OTDR trace for a fiber with multiple sections.

indicate the locations and the strengths of reflections from the connectors, while the splices simply introduce attenuation and the reflection is usually small. In addition, negative attenuations can be found in a number of fiber splices (events 8, 12, 15 and 17 in [Tab. 4.3.1](#)). This is the result of the change in the numerical aperture of the fiber sections. Therefore, more accurate evaluation of splicing loss has to use the measurements from both directions as indicated in [Equation 4.3.16](#).

For accurate fiber attenuation measurement using OTDR, receiver SNR is a very important concern; it limits the total fiber length the OTDR can measure. For example, for a high-quality receiver with noise-equivalent power of $NEP = 0.5 \text{ pW}/\sqrt{\text{Hz}}$, if the bandwidth of the receiver is 1 MHz, the required signal optical power that reaches the receiver is $P_{BS} = 0.5 \text{ nW} = -63 \text{ dBm}$ to achieve $SNR = 1$. Because the noise is random while the signal is deterministic, the SNR can be improved by averaging. In general, the average over N pulses will increase the SNR by simply a factor of $\sqrt{N}$. Therefore, an OTDR measurement usually takes time, from a few seconds to a few minutes, depending on the number of averages required.

4.3.3 Improvement Considerations of OTDR

One possibility to improve the sensitivity of an OTDR is to use coherent detection. Unlike direct detection where the SNR is mainly limited by the receiver thermal noise and the dark current noise of the photodiode, the SNR

Table 4.3.1
Explanation of the OTDR trace in Fig. 4.3.6

	Event (23)	Distance (km)	Attenuation (dB)	Reflectance (dB)	Slope (dB/km)	Rel. Dist. (km)	Link Budget (dB)
1		0.98226	0.545	−41.09		0.98226	0.191
2		7.53062	0.168		0.195	6.54837	2.013
3		13.75157	0.219		0.202	6.22095	3.446
4		16.20721	0.532		0.202	2.45564	4.152
5		20.13623	0.100		0.216	3.92902	5.541
6		26.19347	0.131		0.204	6.05724	6.868
7		30.44991	0.098		0.195	4.25644	7.837
8		32.33257	−0.159		0.187	1.88266	8.280
9		37.32570	0.037		0.209	4.99313	9.174
10		38.63537	0.187		0.214	1.30967	9.492
11		44.93817	0.076	−9.81	0.200	6.30280	10.937
12		51.07727	−0.181		0.194	6.13909	12.195
13		57.38007	0.234		0.202	6.30280	13.295
14		63.60102	0.153		0.211	6.22095	14.843
15		68.92157	−0.066		0.199	5.32055	16.047
16		75.14252	0.188		0.208	6.22095	17.285
17		81.11791	−0.012		0.200	5.97539	18.665
18		98.47108	0.407		0.232	17.35317	22.682
19		110.74927	0.165		0.227	12.27819	25.862
20		119.83513	0.372		0.228	9.08586	28.093
21		122.94560	0.692		0.229	3.11047	29.174
22		129.98510		<31.07	0.360	7.03950	32.422
23							

in coherent detection is primarily limited by the shot noise due to the strong local oscillator. As shown in Figure 2.7.4, the SNR decreases with signal optical power at the rate of −20 dB/decade for direct detection, whereas for coherent detection this slope is only −10 dB/decade. That is why coherent detection is especially advantageous at very low signal optical power levels in comparison with direct detection.

Figure 4.3.8 shows the block diagram of a coherent OTDR, where the output from a single-frequency laser diode is split into two parts. One of them is used as

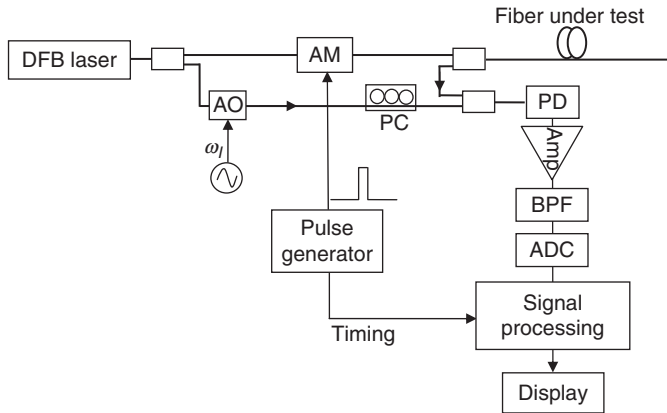


Figure 4.3.8 Block diagram of an OTDR with coherent detection. *AM*: amplitude modulator, *AO*: acousto-optic frequency modulator, *PC*: polarization controller, *PD*: photodetector, *BPF*: RF bandpass filter, *ADC*: analog-to-digital converter.

the stimulating signal that is modulated into a pulse train by an external modulator and sent into the fiber under test. The other part is used as the local oscillator for coherent detection. An acousto-optic (AO) modulator is used to shift the optical frequency of the local oscillator by ω_{IF} , which determines the intermediate frequency (IF) of the coherent heterodyne detection. The backscattered optical signal from the fiber is combined with the frequency-shifted local oscillator through a fiber coupler and detected by a wideband photodiode. A polarization controller is used in the local oscillator branch to match the state of polarization and maximize the coherent detection efficiency. An RF bandpass filter with the central frequency at ω_{IF} is used to select the intermediate frequency component. Generally, in coherent detection the power of the local oscillator is strong enough such that the receiver SNR is mainly limited by the shot noise of the local oscillator.

As an example, assume that the photodiode responsivity is $\Re = 1A/W$ and the receiver bandwidth is 1 MHz. Based on Equation 2.7.16, the required signal optical power is approximately -95 dBm to achieve $SNR = 1$. Compared to the direct detection case discussed at the end of [Section 4.3.2](#) in which the minimum detectable power level was -63 dBm, the OTDR with coherent detection has more than three orders of magnitude sensitivity improvement over its direct detection counterpart. However, the price paid for this sensitivity improvement are increased complexity as well as the requirements of a high-quality single-frequency laser diode and the frequency shifter for the local oscillator.

Spatial resolution is another concern for an OTDR. A wide optical pulse carries high optical energy for a fixed-peak power level, and at the same time the required receiver electrical bandwidth is also relatively low. Both of these factors may contribute to increasing the receiver SNR. However, wide pulse width corresponds to lower spatial resolution in the OTDR measurement. In fact, the spatial resolution of an OTDR is proportional to the pulse width τ by $R_{\text{resolution}} = 0.5v_g\tau$. For example, a pulse width of 100 ns in an OTDR will provide a spatial resolution of approximately 10 m in the fiber. Therefore trade-offs have to be made between the resolution and the SNR.

Dynamic range of the optical receiver in an OTDR is also an important issue. In the example shown in [Figure 4.3.5](#), there is 50 dB reduction in the received optical power level over 100 km single-mode fiber. Theoretically the dynamic range requirement is determined by the roundtrip fiber attenuation over the total length L , which is $\Delta P_{\text{dB}} = 2\alpha_{\text{dB}}L$. Commercial OTDRs usually provide >90 dB dynamic range to allow the measurement of long fibers with high losses.

Another practical issue is that one has to consider the Fresnel reflection at the fiber input terminal. For a well-cleaved fiber end facet, considering the refractive indices between air ($n = 1$) and silica ($n = 1.47$), the Fresnel reflection is approximately 3.6 percent, or -14.4 dB, which is several orders of magnitude higher than the Rayleigh backscattering from inside the fiber. Therefore most OTDR designs have to block this front-end reflection using proper time gating. This usually introduces difficulty in the measurement of fiber characteristics near the input, which is commonly referred to as the *dead zone*. Nevertheless, OTDR is by far the most popular equipment for fiber attenuation measurement, fiber length measurement, and troubleshooting in fiber-optical systems.

4.4 FIBER DISPERSION MEASUREMENTS

The phenomenon of dispersions in optical fibers, their creation mechanisms, and their consequences in fiber-optic systems were discussed briefly in Chapter 1. Fiber dispersion can be categorized into intermodal dispersion and chromatic dispersion. Intermodal dispersion is caused by the fact that different propagation modes in a fiber travel at different speeds. Usually a large number of modes coexist in a multimode fiber; therefore intermodal dispersion is the major source of dispersion in multimode fibers. Chromatic dispersion is originated from the fact that different frequency components in each propagation mode may travel at slightly different speeds and therefore it is the dominant dispersion source in single-mode fibers. Both intermodal dispersion and chromatic dispersion may cause optical signal pulse broadening and waveform distortion in fiber-optic systems.

4.4.1 Intermodal Dispersion and Its Measurement

In a multimode optical fiber, the optical signal is carried by many different propagating modes. Each mode has its own spatial field distribution and its propagation constant in the longitudinal direction. Because different modes have different propagation speeds, an optical pulse has different arrival times after traveling through the fiber.

Figure 4.4.1 shows a ray trace of propagation modes in a multimode fiber, where each mode travels in a different trace angle. According to Fresnel refraction theory, the largest incidence angle at the core/cladding interface is determined by the critical angle, $\theta \geq \theta_c = \sin^{-1}(n_2/n_1)$. Obviously the slowest propagation mode is the one with the incidence angle $\theta = \theta_c$ at the core/cladding interface, whereas the fastest propagation mode is the one that propagates exactly in the direction of the fiber axis ($\theta = \pi/2$). Thus to the first approximation, the maximum intermodal dispersion can be estimated by the propagation delay difference between these two extreme ray traces. The maximum differential delay for a fiber of unit length is

$$D_{\text{mod}} = \frac{n_1}{c} \left(\frac{1}{\sin \theta_c} - 1 \right) = \frac{n_1}{c} \left(\frac{n_1 - n_2}{n_2} \right) \quad (4.4.1)$$

The unit of intermodal dispersion is in [ns/km]. Equation 4.4.1 indicates that intermodal dispersion is proportional to the index difference between fiber core and cladding. For example, for a multimode fiber with a numerical aperture of $NA = 0.2$ and $n_1 = 1.5$, the absolute maximum intermodal dispersion is approximately 45 ns/km. However, in practice most of the optical signal energy in a multimode fiber is carried by lower-order modes and their intermodal dispersion values are much lower than that predicted by Equation 4.4.1. The actual intermodal dispersion value in a multimode fiber is on the order of 200 ps/km.

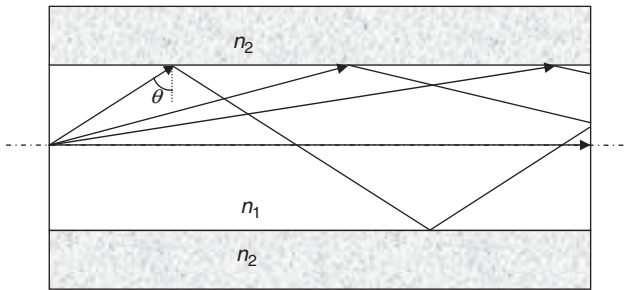


Figure 4.4.1 Geometric representation of propagation modes in a multimode fiber.

Generally, with an arbitrary launching condition, the lightwave signal in a multimode fiber excites only a certain number of modes instead of all the possible guided modes that can be supported by the fiber. Therefore the optical signal energy distribution among various modes is somewhat arbitrary near the input of the fiber which is quite sensitive to the launching condition. This may cause uncertainties in the intermodal dispersion measurements. In practice, an optical fiber is not ideally uniform in geometry. Signal optical energy may randomly couple between modes when there are bending, twisting and other intrinsic and extrinsic perturbations. Thus after a certain transmission distance, energy will be eventually coupled into most of the modes that the fiber can support and reaches to the equilibrium distribution of modes. To speed up this mode redistribution process, mode scrambler is often used in the measurement setups to make the results more stable and reliable. This can be accomplished by twisting or winding the fiber on an array of mechanical bars; therefore the measured intermodal dispersion is less sensitive to other mechanical disturbances.

Intermodal dispersion of a multimode fiber can be measured both in time domain and in frequency domain [16]. Since the dispersion value in a multimode fiber is typically >100 ps/km, the measurement accuracy requirement is generally not as stringent as it is for the chromatic dispersion measurement in single-mode fibers, where the dispersion values are much smaller. However, due to the randomness of mode coupling in a multimode fiber, the measurement only provides a statistic value of intermodal dispersion and a mode scrambler has to be used to ensure the repeatability of the measurement, especially when the fiber is short.

4.4.1.1 Pulse Distortion Method

Figure 4.4.2 shows the block diagram of an intermodal dispersion measurement setup using the pulse distortion method, which is also referred to as the pulse-broadening technique. In this setup, a short optical pulse is generated by a laser diode that is directly modulated by an electrical short pulse train or by active mode locking. The optical pulses are injected into the multimode fiber through a mode scrambler to ensure the equilibrium modal distribution in the fiber. A high-speed photodetector is used at the receiver to convert the received optical pulses into the electrical domain; therefore they can be amplified and the waveforms can be displayed on a sampling oscilloscope. Because of the intermodal dispersion of the fiber, the output pulses will be broader than the input pulses, and the amount of pulse broadening is linearly

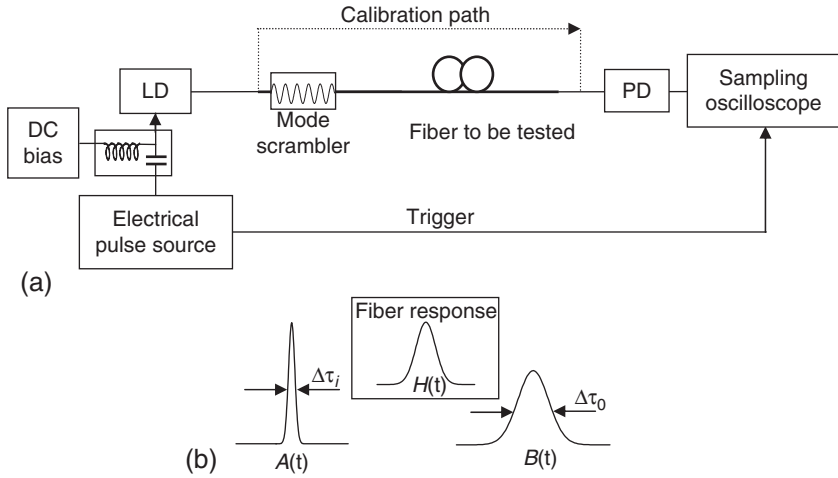


Figure 4.4.2 Pulse-broadening method to measure intermodal dispersion: (a) measurement setup and (b) input and output waveforms.

proportional to the intermodal dispersion of the fiber. In this way, the fiber dispersion characteristics can be directly measured in the time domain.

Assuming that the optical signal power is low enough so that the system operates in the linear regime, the relationship between the input waveform $A(t)$ and the output waveform $B(t)$ is related by

$$B(t) = A(t) \otimes H(t) \quad (4.4.2)$$

where $\otimes$ stands for convolution and $H(t)$ is the time-domain transfer function of the fiber. Consider in a multimode fiber, intermodal dispersion is the dominant source of dispersion effect; thus $H(t)$ essentially represents intermodal dispersion of the fiber. To take into account the finite bandwidth of the measurement system other than the fiber, we need a system calibration that directly connects the transmitter to the receiver, as illustrated by the calibration path in Figure 4.4.2. Since this is a linear system in which superposition is valid, the waveform measured in the calibration process can be regarded as the input waveform $A(t)$ in the calculation of the fiber transfer function $H(t)$. If we use Gaussian approximations for both the input and output signal waveforms and the fiber transfer function, the bandwidth of the fiber can be simply evaluated by the pulse width difference between the input $A(t)$ and the output $B(t)$. If the input pulse width is $\Delta\tau_i$ and the output pulse width is $\Delta\tau_0$, the response time of the fiber will be $\Delta\tau_H = \Delta\tau_0 - \Delta\tau_i$.

4.4.1.2 Frequency Domain Measurement

Although the time-domain pulse-broadening technique is easily understandable, a more straightforward way to find the fiber transfer function is in frequency domain. By converting the time-domain waveforms $A(t)$ and $B(t)$ into frequency-domain $A(\omega)$ and $B(\omega)$ through Fourier transform, the frequency domain transfer function of the fiber can be directly obtained by

$$H(\omega) = \frac{B(\omega)}{A(\omega)} \quad (4.4.3)$$

Figure 4.4.3 shows the block diagram of intermodal dispersion measurement in the frequency domain using an electrical network analyzer, where the network analyzer is set in the S_{21} parameter mode. A laser diode is intensity-modulated by the output of the network analyzer, which is a frequency-swept RF source. The DC bias and the RF modulation to the laser are combined through a bias-tee. The optical signal is launched into the fiber under test through a mode scrambler. A wideband photodiode is used to convert the received optical signal into electrical domain and the RF signal is then fed to the receiving terminal of the network analyzer. Because the effect of fiber dispersion is equivalent to a lowpass filter, by comparing the transfer functions

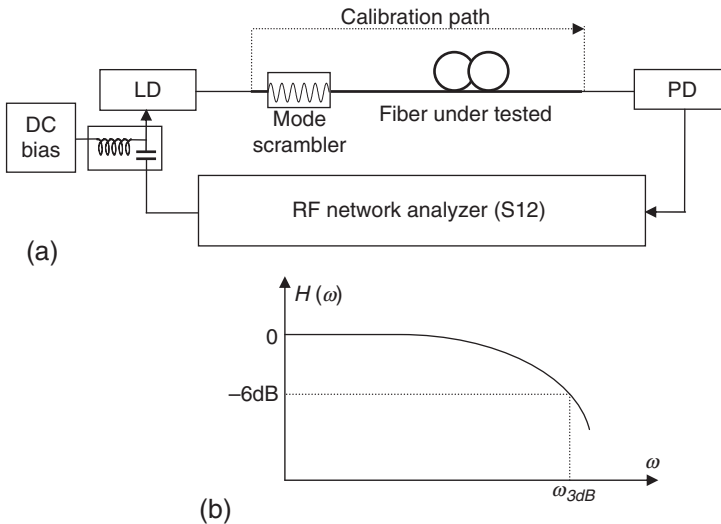


Figure 4.4.3 Frequency-domain method to measure intermodal dispersion: (a) measurement setup and (b) electrical power transfer function.

with and without the fiber in the system, the net effect of bandwidth limitation caused by the fiber can be obtained. The use of an RF network analyzer makes this measurement very simple and accurate. Compared to the time-domain measurement discussed previously, this frequency-domain technique does not require a short pulse generator and a high-speed oscilloscope. The fiber transfer function $H(f)$ is measured directly without the need for additional calculation using FFT.

The measurement procedure can be summarized as follows:

1. Remove the fiber and use the calibration path to measure the frequency response of the measurement system (transmitter/receiver pair and other electronic circuitry), which is recorded as $H_{cal}(\omega)$.
2. Insert the optical fiber and measure the frequency response again, which is recorded as $H_{sys}(\omega)$. Then the net frequency response of the fiber is

$$H_{fib}(\omega) = \frac{H_{sys}(\omega)}{H_{cal}(\omega)} \quad (4.4.4)$$

It is important to note that in this setup, the S_{21} parameter of the electrical power spectral transfer function is measured by the RF network analyzer. We know that at the output of the photodiode, signal electrical power is proportional to the square of the received optical power; therefore 3 dB optical bandwidth is equivalent to 6 dB bandwidth of the electrical transfer function of $H_{fib}(\omega)$. Or simply, a 6 dB change in the measured $H_{fib}(\omega)$ is equivalent to a 3 dB change in the optical power level.

Obviously, both the pulse-broadening method and the frequency-domain method are very effective in characterizing fiber dispersion by measuring the dispersion-limited information bandwidth. In general, the pulse-broadening method is often used to characterize large-signal response in the time domain, whereas the frequency-domain method is only valid for small-signal characterization. However, in fiber dispersion characterization, especially in the measurement of multimode fibers, the nonlinear effect is largely negligible, and thus the time-domain and frequency-domain methods should result in the same transfer function.

Another important note is that the unit of intermodal dispersion is in [ps/km], whereas the unit of chromatic dispersion is in [ps/nm/km]. The reason is that pulse broadening caused by intermodal dispersion is independent of the spectral bandwidth of the optical source and it is only proportional to the fiber length. Because the value of intermodal dispersion in multimode fibers is on the order of 100 ps/km, the RF bandwidth requirement of the measurement equipment is not very stringent. Typically a measurement setup with 15 GHz RF bandwidth is sufficient to characterize a multimode fiber of >2 km in length.

4.4.2 Chromatic Dispersion and Its Measurement

Chromatic dispersion is introduced because different frequency components within an optical signal may travel in different speed, which is the most important dispersion effect in single-mode fibers. But this does not mean a multimode fiber would not have chromatic dispersion. Since chromatic dispersion is usually much smaller than intermodal dispersion for the majority of multimode fibers, its effect is often negligible. From a measurement point of view, because of the relatively small chromatic dispersion values, the dispersion-limited information bandwidth is usually very wide; it is rather difficult to directly measure this equivalent bandwidth. For example, for a standard single-mode fiber operating in the 1550 nm wavelength window, the chromatic dispersion is approximately 17 ps/nm/km. It would require a picosecond level of pulse width for the optical source and >100 GHz optical bandwidth in the receiver to directly measure the pulse broadening in a 1 km fiber, which would be extremely challenging. Therefore alternative techniques have to be used to characterize chromatic dispersion in single-mode fibers. There are a number of different methods for this purpose, but in this section we will discuss the two most popular techniques: (1) the phase shift method, which operates in the time domain, and (2) the AM response method, which operates in the frequency domain.

4.4.2.1 Modulation Phase Shift Method

Figure 4.4.4 shows the measurement setup of the modulation phase shift method to characterize fiber chromatic dispersion. A tunable laser is used as the light source whose wavelength can be adjusted to cover the wavelength window in which the chromatic dispersion needs to be measured. An electro-optic intensity modulator converts a sinusoidal electrical signal at frequency f_m , into

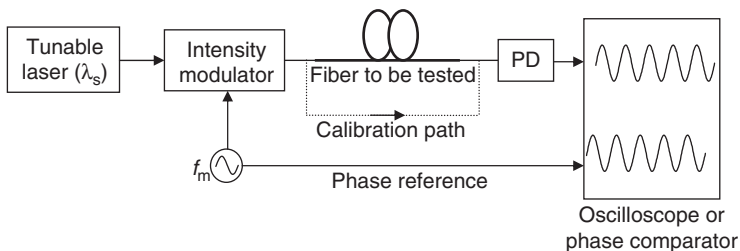


Figure 4.4.4 Experimental setup of the modulation phase shift method to measure fiber chromatic dispersion.

optical domain. After the modulated optical signal passes through the optical fiber, it is detected by a photodiode, and a sinusoidal electrical signal is recovered that is then amplified and sent to an oscilloscope. The waveform of the same sinusoid source at frequency f_m is also directly measured by the same oscilloscope for phase comparison.

Because of the chromatic dispersion in the fiber, the propagation delay of the optical signal passing through the fiber is different at different wavelengths: $\tau_g = \tau_g(\lambda)$. To measure this wavelength-dependent propagation delay, the optical source is modulated by a sinusoidal wave and the propagation delay can be evaluated by the relative phase retardation of the received RF signal. By varying the wavelength of the tunable laser, the RF phase delay as a function of the source wavelength, $\phi = \phi(\lambda)$, can be obtained [17, 18].

If the modulating frequency of the sinusoidal wave is f_m , a relative phase delay of $\Delta\phi = 360^\circ$ between two wavelength components corresponds to a group delay difference of $\Delta\tau_g = 1/f_m$ between them. Therefore the group delay versus optical signal wavelength λ can be expressed as

$$\Delta\tau_g(\lambda) = \frac{\phi(\lambda) - \phi(\lambda_r)}{360^\circ} \frac{1}{f_m} \quad (4.4.5)$$

where $\phi(\lambda) - \phi(\lambda_r)$ is the RF phase difference measured between wavelength λ and a reference wavelength λ_r . With the knowledge of group delay as a function of wavelength, $\phi(\lambda)$, the chromatic dispersion coefficient $D(\lambda)$ can be easily derived as

$$D(\lambda) = \frac{1}{L} \frac{d(\Delta\tau_g(\lambda))}{d\lambda} = \frac{1}{360^\circ L f_m} \frac{d\phi(\lambda)}{d\lambda} \quad (4.4.6)$$

where L is the fiber length. Since the value of chromatic dispersion is determined by the derivative of the group delay instead of its absolute value, the reference phase delay $\phi(\lambda_r)$ and the choice of reference wavelength are not important. In comparison, the selection of the RF modulation frequency may have a much bigger impact on the measurement accuracy. In general, a high modulation frequency helps increase phase sensitivity in the measurement. For a given fiber dispersion, the phase delay per unit wavelength change is linearly proportional to the modulation frequency: $d\phi(\lambda)/d\lambda = 360^\circ D(\lambda) L f_m$.

However, if the modulation frequency is too high, the measured phase variation may easily exceed 360° within the wavelength of interest; therefore the measurement system has to track the number of full cycles of 2π phase shift. In addition, the wavelength tuning step of the source needs to be small enough such that there is enough number of phase measurements within each 2π cycle. Therefore, the selection of the modulation frequency depends on the chromatic dispersion value of the fiber under test and the wavelength window the measurement needs to cover.

The tunable laser in Figure 4.4.4 can also be replaced by the combination of a wideband light source, such as superluminescence LED (SLED), and a tunable bandpass optical filter. The SLED can be directly modulated by injection current, and the bandpass optical filter can be placed either immediately after the SLED (before the fiber) or at the receiver side before the photodiode to select the signal wavelength.

As an example, a spool of 40 km standard single-mode fiber needs to be characterized in the wavelength window between 1530 nm and 1560 nm. We know roughly that the zero-dispersion wavelength of the fiber is around $\lambda_0 = 1310$ nm and the dispersion slope is approximately $S_0 = 0.09$ ps/nm²-km. Using the Sellmeier equation shown in Equation 1.3.84, the wavelength-dependent group delay is expected to be

$$\tau_g(\lambda) = L \int_{\lambda_r}^{\lambda} D(\lambda) d\lambda = \frac{S_0 L}{8} \left(\lambda^2 + \frac{\lambda_0^4}{\lambda^2} \right) + \tau_g(\lambda_r) \quad (4.4.7)$$

Figure 4.4.5 shows the expected relative group delay $\Delta\tau_g(\lambda) = \tau_g(\lambda) - \tau_g(\lambda_r)$ and the number of full 2π cycles of RF phase shifts for the modulation frequencies of 1 GHz, 5 GHz, and 10 GHz. To obtain this plot, the reference wavelength was set at 1530 nm. In terms of the measurement setup, the speed requirement of the electronic and opto-electronic devices and instrument is less than 10 GHz, which is usually easy to obtain. The measurement accuracy may be limited by the wavelength accuracy of the tunable laser and the step size of wavelength tuning. It may also be limited by noises in the measurement system when the signal level is low. Since this is a narrowband measurement, an RF bandpass filter at the modulation frequency can be used at the receiver to significantly reduce the noise level and improve the measurement accuracy.

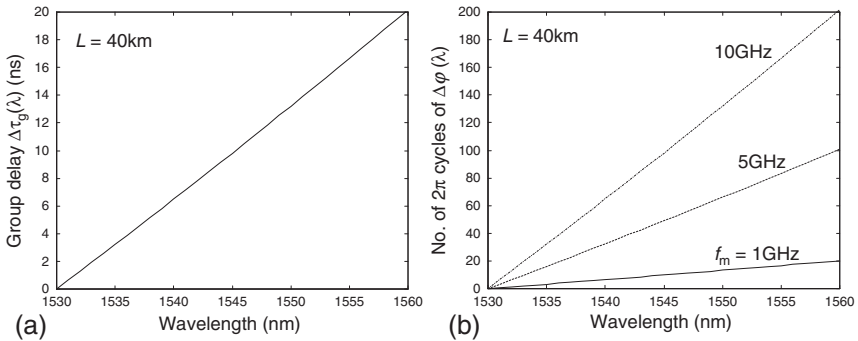


Figure 4.4.5 (a) Relative group delay and (b) the number of full 2π cycles of phase shift for the modulation frequencies of 1 GHz, 5 GHz, and 10 GHz. Fiber length is 40 km.

4.4.2.2 Baseband AM Response Method

The modulation phase shift method previously discussed relies on the RF phase comparison between the reference path and the signal passing through the fiber under test. An oscilloscope or a digitizing system may be used to perform waveform comparison. In this section we discuss the baseband AM response method, which measures chromatic dispersion based on the interference between modulation sidebands [19, 20]. Because of the chromatic dispersion, different modulation sidebands may experience different phase delays, and their interferences at the receiver can be used to predict the dispersion value.

Figure 4.4.6 shows the experimental block diagram of the AM response method to measure fiber chromatic dispersion. A tunable laser is used as the optical source whose output is intensity modulated by an external electro-optic modulator. The modulator is driven by the output from an RF network analyzer operating in the S_{21} mode. The network analyzer provides a frequency-swept RF signal to modulate the optical signal. After passing through the optical fiber under test, the optical signal is detected by a wideband photodiode. Then the weak RF signal is amplified and fed to the receiving port of the network analyzer. In fact, this measurement technique is similar to that shown in Figure 3.2.9 in Chapter 3, where the modulation chirp of an optical transmitter was characterized.

Through the analysis in Section 3.2 we know that the AM response of the fiber system shown in Figure 4.4.6 is not only determined by the transmitter modulation chirp; it also depends on the chromatic dispersion of the fiber. According to the AM modulation response given by Equation 3.2.22 the photocurrent in the receiver is,

$$I(f) \propto \cos \left[\frac{\pi \lambda_s^2 D(\lambda_s) f^2 L}{c} + \tan^{-1}(\alpha_{hw}) \right] \quad (4.4.8)$$

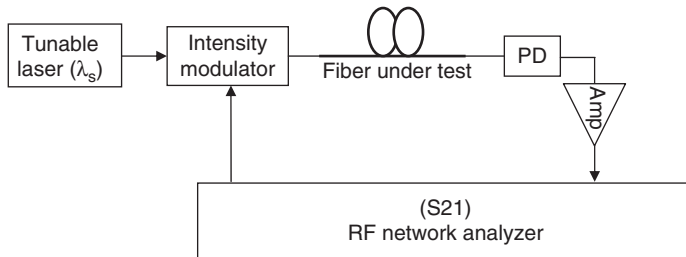


Figure 4.4.6 Experimental block diagram of AM response method to measure fiber chromatic dispersion.

where α_{hw} is the modulator chirp parameter and f is the modulation frequency. An important feature of this AM modulation response is the resonance zeroes at frequencies determined by the following equation:

$$f_k = \sqrt{\frac{c}{2D(\lambda_s)L\lambda_s^2} \left(1 + 2k - \frac{2}{\pi} \tan^{-1}(\alpha_{hw}) \right)} \quad (4.4.9)$$

In Chapter 3, the purpose of the experiment was to measure the modulator chirp parameter; we assumed that the fiber dispersion parameter D and the fiber length L were known. In this section, the purpose is to measure fiber chromatic dispersion, so we need to eliminate the impact of modulator chirp of the transmitter. From Equation 4.4.9, we can find that two consecutive zeroes in the AM response, f_k and f_{k+1} , are related only to the accumulated chromatic dispersion:

$$D(\lambda_s) \cdot L = \frac{c}{(f_{k+1}^2 - f_k^2)\lambda_s^2}, \quad (4.4.10)$$

This allows the accurate determination of fiber dispersion by measuring the spectrum of AM modulation response at each wavelength of the laser source. By adjusting the wavelength λ_s of the laser source, the wavelength-dependent chromatic dispersion parameter $D(\lambda_s)$ can be obtained. Compared to the modulation phase shift method, this AM response technique directly measures dispersion parameter $D(\lambda_s)$ at each source wavelength λ_s without the need to measure group delay around this wavelength. Since the zeroes in the spectrum of the AM response are sharp enough, as indicated by Figure 3.2.12, the measurement can be quite accurate. However, one practical concern might be the required RF bandwidth of the electro-optic devices as well as the RF network analyzer. In general, short fibers and low chromatic dispersions in the fiber require wide bandwidth of the measurement system. As an example, Figure 4.4.7(a) shows the normalized AM response for 100 km standard single-mode fiber with a 17 ps/nm/km dispersion parameter and 100 km True-wave fiber (TWF) with a 2 ps/nm/km dispersion parameter. The source modulation chirp is assumed to be zero. Figure 4.4.7(b) shows the first three frequencies of resonance zero in AM response as the function of the accumulated fiber dispersion $D \cdot L$. Using a network analyzer with 20 GHz bandwidth, the minimum accumulated dispersion the system can measure is about 500 ps/nm; this is because at least two resonance zeroes f_0 and f_1 need to be observed within the 20 GHz network analyzer bandwidth. 500 ps/nm corresponds to approximately 29 km of standard single-mode fiber with 17 ps/nm/km dispersion parameter.

From our discussion on chromatic dispersion measurement so far, we've found both modulation phase shift and AM modulation response techniques suitable for measuring fibers with relatively large accumulated dispersion. If

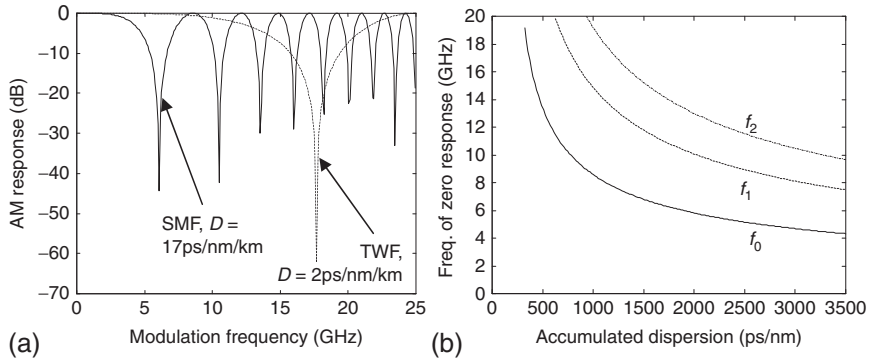


Figure 4.4.7 (a) Normalized AM responses of SMF with $D = 17$ ps/nm/km and TWF with $D = 2$ ps/nm/km. $L = 100$ km and $\alpha_{fw} = 0$ are assumed. (b) Frequencies of resonance zeroes in the AM response versus the accumulated chromatic dispersion DL .

a fiber sample is very short, for example, only a few meters, the required RF bandwidth for both of these techniques would have to be extremely wide. In this case, alternative techniques have to be used such as interferometric method.

4.4.2.3 Interferometric Method

The basic block diagram of the interferometric method for the measurement of chromatic dispersion is shown in Figure 4.4.8 [21,22]. A tungsten lamp is usually used as a wideband light source that can cover a very wide wavelength window from 500 nm to 1700 nm. The signal wavelength is selected by a tunable optical bandpass filter with the FWHM bandwidth of $\Delta\lambda$. The wavelength selection can also be accomplished with a monochromator before the photodetector. In that case, the bandwidth can be adjusted by changing the resolution of the monochromator. The optical signal is then fed into a Mach-zehnder interferometer with the fiber-under-test put in one arm while the length of the other arm is continuously variable. A photodiode at the output of the interferometer detects the optical signal and the waveform is displayed on an oscilloscope.

For the measurement using a Mach-zehnder interferometer, as discussed in Section 3.2.3, if the length difference between the two arms is much shorter than the source coherence length, coherent interference happens, and the photocurrent generated in the photodiode is

$$i(t) = \Re \left\{ P_1 + P_2 + 2\sqrt{P_1 P_2} \cos[\Delta\phi(\Delta L)] \right\} \quad (4.4.11)$$

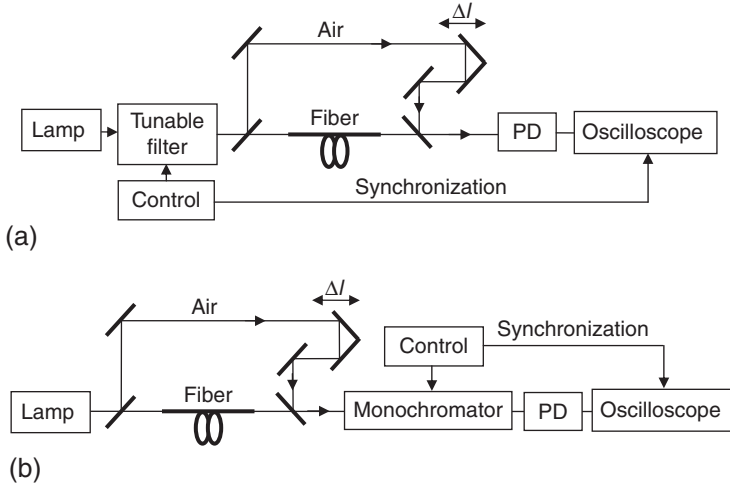


Figure 4.4.8 Interferometric technique to measure fiber chromatic dispersion: (a) using a tunable filter and (b) using a monochromator.

where Δl is the arm length difference, P_1 and P_2 are the optical powers in the two arms, $\Re$ is the photodiode responsivity, and $\Delta\varphi(\Delta l)$ is the relative optical phase difference between the two arms. On the other hand, if the length difference between the two interferometer arms is much longer than the source coherence length, the interference is incoherent and the photocurrent generated in the photodiode is simply proportional to the power addition of the two arms, that is,

$$i(t) = \Re\{P_1 + P_2\} \quad (4.4.12)$$

For the tunable optical filter with the FWHM bandwidth of $\Delta\lambda$, the coherence length of the selected optical signal is approximately

$$L_c = \frac{\lambda^2}{n_g \Delta\lambda} \quad (4.4.13)$$

where n_g is the group index of the material, and λ is the central wavelength selected by the optical filter. For example, for a signal optical bandwidth of $\Delta\lambda = 1$ nm in a 1550 nm wavelength region, the coherence length is approximately 2.4 mm in air.

Obviously, when the two interferometer arms have the same group delay, a slight change in the differential length Δl will introduce the maximum variation of photocurrent. Though outside the coherence length, the photocurrent is independent of the differential arm length Δl , as illustrated in [Figure 4.4.9](#).

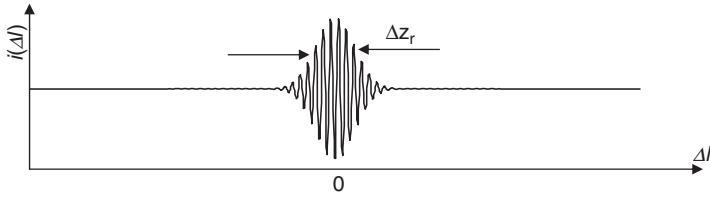


Figure 4.4.9 Illustration of the photocurrent versus the arm length difference.

The width of the envelope of the coherent interference pattern is approximately

$$\Delta z_r = \frac{\lambda^2}{n_g \Delta \lambda} \quad (4.4.14)$$

A narrower width of Δz_r is desired to help accurately determine the differential arm delay of the interferometer. According to the definition, chromatic dispersion is proportional to the derivative of the group delay:

$$D(\lambda) = \frac{1}{L} \frac{d(\tau_g(\lambda))}{d\lambda} \quad (4.4.15)$$

where L is the fiber length and $\tau_g(\lambda)$ is the wavelength-dependent group delay. To measure the chromatic dispersion in a fiber, its wavelength dependency of the group delay is the key parameter that needs to be evaluated. This can be accomplished by sweeping the wavelength of the tunable optical filter and measuring the relative delay as a function of the wavelength.

In the interferometer configuration shown in [Figure 4.4.8](#), the reference arm for which group delay is wavelength-independent is in the air, and therefore the important parameter to evaluate is the differential arm length difference as a function of wavelength $\Delta l(\lambda)$, which indicates the chromatic dispersion of the fiber under test.

$$D(\lambda) = \frac{1}{Lc} \frac{d(\Delta l(\lambda))}{d\lambda} \quad (4.4.16)$$

As illustrated in [Figure 4.4.10](#), when the tunable optical filter sweeps across the wavelength, the peak of the interference envelope will move, providing $\Delta l(\lambda)$. In practical measurements, the resolution of $\Delta l(\lambda)$ measurement is determined by the sharpness of the interference pattern Δz_r , as indicated by [Equation 4.4.14](#), while the minimum wavelength step is limited by the bandwidth of the tunable filter $\Delta \lambda$. Although it is desirable for both of them to be small, [Equation 4.4.14](#) indicated that their product is only determined by the square of the wavelength. As an example, if the filter bandwidth is 2 nm in a 1550 nm wavelength

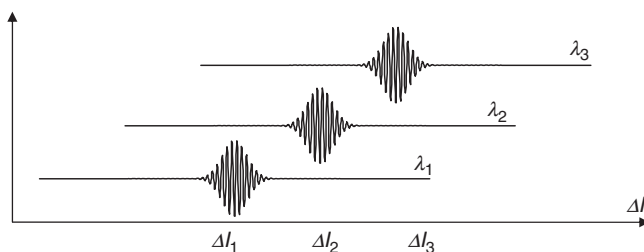


Figure 4.4.10 Illustration of measured arm length difference as a function of the filter wavelength.

window, the spatial resolution Δz_r is approximately 0.8 mm in the fiber, which corresponds to a relative time delay of 2.7 ps in air. The light source used in this technique can have a very wide wavelength window in which the measurement can take place. This fiber dispersion characterization technique covers wide bandwidth. In practice, the wavelength-dependent group delay and thus the chromatic dispersion can be measured in the wavelength window between 500 nm and 1700 nm by fitting the measured dispersion characteristic with the well-known three-term Sellmeier equation for standard single-mode fibers. It allows the precise determination of chromatic dispersion, even for short pieces of fiber samples.

However, the major disadvantage of this technique is that the fiber length cannot be too long. The major reason is that the reference arm is in free space so that a long reference arm may introduce mechanical instability and thus excessive coupling loss. Because the reference arm length has to be roughly equal to the length of the fiber sample, the typical fiber length in the interferometric measurement setup should not be longer than one meter. For fibers with very small dispersion parameters, or if the dispersion is a strong function of the wavelength that cannot be fitted by the Sellmeier equation, such as dispersion shifted or dispersion flattened fibers, it would be quite difficult to ensure the accuracy of the measurement.

To overcome the reference arm length limitation, an all-fiber setup can be useful for measuring the characteristics of long fibers. In the all-fiber setup, a foldback interferometer using a 3-dB fiber coupler may be employed with total reflection at the end of the reference fiber arm, as illustrated in Figure 4.4.11. This all-fiber setup is simple and easy to align; however, we must know precisely the dispersion characteristic of the fiber in the reference arm before the measurement. Another difficulty is that the bandwidth of a 3-dB fiber coupler is usually narrow compared to a thin film-based free-space beam splitter, and therefore the measurement may be restricted to a certain wavelength subband.

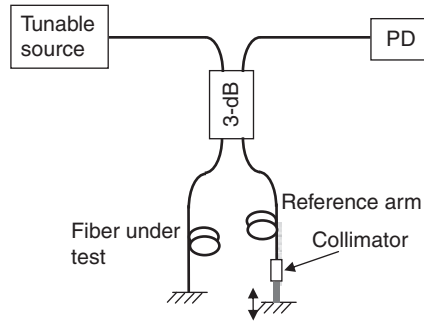


Figure 4.4.11 Dispersion measurement using a foldback all-fiber interferometer.

4.5 POLARIZATION MODE DISPERSION (PMD) MEASUREMENT

Unlike chromatic dispersion, which represents wavelength-dependent differential group delay, polarization mode dispersion is a special type of intermodal dispersion in single-mode optical fibers. It is well known that in a “single-mode” fiber there are actually two propagation modes co-exists, and they are orthogonally polarized; they’re commonly referred to as HE_{11}^x and HE_{11}^y . In an ideal cylindrically symmetrical optical fiber, these two modes are degenerate and they have identical propagation constant. However, in practical optical fibers, the cylindrical symmetry may not be precise; as a result, these two orthogonal propagation modes may travel in different group velocities due to the birefringence in the fiber.

4.5.1 Representation of Fiber Birefringence and PMD Parameter

The birefringence in an optical fiber is generally introduced through intrinsic or extrinsic perturbations. *Intrinsic perturbation* is related to permanent features in the fiber due to errors in the manufacturing process. Manufacturing errors during fiber production may cause noncircular fiber core, which is usually referred to as *geometric birefringence*, and nonsymmetric stress due to the material in the fiber perform, which is known as *stress birefringence*. On the other hand, *extrinsic perturbation* is often introduced due to external forces in the cabling and installation process. Both of these two types of perturbations contribute to the effect of birefringence; as a consequence, the two orthogonally polarized propagation modes exhibit different propagation constants [Ulrich 1980].

Assume that the propagation constant of the vertically polarized propagation mode is $\beta_{\perp} = \omega n_{\perp}/c$, the propagation constant of the horizontally polarized propagation mode is $\beta_{//} = \omega n_{//}/c$ where ω is the optical frequency, $n_{\perp}$ and $n_{//}$ are the effective refractive indices of the two polarization modes, and c is the speed of light. Because of the birefringence in the fiber, $n_{\perp}$ and $n_{//}$ are not equal; therefore $\beta_{\perp} \neq \beta_{//}$. Their difference is

$$\Delta\beta = \beta_{//} - \beta_{\perp} = \frac{\omega(n_{//} - n_{\perp})}{c} = \frac{\omega}{c} \Delta n_{eff} \quad (4.5.1)$$

where $\Delta n_{eff} = n_{//} - n_{\perp}$ is the effective differential refractive index. Typically Δn_{eff} is on the order of 10^{-5} to 10^{-7} in coiled standard single-mode fibers [24]. Then for a fiber of length L , the relative propagation delay between the two orthogonally polarized modes is

$$\Delta\tau = \frac{(n_{//} - n_{\perp})}{c} L = \frac{L \Delta n_{eff}}{c} \quad (4.5.2)$$

This is commonly referred to as *differential group delay* (DGD) of the fiber.

Because of fiber birefringence, the state of polarization (SOP) of the optical signal will rotate when propagating in the fiber. For a total phase walk-off $\Delta\beta L = 2\pi$ between the two orthogonal polarization modes, the SOP of the signal at the fiber output completes a full 2π rotation. According to Equations 4.5.1 and 4.5.2, this polarization rotation can be induced by the changes of the fiber length L , the differential refractive index Δn_{eff} , or the signal optical frequency ω .

As a simple example, consider an optical signal with two frequency components ω_1 and ω_2 . If these two frequency components are copolarized when the optical signal enters an optical fiber, then at the fiber output their polarization states will walk off from each other by an angle:

$$\Delta\theta = \frac{\Delta\omega \cdot \Delta n_{eff}}{c} L = \Delta\omega \cdot \Delta\tau \quad (4.5.3)$$

where $\Delta\omega = \omega_2 - \omega_1$. In other words, the fiber DGD value $\Delta\tau$ can be measured through the evaluation of differential polarization walk-off between different frequency components of the optical signal. In fact, this is the basic concept of the frequency domain PMD measurement technique, discussed later in this section.

The concept of birefringence is relatively straightforward in a short fiber where refractive indices are slightly different in the two orthogonal axes of the transversal cross-section. However, if the fiber is long enough, the birefringence axes may rotate along the fiber due to banding, twisting, and nonuniformity. Meanwhile, there is also energy coupling between the two orthogonally polarized propagation modes in the fiber. In general, both the birefringence axis rotation

and the mode coupling are random and unpredictable, which make polarization mode dispersion a complex problem to understand and to solve [25].

The following are a few important parameters related to fiber polarization mode dispersion:

1. *Principal state of polarization (PSP)* indicates two orthogonal polarization states corresponding to the fast and slow axes of the fiber. Under this definition, if the polarization state of the input optical signal is aligned with one of the two PSPs of the fiber, the output optical signal will keep the same SOP. In this case, the PMD has no impact in the optical signal, and the fiber only provides a single propagation delay. It is important to note that PSPs exist not only in “short” fibers but also in “long” fibers. In a long fiber, if 2nd order PMD is negligible, although the birefringence along the fiber is random and there is energy coupling between the two polarization modes, an equivalent set of PSPs can always be found. Again, PMD has no impact on the optical signal if its polarization state is aligned to one of the two PSPs of the fiber. However, in practical fibers the orientation of the PSPs usually change with time, especially when the fiber is long. The change of PSP orientation over time is originated from the random changes in temperature and mechanical perturbations along the fiber.

Using Stokes parameter representation on the Poincare sphere, the two PSPs are represented as two vectors, which start from the origin and point to the two opposite extremes on the Poincare sphere as shown [Figure 4.5.1](#). In a short fiber without energy coupling between the two polarization modes, the PSP of the fiber is fixed and is independent of optical signal frequency. In [Figure 4.5.1](#), different circles represent different input polarization states of the optical signal. For each input polarization state, the output polarization vector rotates on the Poincare sphere in a regular circle when varying the optical signal frequency. If the fiber is long enough, the PSP will be largely dependent on the signal

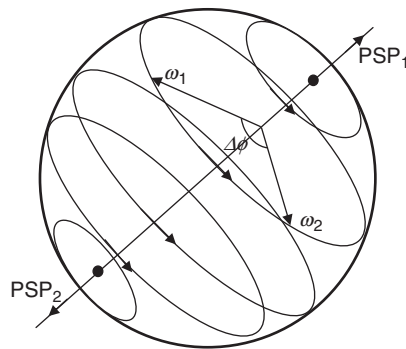


Figure 4.5.1 Definition of PSPs on Poincare sphere representation.

optical frequency, and the regular circles shown in Figure 4.5.1 will no longer exist.

2. *Differential group delay* (DGD) indicates the propagation delay difference between the signals carried by the two PSPs. In general, DGD is a random process due to the random nature of the birefringence in the fiber. The probability density function (PDF) of DGD in an optical fiber follows a Maxwellian distribution [25]:

$$P(\Delta\tau) = \sqrt{\frac{2}{\pi}} \frac{\Delta\tau^2}{\alpha^3} \exp\left(-\frac{\Delta\tau^2}{2\alpha^2}\right) \quad (4.5.4)$$

where $P(\Delta\tau)$ is the probability that the DGD of the fiber is $\Delta\tau$ and α is a parameter related to the mean DGD. In fact, according to the Maxwellian distribution given in Equation 4.5.4, the average value of DGD is

$$\langle\Delta\tau\rangle = \alpha\sqrt{\frac{8}{\pi}} \quad (4.5.5)$$

This is usually referred to as the mean DGD, while the root mean square (RMS) value of DGD is $\alpha\sqrt{3}$. A Maxwellian distribution with $\alpha = 0.5$ ps is shown in Figure 4.5.2, which is plotted both in the linear scale and in the logarithm scale. It is important to notice that even though the average DGD is only 0.8 ps in this case, low-probability events may still happen; for example, the instantaneous DGD may be 2 ps but at a low probability of 10^{-5} . This is an important consideration in calculating the outage probability of optical communication systems.

3. *Mean DGD*, which is the average DGD, $\langle\Delta\tau\rangle$ as defined in Equation 4.5.5. The unit of mean DGD is in picoseconds, mean DGD in a fiber is usually counted as the average value of DGD over certain ranges of wavelength,

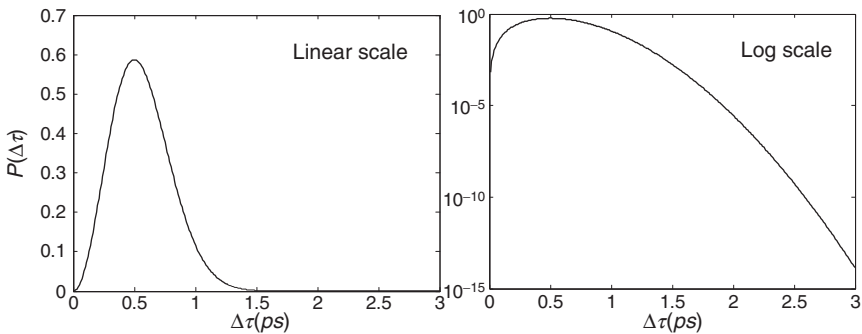


Figure 4.5.2 Probability of fiber DGD at the value of $\Delta\tau$, which has a Maxwellian distribution with $\alpha = 0.5$ ps and a mean DGD of 0.8 ps.

time, and temperature. For short fibers, mean DGD is proportional to the fiber length L : $\langle \Delta\tau \rangle \propto L$, whereas for long fibers, the mean DGD is proportional to the square root of fiber length $\sqrt{L}$: $\langle \Delta\tau \rangle \propto \sqrt{L}$. This is due to the random mode coupling and the rotation of birefringence axes along the fiber. The mean DGD is an accumulated effect over fiber; therefore random mode coupling makes the average effect of PMD smaller. However, due to the random nature of mode coupling, the instantaneous DGD can still be quite high at the event when the coupling is weak.

4. *PMD parameter* is defined as the mean DGD over a unit length of fiber. For the reasons we've discussed, for short fibers, the unit of PMD parameter is [ps/km], whereas for long fibers, the unit of PMD parameter becomes [ps/ $\sqrt{\text{km}}$]. Because of the time-varying nature of DGD, the measurement of PMD is not trivial. In the following sections, we discuss several techniques that are often used to measure PMD in fibers.

4.5.2 Pulse Delay Method

The simplest technique to measure DGD in an optical fiber is the measurement of differential delay of short pulses that are simultaneously carried by the two polarization modes [26]. Figure 4.5.3(a) shows a block diagram of the measurement setup. Typically a mode-locked laser source is used to provide short optical pulses and is injected into the fiber under test through a polarization controller. A high-speed photodiode converts the output optical signal. A high-speed photodiode converts the output optical signal

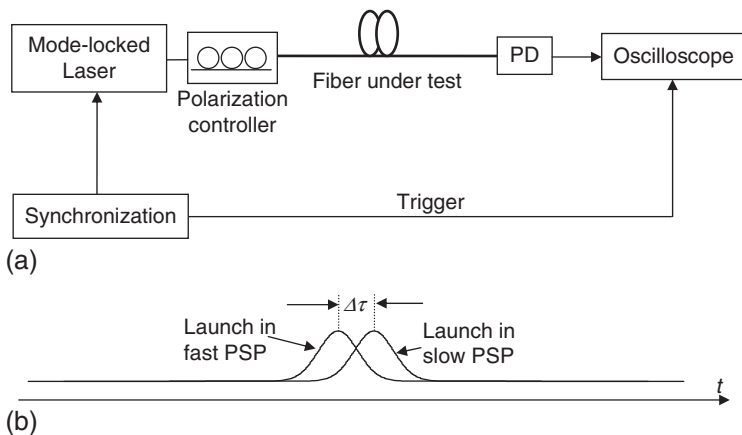


Figure 4.5.3 PMD measurement using differential pulse delay technique: (a) experimental setup and (b) illustration of differential pulse delay.

from the fiber into electrical domain, and the pulse waveform is displayed by an oscilloscope.

The propagation delay of the optical pulse depends on the group velocity of the optical signal that propagates in the fiber. Since the DGD is defined as the group delay difference between the fast axis and the slow axis of the fiber, it can be evaluated by directly measuring the fastest and the slowest arrival times of the optical pulses at the receiver, as shown in Figure 4.5.3(b). Since the optical signal from the source laser is polarized, the search for the fastest and the slowest group velocities in the fiber is accomplished by adjusting the polarization controller before the optical signal is launched into the fiber.

The underlying principle of the differential pulse-delay technique is straightforward and easy to understand. But the accuracy of this measurement, to a large extent, depends on the temporal width of the optical pulses used. Narrower optical pulses are obviously desired to measure fibers with low levels of DGD. However, chromatic dispersion in the fiber will broaden and distort the optical pulses. For example, to measure a 10 km standard single-mode fiber with 17 ps/nm/km dispersion parameter in a 1550 nm wavelength window, if the input optical pulse width is 10 ps, the optical spectral bandwidth is approximately 0.8 nm. The pulse width at the output of the fiber will be about 136 ps only due to chromatic dispersion. Therefore it would not be helpful to use very narrow pulses for DGD measurement when the fiber has significant chromatic dispersion. In fact, if the width of the input optical pulse is Δt_{in} , the output pulse width can be calculated by

$$\Delta t_{out} = \Delta t_{in} + \frac{D\lambda^2}{c\Delta t_{in}}L \quad (4.5.6)$$

where D is the fiber chromatic dispersion parameter, λ is the operation wavelength, and L is the fiber length. If we assume that $D = 17$ ps/nm/km and $L = 10$ km, the relationship between the widths of the output and the input pulses is shown in Figure 4.5.4. The minimum width of the output pulse is about 75 ps; obviously, it is not feasible to measure DGD values of less than a few picoseconds without proper dispersion compensation. On the other hand, if dispersion compensation is applied, the DGD of the dispersion compensator has to be taken into account and its impact in the overall result of measurement may become complicated.

To summarize, the differential pulse delay method is a time-domain technique that directly measures the DGD in a fiber, and the explanation of the results is rather straightforward. By changing the wavelength of the optical source, the wavelength dependency of DGD can be evaluated; thus the PMD parameter can also be obtained. Although the operating principle of this technique is simple; its accuracy may be limited by the achievable pulse width. This technique is usually used to measure fibers with low chromatic dispersion and relatively high levels of DGD.

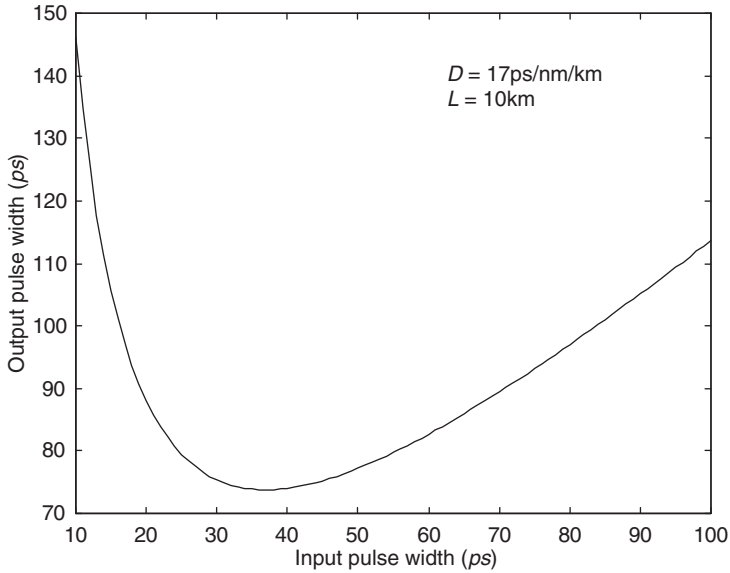


Figure 4.5.4 Output pulse width versus input pulse width in a 10 km fiber with 17 ps/nm/km chromatic dispersion parameter.

4.5.3 The Interferometric Method

As we just discussed in the last section, the major difficulty of the differential pulse delay method is the requirement of short optical pulses and ultrafast detection. These can be avoided using the optical interferometric technique. The principle of the interferometric method is based on the measurement of the differential delay between the signals carried by two PSPs using the low-coherent interferometer technique [27, 28].

Figure 4.5.5 shows the schematic diagram of a DGD measurement setup using the interferometric technique. In this system, a wideband lightwave source, such as a halogen lamp, is used. A monochromator selects the wavelength of the optical signal and the spectral bandwidth. If the spectral bandwidth of the light source is wide enough, it has a very short coherence length, and the optical signals reflected from the two interferometer arms of the Michelson interferometer are coherent only when the two arms have almost the same length. Assume that the spectral bandwidth selected from the light source is $\Delta\lambda$ and the center wavelength is λ ; its coherence length is approximately

$$\Delta l = \frac{1}{2n} \frac{\lambda^2}{\Delta\lambda} \quad (4.5.7)$$

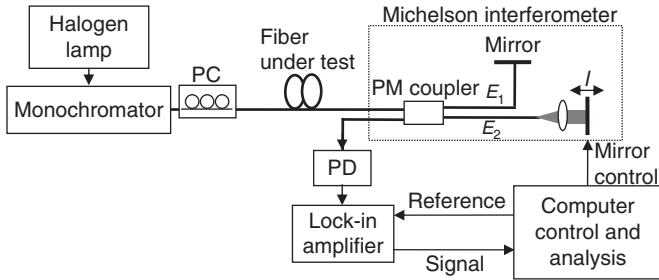


Figure 4.5.5 Optical setup for DGD measurement using interferometry method.

where n is the refractive index. In the Michelson interferometer, if the arm length difference is shorter than the source coherence length, the mixing between the optical fields coming from the two arms is coherent in the photodetector. On the other hand, if the arm length difference is longer than the coherence length, there is no phase relation between optical fields coming from the two arms, and therefore the optical powers add up at the photodetector incoherently. To demonstrate the operation principle, we assume that the polarization-maintaining fiber coupler has a 50 percent splitting ratio and the optical signal amplitudes in the two arms are equal, i.e., $|E_1| = |E_2|$. Within coherence length the mixed optical power at the photodiode will oscillate between zero (destructive interference) and $2|E_1|^2$ (constructive interference), depending on the phase relation between the two arms. Outside the coherence length, the total optical power will simply be a constant $|E_1|^2$, as illustrated in Figure 4.5.6.

When a birefringent optical fiber is inserted between the light source and the interferometer, the optical signal will be partitioned into the fast axis and the slow axis. At the fiber output, the optical field is

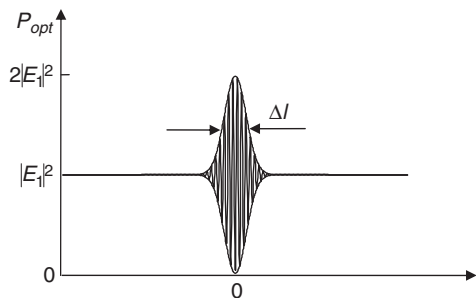


Figure 4.5.6 Illustration of interference pattern in a low-coherence interferometer. $|E_1|$ is the optical field in each of the two interferometer arms and l is the arm length difference.

$$E = (E_x e^{j\beta_x L} + E_y e^{j\beta_y L}) \quad (4.5.8)$$

where x and y are the two orthogonal PSPs of the fiber, β_x and β_y are the propagation constants of the fast and the slow axis, E_x and E_y are the amplitude of the optical signals carried by these two orthogonal modes, and L is the fiber length. Then, after a roundtrip through the Michelson interferometer, the optical signal at the photodiode is

$$E_0 = \frac{1}{2} (E_x e^{j\beta_x L} + E_y e^{j\beta_y L}) (e^{j\beta l} + 1) \quad (4.5.9)$$

where l is the differential delay between the two interferometer arms and $\beta = 2\pi n/\lambda$ is the propagation constant in the interferometer arm. We have assumed that the power-splitting ratio of the optical coupler is 50 percent. Then the square-law detection of the photodiode generates a photocurrent, which is

$$I = \eta |E_0|^2 = \frac{\eta}{4} \left| E_x (e^{j\beta_x L} + e^{j(\beta_x L + \beta l)}) + E_y (e^{j\beta_y L} + e^{j(\beta_y L + \beta l)}) \right|^2 \quad (4.5.10)$$

As illustrated in [Figure 4.5.7\(a\)](#), a strong coherent interference peak is generated when $l = 0$, which is mainly caused by the self-mixing terms of $|E_x|^2$ and $|E_y|^2$ in [Equation 4.5.10](#). In addition, two satellite peaks are generated when the differential arm length l satisfies

$$(\beta_y - \beta_x)L \pm \beta l = 0 \quad (4.5.11)$$

or approximately

$$l = \pm \Delta\tau \cdot c \quad (4.5.12)$$

where $\Delta\tau$ is the DGD of the fiber under test. These two coherent interference peaks represent the contribution of the cross-mixing between E_x and E_y terms

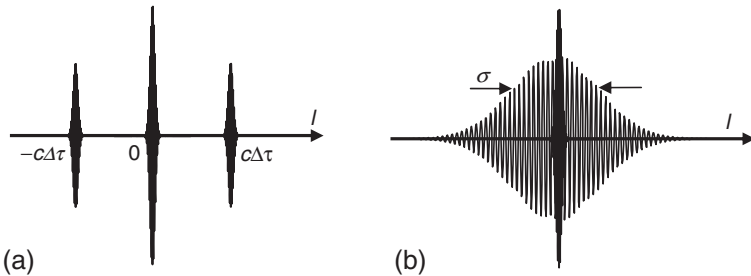


Figure 4.5.7 Interference pattern as the function of differential delay l of the interferometer: (a) short fiber and (b) long fiber with random mode coupling.

as given by Equation 4.5.10, and the width of each interference peak is determined by the spectral width of the source. By measuring the location of these two secondary interference peaks, the fiber DGD can be estimated with Equation 5.4.12. A good measurement resolution (and thus good accuracy) requires narrow interference peaks, which corresponds to a wide optical bandwidth used.

On the other hand, if the fiber is long enough, random-mode coupling exist between the two polarization modes and the PSPs vary along the fiber. In this case, the interference pattern consists of a stable central peak and randomly scattered sidebands due to random-mode coupling. The overall envelope of the energy distribution is generally Gaussian, as shown in Figure 4.5.7(b). Suppose the standard deviation of this Gaussian fitting is σ ; then the mean DGD can be calculated as $\langle \Delta\tau \rangle = \sigma\sqrt{2/\pi}$, whereas the RMS value of DGD is $\langle \Delta\tau^2 \rangle^{1/2} = \sigma\sqrt{3/4}$ [29].

In the long fiber situation, due to mode coupling in the fiber, the DGD value is generally a function of signal wavelength. This measurement essentially provides an averaged DGD over the signal spectral bandwidth $\Delta\lambda$. Referring to Figure 4.5.5, the wavelength resolution of the DGD measurement can be selected by choosing a proper width of the exit slit of the monochromator.

4.5.4 Poincare Arc Method

As discussed in Section 4.5.1, PMD is originated from the birefringence of the fiber, which can be measured by the frequency dependency of polarization rotation. A more sophisticated representation of the SOP is the use of Stokes parameters, in which a polarized optical signal is represented as a vector $\vec{S}$ on the Poincare sphere, as discussed in Section 2.6.2. After propagating through a birefringent fiber, the $\vec{S}$ vector will rotate around the PSP on the Poincare sphere when the wavelength (or the frequency) is changed.

In a *short fiber* without mode coupling, the PSP of the fiber is stable and independent of optical signal frequency. In this case the SOP vector of the optical signal rotates on the Poincare sphere in a regular circle around the PSP when the signal optical frequency is swept as shown in Figure 4.5.8(a). On the other hand, in a *long fiber*, where random mode coupling is significant, the PSP is no longer stable and is a function of signal optical frequency. In this case each frequency change of the optical signal has its corresponding PSP and the SOP vector wanders on the Poincare sphere irregularly with the change of signal optical frequency, as shown in Figure 4.5.8(b). By definition, the PMD vector Ω originates from the center of the Poincare sphere and points toward the PSP.

In either case in Figure 4.5.8, with the change of optical signal frequency ω , the signal polarization state at fiber output will change. This is represented by

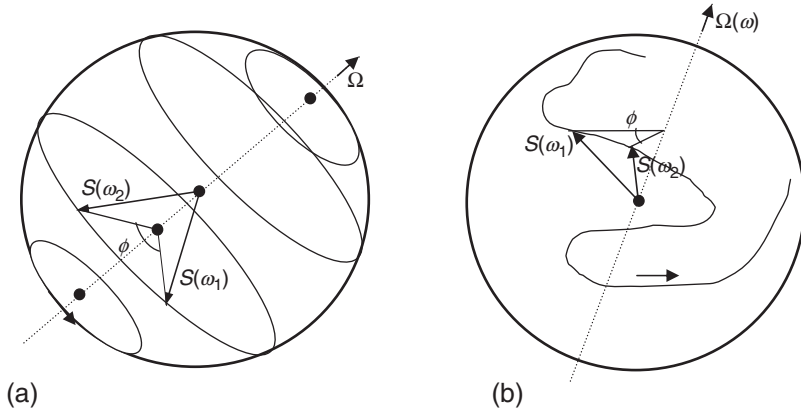


Figure 4.5.8 Illustration of traces of signal polarization vector on Poincare spheres when the optical frequency is varied (a) for a short fiber and (b) for a long fiber.

the change of the SOP vector $\vec{S}$ versus ω . Obviously, the amount of this change is directly proportional to fiber birefringence (or PMD). For an infinitesimal change of the signal optical frequency $d\omega$, this vectorial relationship is

$$\frac{d\vec{S}}{d\omega} = \Omega \times \vec{S} \quad (4.5.13)$$

For a step change in the optical frequency, $\Delta\omega = \omega_2 - \omega_1$, it is also convenient to use the scalar relationship [29]:

$$\phi = \Delta\tau \cdot \Delta\omega \quad (4.5.14)$$

where ϕ is the angular change of the polarization vector $\vec{S}$ in radians on the plan perpendicular to the PSP, as shown in Figure 4.5.8, and $\Delta\tau$ is the DGD between the two PSP components. Obviously, the simple relation given in Equation 4.5.14 can be used to measure fiber DGD if a polarimeter is available and the variation of Stokes parameters versus signal frequency can be measured.

Figure 4.5.9 shows the block diagram of the measurement setup using Poincare arc technique, where a tunable laser is used which provides a source whose optical frequency can be swept. A polarization controller is placed before the optical fiber

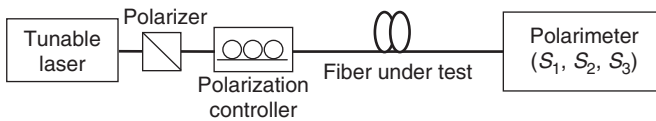


Figure 4.5.9 Block diagram of DGD measurement using a polarimeter.

under test to explore various launching SOPs of the optical signal into the fiber. A polarimeter is used to measure the Stokes parameters corresponding to each frequency of the optical source, $\vec{S}(\omega) = \vec{a}_x S_1(\omega) + \vec{a}_y S_2(\omega) + \vec{a}_z S_3(\omega)$, where $\vec{a}_x$, $\vec{a}_y$, and $\vec{a}_z$ are unit vectors. For a small wavelength increment $\Delta\omega = \omega_2 - \omega_1$, the angular rotation of the polarization vector on the plan perpendicular to the PSP vector can be evaluated by

$$\phi = \cos^{-1} \left\{ \frac{(\vec{S}(\omega_1) \times \vec{\Omega}(\omega)) \cdot (\vec{S}(\omega_2) \times \vec{\Omega}(\omega))}{|\vec{S}(\omega_1) \times \vec{\Omega}(\omega)| |\vec{S}(\omega_2) \times \vec{\Omega}(\omega)|} \right\} \quad (4.5.15)$$

Then the DGD value at this frequency can be obtained by

$$\Delta\tau(\omega) = \frac{\phi}{\Delta\omega} \quad (4.5.16)$$

where $\omega = (\omega_2 + \omega_1)/2$ is the average frequency. Overall, the Poincare arc measurement technique is easy to understand and the measurement setup is relatively simple. However, it requires a polarimeter, which is a specialized instrument. In addition, the frequency tuning of the tunable laser has to be continuous to provide the accurate trace of the polarization rotation, as illustrated in [Figure 4.5.8\(b\)](#).

4.5.5 Fixed Analyzer Method

Compared to the Poincare arc measurement technique, the fixed analyzer method replaces the polarimeter with a fixed polarizer; thus it is relatively easy to implement because of the reduced requirement of specialized instrumentation. [Figure 4.5.10](#) shows the schematic diagrams of two equivalent versions of the fixed

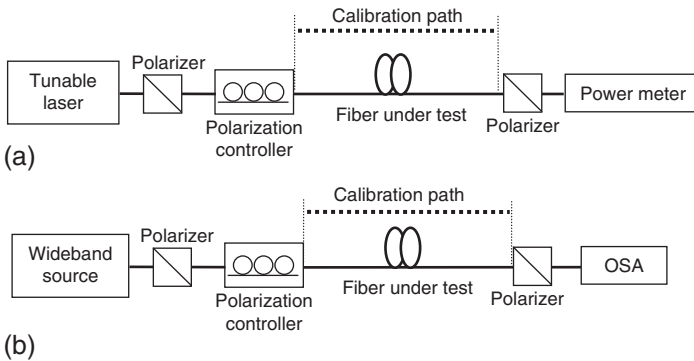


Figure 4.5.10 Block diagram of PMD measurement using a fixed analyzer method: (a) tunable laser + power meter combination and (b) wideband source + OSA combination.

analyzer measurement apparatus [30]. Figure 4.5.10(a) uses a tunable laser as the optical source, while at the receiver side a fixed polarizer is followed by an optical power meter. By varying the frequency of the tunable optical source, the signal polarization state at the fiber output changes, and the polarizer converts this signal SOP change into an optical power change, which is then detected by the power meter. The system shown in Figure 4.5.10(b) uses a wideband source that provides a signal with a broad optical spectrum. Because of the birefringence in the fiber, different signal frequency components will exhibit different polarization states at the output of the fiber, and the polarizer converts this frequency-dependent polarization rotation into a frequency-dependent optical power spectral density, which can be accurately measured by an optical spectrum analyzer. In both implementations, a polarizer immediately after the source is used to make sure the source has a fixed polarization state. The polarization controller enables the change of the polarization state of the optical signal that is injected into the fiber.

In the Poincare sphere representation, the power transfer function through a perfect polarizer can be expressed as

$$T(\omega) = \frac{P_{out}}{P_{in}} = \frac{1}{2} [1 + \hat{s}(\omega) \cdot \hat{p}] \quad (4.5.17)$$

where $\hat{s}(\omega)$ is the unit vector representing the polarization state of the input optical signal into the polarizer and $\hat{p}$ is the unit vector representing the high transmission state of the polarizer.

Because of the birefringence in the fiber, the SOP of the optical signal at fiber output will rotate around the PSP on the Poincare sphere when the optical signal frequency is varied. In a long optical fiber, the birefringence axis is randomly oriented along the fiber and the mode coupling may also be significant; therefore the polarization vector may essentially walk all over the Poincare sphere with the change of the signal optical frequency. As a result, the power transmission efficiency $T(\omega)$ through the fixed polarization analyzer at the fiber output can have all possible values between 0 and 1 as illustrated Figure 4.5.11. If the birefringence orientation along the fiber is truly random, the transfer function $T(\omega)$ will have a uniform probability distribution between 0 and 1 with the average value of 0.5.

If we define

$$T' = \frac{dT(\omega)}{d\omega} \quad (4.5.18)$$

as the frequency derivative of the power transfer function of the fixed polarizer, then statistically, the expected value of $|T'|$ is $E\{|T'|, T = \langle T \rangle\}$, which is obtained under the condition that the transfer function is at its mean level $T(\omega) = \langle T(\omega) \rangle$. According to the fundamental rules of statistics, the mean-level

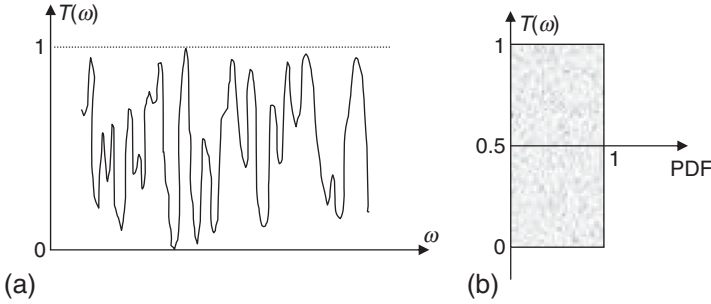


Figure 4.5.11 (a) Illustration of fixed polarizer transfer function $T(\omega)$ with random input polarization state of the signal and (b) probability distribution of $T(\omega)$.

crossing density, which specifies how often the random variable $T(\omega)$ crosses through its average value per unit frequency interval, is

$$\gamma_m = f_T(\langle T \rangle) (E\{|T'\}|, T = \langle T \rangle) \quad (4.5.19)$$

where $f_T(\langle T \rangle)$ is the probability density function (PDF) of the transfer function $T(\omega)$ evaluated at its average level $\langle T(\omega) \rangle$. In this specific case of uniform distribution, since $\langle T(\omega) \rangle = 0.5$ and $f_T(\langle T \rangle) = f_T(0.5) = 1$, Equation 4.5.19 can be simplified into

$$\gamma_m = E\{|T'|, T = \langle T \rangle\} \quad (4.5.20)$$

Use the definition of $T(\omega)$ shown in Equation 4.5.17, its derivative shown in Equation 4.5.18 can be expressed as

$$T'(\omega) = \frac{1}{2} \left[\frac{d\hat{s}(\omega)}{d\omega} \cdot \hat{p} \right] \quad (4.5.21)$$

Meanwhile, Equation 4.5.13 in the last section indicated that

$$\frac{d\hat{s}(\omega)}{d\omega} = \Omega \times \hat{s}(\omega) \quad (4.5.22)$$

where Ω is the PMD vector. Then one can find:

$$T'(\omega) = \frac{1}{2} [(\Omega \times \hat{s}) \cdot \hat{p}] = \frac{1}{2} [(\hat{s} \times \hat{p}) \cdot \Omega] \quad (4.5.23)$$

According to Equation 4.5.17, at the mean value of $T(\omega) = 0.5$, $\hat{s}(\omega) \cdot \hat{p}$ has to be equal to zero, which is equivalent to $\hat{s}(\omega) \times \hat{p} = \hat{1}$, where $\hat{1}$ is a unit vector and therefore, $T'(\omega) = 0.5[\hat{1} \cdot \Omega]$. Then Equation 4.5.20 can be expressed as

$$\gamma_m = \frac{1}{2} \langle |\hat{1} \cdot \Omega| \rangle = \frac{1}{2} \langle |\Omega| \rangle \langle |\cos \theta| \rangle \quad (4.5.24)$$

In a long fiber with truly random mode coupling, $\cos \theta$ is uniformly distributed between $-1 \leq \cos \theta \leq 1$ so that $\langle |\cos \theta| \rangle = 0.5$. Also, by definition, the magnitude of the PMD vector Ω is equal to the DGD of the fiber, $\Delta\tau = |\Omega|$; therefore

$$\gamma_m = \frac{\langle \Delta\tau \rangle}{4} \quad (4.5.25)$$

Recall that the physical meaning of γ_m can be explained by how often the transfer function $T(\omega)$ crosses through its average value per unit frequency interval. Then within a frequency interval $\Delta\omega$, if the average number of cross-overs is $\langle N_m \rangle$, we will have $\gamma_m = \langle N_m \rangle / \Delta\omega$, that is,

$$\langle \Delta\tau \rangle = 4 \frac{\langle N_m \rangle}{\Delta\omega} \quad (4.5.26)$$

In this way, by scanning the signal frequency and counting the number of transmission crossovers through its average value within a certain frequency interval, the average DGD can be evaluated.

Similarly, the measurement can also be performed by counting the average number of extrema (maximum + minimum) within a frequency interval to decide the average DGD value. Without further derivation, the equation to calculate the DGD can be found as [30]

$$\langle \Delta\tau \rangle = 0.824\pi \frac{\langle N_e \rangle}{\Delta\omega} \quad (4.5.27)$$

where $\langle N_e \rangle$ is the number of extrema within a frequency interval $\Delta\omega$.

If we choose to use a wavelength interval instead of a frequency interval, Equation 4.5.27 can be expressed as

$$\langle \Delta\tau \rangle = \frac{0.412 \langle N_e \rangle \lambda_{start} \lambda_{stop}}{c(\lambda_{stop} - \lambda_{start})} \quad (4.5.28)$$

where λ_{start} and λ_{stop} are the start and stop wavelengths of the measurement.

One practical consideration using the fixed polarizer method is how small the frequency step size should be used for the tunable laser if the setup shown in Figure 4.5.10(a) is implemented. This question is equivalent to the selection of the spectral resolution for the OSA if the setup shown in Figure 4.5.10(b) is used. Assuming that we need at least three data points between adjacent transmission extrema, according to Equation 4.5.28 the wavelength step size has to be

$$\delta\lambda < \frac{0.412\lambda^2}{4c\langle \Delta\tau \rangle} \quad (4.5.29)$$

Approximately $\delta\lambda < \lambda^2 / (8c\langle \Delta\tau \rangle)$ can be used as a rule of thumb to determine the size of the wavelength step. For example, in the 1550 nm wavelength window,

$\lambda^2/(8c) \approx 10^{-21}$; therefore the wavelength step size of the source (or the spectral resolution of the OSA) has to be $\delta\lambda < 10^{-21}/\langle\Delta\tau\rangle$, where $\langle\Delta\tau\rangle$ is the expected DGD in seconds and $\delta\lambda$ is the wavelength step size in meters. For convenience, if we use $\langle\Delta\tau\rangle$ in the unit of picoseconds and $\delta\lambda$ is in the unit of nanometers, this requirement becomes very simple:

$$\delta\lambda < \frac{1}{\langle\Delta\tau\rangle} \quad (4.5.30)$$

For instance, for a fiber with the mean DGD of $\langle\Delta\tau\rangle = 20$ ps, the source wavelength step size (or the OSA spectral resolution) must be better than 0.05 nm.

4.5.6 The Jones Matrix Method

The Jones Matrix method is a comprehensive PMD measurement technique, although its measurement setup, as shown in Figure 4.5.12, is similar to the Poincare arc technique discussed previously, the sequence of the input SOP of the optical signal is sequentially adjusted and the response at the fiber output is systematically measured [31, 32].

To explain the measurement principle of the Jones Matrix method, it is useful to review what the Jones Matrix is about. In general, a Jones Matrix has been widely used to represent the polarization state of a lightwave signal or the transfer matrix of a passive optical device. For example, the transversal electrical field of a polarized light can be represented by a two-element complex vector in the Cartesian coordinate system. These two vectors specify the magnitude and the phase of the x and y components of the electrical field at a particular point in space [33]:

$$\vec{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} E_{x0}e^{j\varphi_x} \\ E_{y0}e^{j\varphi_y} \end{pmatrix} \quad (4.5.31)$$

where φ_x and φ_y represent the phase of the two elements.

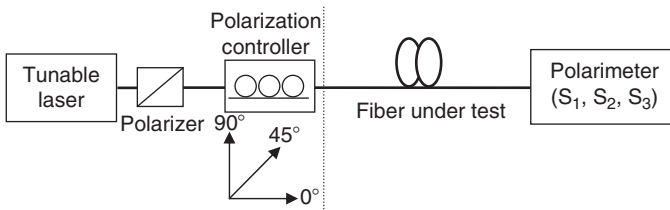


Figure 4.5.12 Block diagram of the Jones Matrix technique to measure fiber PMD.

On the other hand, the transfer function of a passive optical device can be written as a four-element matrix in the form:

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \quad (4.5.32)$$

The transfer matrix of an optical device specifies the relationship between the input and output Jones vectors of the lightwave signal. This transfer matrix can be characterized by measuring three output Jones vectors in response to three known input Jones vectors. It is worthwhile to note that although it is measured with a few specific polarization states of the optical signal, a Jones Matrix describes a passive optical device such as an optical fiber and is independent of the input launching condition of the optical signal.

Suppose the input Jones vector of the optical signal is $\vec{E}_{in} = \begin{pmatrix} E_{x,in} \\ E_{y,in} \end{pmatrix}$ and the corresponding output Jones vector is $\vec{E}_{out} = \begin{pmatrix} E_x \\ E_y \end{pmatrix}$. The Jones Matrix of the optical device must be

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} E_{x,in} \\ E_{y,in} \end{bmatrix} \quad (4.5.33)$$

To obtain the transfer function of a passive optical device, it is convenient to use three input stimuli with linear polarization states oriented at 0° , 45° , and 90° . In the following, we explain how the four matrix elements can be obtained through the three independent measurements.

1. The input optical signal is linearly polarized at 0° (linear horizontal). That is,

$$\vec{E}_{in} = \begin{pmatrix} E_{x,in} \\ 0 \end{pmatrix} = E_{x,in} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.5.34)$$

If the measured output Jones vector is

$$\vec{E}_{out1} = \begin{pmatrix} E_{x1} \\ E_{y1} \end{pmatrix} \quad (4.5.35)$$

then the input/output relationship is

$$\begin{bmatrix} E_{x1} \\ E_{y1} \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} E_{x,in} \quad (4.5.36)$$

which is equivalent to

$$E_{x1} = M_{11} E_{x,in} \quad (4.5.37a)$$

$$E_{y1} = M_{21} E_{x,in} \quad (4.5.37b)$$

Now we can define a new parameter,

$$k_1 = \frac{E_{x1}}{E_{y1}} = \frac{M_{11}}{M_{21}} \quad (4.5.38)$$

Since both E_{x1} and E_{y1} have already been determined by the measurement, k_1 can be easily obtained.

2. The input is linearly polarized at 90° (linear vertical). In this case,

$$\vec{E}_{in2} = \begin{pmatrix} 1 \\ E_{y,in2} \end{pmatrix} = E_{y,in2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (4.5.39)$$

Suppose the measured output Jones vector corresponding to this input is

$$\vec{E}_{out2} = \begin{pmatrix} E_{x2} \\ E_{y2} \end{pmatrix} \quad (4.5.40)$$

The input/output relation yields

$$\begin{bmatrix} E_{x2} \\ E_{y2} \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} E_{x,in2} \quad (4.5.41)$$

which is equivalent to

$$E_{x2} = M_{12} E_{x,in2} \quad (4.5.42a)$$

$$E_{y2} = M_{22} E_{x,in2} \quad (4.5.42b)$$

We can then define a new parameter:

$$k_2 = \frac{E_{x2}}{E_{y2}} = \frac{M_{12}}{M_{22}} \quad (4.5.43)$$

Again, k_2 can be determined through the measurement of E_{x2} and E_{y2} .

3. The input is linearly polarized at 45° . In this case $E_{y,in} = E_{x,in}$, so that

$$\vec{E}_{in3} = \frac{E_{y,in3}}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.5.44)$$

Suppose the measured output Jones vector corresponding to this input is

$$\vec{E}_{out3} = \begin{pmatrix} E_{x3} \\ E_{y3} \end{pmatrix} \quad (4.5.45)$$

The input/output relationship is

$$\begin{bmatrix} E_{x3} \\ E_{y3} \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \frac{E_{x,in3}}{\sqrt{2}} \quad (4.5.46)$$

which is equivalent to

$$E_{x3} = (M_{11} + M_{12})E_{x,in3}/\sqrt{2} \quad (4.5.47a)$$

$$E_{y3} = (M_{21} + M_{22})E_{x,in3}/\sqrt{2} \quad (4.5.47b)$$

Then the third new parameter can be introduced as

$$k_3 = \frac{E_{x3}}{E_{y3}} = \frac{M_{11} + M_{12}}{M_{21} + M_{22}} \quad (4.5.48)$$

which can be determined through the measurement of E_{x3} and E_{y3} .

If we neglect the absolute propagation delay and the attenuation of the optical device under test, the four elements in the Jones Matrix will not be independent and we can arbitrarily set $M_{22} = 1$. Then the rest of the three matrix elements can be found using the following three equations:

$$k_1 = \frac{M_{11}}{M_{21}} \quad (4.5.49a)$$

$$k_2 = \frac{M_{12}}{M_{22}} = M_{12} \quad (4.5.49b)$$

$$k_3 = \frac{M_{11} + M_{12}}{M_{21} + M_{22}} = \frac{M_{11} + M_{12}}{M_{21} + 1} \quad (4.5.49c)$$

The solutions to this set of equations are

$$M_{11} = k_1 \frac{k_2 - k_3}{k_3 - k_1} = k_1 k_4 \quad (4.5.50a)$$

$$M_{12} = k_2 \quad (4.5.50b)$$

$$M_{21} = \frac{k_2 - k_3}{k_3 - k_1} = k_4 \quad (4.5.50c)$$

where k_4 is defined as

$$k_4 = \frac{k_2 - k_3}{k_3 - k_1} \quad (4.5.51)$$

Finally, the Jones Matrix of the optical device can be represented as

$$[M] = C \begin{bmatrix} k_1 k_4 & k_2 \\ k_4 & 1 \end{bmatrix} \quad (4.5.52)$$

where all the elements can be determined by the three independent measurements described previously. A multiplicative complex constant C is introduced that represents the absolute propagation delay and the attenuation of the

optical device. In the following, we discuss how the DGD of an optical fiber can be represented with the Jones Matrix.

As discussed earlier, due to the birefringence in a fiber, the signal SOP at the fiber output is a function of the optical frequency ω . Thus, the Jones Matrix of the fiber also must be a function of the optical frequency, $M = M(\omega)$. For an arbitrary input optical field vector $E_{in}(\omega)$, if the output optical field vector is measured as $E_{out}(\omega)$, the input/output relation is

$$E_{out}(\omega) = M(\omega)E_{in}(\omega) \quad (4.5.53)$$

The output field vector has an amplitude $A(\omega)$ and a phase $\phi(\omega)$

$$E_{out}(\omega) = A(\omega)e^{j\phi(\omega)} \quad (4.5.54)$$

Then the frequency derivative of Equation 4.5.53 yields

$$E'_{out}(\omega) = M'(\omega)E_{in}(\omega) = \left[\frac{A'(\omega)}{A(\omega)} + j\phi'(\omega) \right] E_{out}(\omega) \quad (4.5.55)$$

where the primes denote differentiation with respect to the frequency ω , such as $E'_{out}(\omega) = dE_{out}(\omega)/d\omega$.

If $M^{-1}(\omega)$ represents the inverse Jones Matrix, Equation 4.5.53 is also equivalent to

$$E_{in}(\omega) = M^{-1}(\omega)E_{out}(\omega) \quad (4.5.56)$$

Combining Equations 4.5.55 and 4.5.56, we find

$$M'(\omega)M^{-1}(\omega)E_{out}(\omega) = \left[\frac{A'(\omega)}{A(\omega)} + j\phi'(\omega) \right] E_{out}(\omega) \quad (4.5.57)$$

which is equivalent to a more convenient form,

$$\left\{ M'(\omega)M^{-1}(\omega) - \left[\frac{A'(\omega)}{A(\omega)} + j\phi'(\omega) \right] I \right\} E_{out}(\omega) = 0 \quad (4.5.58)$$

where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the unit matrix.

Recall that the physical meaning of the differential phase

$$\phi'(\omega) = \frac{d\phi(\omega)}{d\omega} = \tau_g \quad (4.5.59)$$

represents the group velocity of the optical signal in the fiber. For a small frequency interval $\Delta\omega$, a linear approximation can be made such that

$$M'(\omega) = \frac{dM(\omega)}{d\omega} \approx \frac{M(\omega + \Delta\omega) - M(\omega)}{\Delta\omega} \quad (4.5.60)$$

and thus

$$\begin{aligned} M'(\omega)M^{-1}(\omega) &\approx \frac{[M(\omega + \Delta\omega)M^{-1}(\omega) - M(\omega)M^{-1}(\omega)]}{\Delta\omega} \\ &= \frac{[M(\omega + \Delta\omega)M^{-1}(\omega) - I]}{\Delta\omega} \end{aligned} \quad (4.5.61)$$

If we neglect the frequency dependency of the optical loss, which is often a valid assumption for an optical fiber within a relatively narrow optical band, that is $A'(\omega) = dA(\omega)/d\omega \approx 0$, the combination of [Equations 4.5.58](#) and [4.5.61](#) becomes

$$\left\{ M(\omega + \Delta\omega)M^{-1}(\omega) - (1 + j\tau_g\Delta\omega)I \right\} E_{out}(\omega) = 0 \quad (4.5.62)$$

In addition, assuming that $\Delta\omega$ is small enough such that $\tau_g\Delta\omega \ll 1$, the term $(1 + j\tau_g\Delta\omega)$ in [Equation 4.5.62](#) can be replaced by $\exp(j\tau_g\Delta\omega)$; therefore we have

$$\left\{ M(\omega + \Delta\omega)M^{-1}(\omega) - \exp(j\tau_g\Delta\omega)I \right\} E_{out}(\omega) = 0 \quad (4.5.63)$$

In order to satisfy the Eigen-equation [4.5.63](#), the output optical field $E_{out}(\omega)$ has to be the Eigenvector which is the output PSP. $\exp(j\tau_g\Delta\omega)$ correspond to the Eigen values,

$$\rho_1 = \exp(j\tau_{g1}\Delta\omega) \quad (4.5.64a)$$

$$\rho_2 = \exp(j\tau_{g2}\Delta\omega) \quad (4.5.64b)$$

where, τ_{g1} and τ_{g2} represent the group delays corresponding to the two fiber PSPs, their ratio is,

$$\rho_1/\rho_2 = \exp[j(\tau_{g1} - \tau_{g2})\Delta\omega] \quad (4.5.65)$$

Define a matrix

$$M(\omega + \Delta\omega)M^{-1}(\omega) = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \quad (4.5.66)$$

for [Equation 4.5.63](#) to be nontrivial, the following relation must be satisfied

$$\begin{vmatrix} m_{11} - \exp(j\tau_g\Delta\omega) & m_{12} \\ m_{21} & m_{22} - \exp(j\tau_g\Delta\omega) \end{vmatrix} = 0 \quad (4.5.67)$$

Thus, the two Eigen-values can be calculated with

$$\rho_{1,2} = \frac{(m_{11} + m_{22}) \pm \sqrt{(m_{11} + m_{22})^2 + 4(m_{12}m_{21} - m_{11}m_{22})}}{2} \quad (4.5.68)$$

Therefore, the fiber DGD can be expressed as the group delay difference as the function of the optical frequency:

$$\Delta\tau(\omega) = \left| \tau_{g1} - \tau_{g2} \right| = \left| \frac{\text{Arg}(\rho_1/\rho_2)}{\Delta\omega} \right| \quad (4.5.69)$$

Where, $\text{Arg}(\rho_1/\rho_2)$ denotes the phase angle of (ρ_1/ρ_2) . Equation 4.5.69 is the major result of this derivation which underlines the principle of Jones Matrix technique for fiber DGD measurement. Based on Equation 4.5.69, the measurement procedure can be outlined in the following steps:

1. Measure Jones Matrix elements at different optical signal frequencies: $M(\omega_1)$, $M(\omega_2)$, $M(\omega_3)$, $M(\omega_4)$... This is accomplished by using three independent launching conditions at each frequency and measuring their corresponding output SOPs, as described by Equations 4.5.32–4.5.52.
2. Convert the results of step 1 into a series of new matrices shown in Equation 4.5.66 for different wavelengths.
3. Find the fiber DGD value at each optical frequency using Equations 4.5.68 and 4.5.69.

As a practical issue, it is important to decide the frequency step size $\Delta\omega$ for the measurement. On one hand, the step size $\Delta\omega$ should not be too small. If it is too small, not only the measurement would take an unnecessarily long time for the large number of measurements; any fiber path instability, wavelength inaccuracy of laser source, or polarimeter accuracy would have strong impact in the measured results. On the other hand, $\Delta\omega$ should not be too large. As a rule of thumb, the output SOP should not rotate for more than 45° over each frequency step; otherwise the measurement would produce inaccurate results. With these considerations in mind, for a fiber system with an expected DGD of $\Delta\tau$, the maximum frequency step size is limited by $\Delta\omega < \pi/(4\Delta\tau)$. Since $\Delta\omega \approx 2\pi c\Delta\lambda/\lambda^2$, the corresponding wavelength step size is

$$\Delta\lambda < \frac{\lambda^2}{8c\Delta\tau} \quad (4.5.70)$$

In 1550 nm wavelength window, this step size limitation is $\Delta\lambda \cdot \Delta\tau \leq 1$, where $\Delta\lambda$ is in nanometers and $\Delta\tau$ is in picoseconds. Obviously, for fibers with large DGD, the frequency step size must be made small. As a simple example, for a fiber with an expected DGD of $\Delta\tau = 20$ ps, the source wavelength step size must be smaller than 0.05 nm, which is approximately the same as the requirement of the fixed analyzer technique described in the last section.

Due to the statistical nature of fiber PMD, in practice, each measurement will likely result in a different DGD value, that is, N independent measurements may result in N different DGD results. The RMS value of the N measurements is $\sqrt{\frac{1}{N} \sum_{i=1}^N \Delta\tau_i^2}$ since PMD generally has Maxwellian distribution as

defined by Equation 4.5.4. This distribution can be specified by a single parameter α ., which is related to its RMS value $\alpha\sqrt{3}$. Therefore the α value can be found through N independent measurements and

$$\alpha^2 = \frac{1}{3N} \sum_{i=1}^N \Delta\tau_i^2 \quad (4.5.71)$$

where $\Delta\tau_i$ is the DGD of each measurement.

To summarize, the Jones Matrix technique has a number of advantages compared to other DGD measurement methods. First, it needs only a small wavelength window to perform a measurement. As a comparison, a fixed analyzer method needs a much larger wavelength window to have enough number of zero-crossings to obtain an accurate average. From this point of view, the Jones Matrix method is more suitable to evaluate the detailed wavelength dependency of PMD. Second, Jones Matrix measurement can be made fast using automated procedures of polarization controller and a polarimeter. This allows both time-dependent and wavelength-dependent PMD measurements. Third, the accuracy of the Jones Matrix technique is considered the best compared to other techniques because of the efficiently arranged SOP adjustment procedure and the coordinated polarimeter testing.

4.5.7 The Mueller Matrix Method

The Mueller Matrix method (MMM) is another technique that can be used to characterize the wavelength-dependent polarization rotation of an optical fiber. In general, the Jones Matrix method described in the last section requires small enough wavelength steps, as indicated in Equation 4.5.70, for accurate measurement of DGD. These small wavelength steps, in turn, demand relatively high SNR to ensure the measurement accuracy. On the contrary, the Mueller Matrix method does not have this limitation and relatively large frequency steps are allowed. In addition, the Mueller Matrix method can be used to measure polarization-dependent loss (PDL) in an optical fiber, which is discussed separately in the next section.

A Mueller Matrix represents the polarization rotation characteristics of an optical device such as an optical fiber, which is determined by the relationship between a set of input polarization vectors and their corresponding output polarization vectors,

$$\begin{bmatrix} s_{out1} \\ s_{out2} \\ s_{out3} \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \begin{bmatrix} s_{in1} \\ s_{in2} \\ s_{in3} \end{bmatrix} \quad (4.5.72)$$

where $S_{in} = [s_{in1} \ s_{in2} \ s_{in3}]^T$ is the input polarization vector, $S_{out} = [s_{out1} \ s_{out2} \ s_{out3}]^T$ is the output polarization vector, and $M = [m_{ij}]$ is

the Mueller Matrix, which relates the output vector to the input vector. The superscript T stands for matrix transpose. In this section, we assume that the fiber has no PDL and normalized vectors are used with unity magnitude for both the input and the output optical fields.

For three orthogonal input polarization states, $S_{in}^1 = [1 \ 0 \ 0]^T$, $S_{in}^2 = [0 \ 1 \ 0]^T$, and $S_{in}^3 = [0 \ 0 \ 1]^T$, suppose the corresponding polarization vectors of the output fields are

$$\begin{bmatrix} S_{out,11} & S_{out,21} & S_{out,31} \\ S_{out,12} & S_{out,22} & S_{out,32} \\ S_{out,13} & S_{out,23} & S_{out,33} \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4.5.73)$$

The Mueller Matrix elements can be determined by measuring the output polarization states so that [34]

$$[m_{ij}] = [S_{out,ji}] \quad (4.5.74)$$

In fact, if there is no PDL, two independent input polarization vectors S_{in}^1 , and S_{in}^a are sufficient to determine the Mueller Matrix. In this measurement, S_{in}^a does not even have to be orthogonal to S_{in}^1 . If the corresponding output polarization vectors are S_{out}^1 and S_{out}^a , respectively, a third polarization vector can be found through normalization:

$$S_{out}^3 = S_{out}^1 \times S_{out}^a \quad (4.5.75)$$

To complete a three-dimensional Cartesian coordinate system, another vector that is orthogonal to both S_{out}^1 and S_{out}^3 can be obtained as

$$S_{out}^2 = S_{out}^3 \times S_{out}^1 \quad (4.5.76)$$

where both the input and the output vectors are normalized such that $|S_{in}^1| = |S_{in}^a| = 1$ and $|S_{out}^1| = |S_{out}^2| = |S_{out}^3| = 1$.

The experimental setup of Mueller Matrix measurement, shown in Figure 4.5.13, is similar to that of the Jones Matrix technique. The only difference

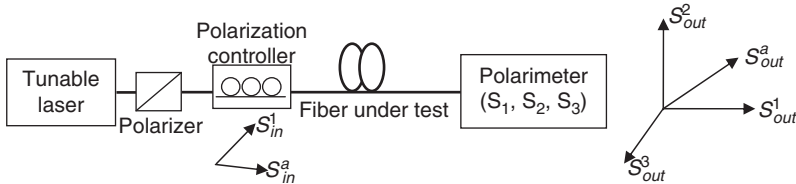


Figure 4.5.13 Block diagram of the Mueller Matrix method to measure fiber PMD.

is that in Mueller Matrix measurement, only two independent linear input polarization states are required, and they do not have to be mutually orthogonal.

Again, since the PMD in a fiber is a frequency-dependent polarization rotation, as indicated in Equation 4.5.13,

$$\frac{dS_{out}}{d\omega} = \Omega \times S_{out} \quad (4.5.77)$$

where Ω is the PMD vector whose magnitude is equal to the DGD of the fiber, which is $\Delta\tau = d\phi/d\omega = |\Omega|$. Performing the measurements at two frequencies ω_1 and $\omega_2 = \omega_1 + \Delta\omega$, we can find two Mueller matrices, $[M(\omega_1)]$ and $[M(\omega_2)]$, through

$$[S_{out}(\omega_1)] = [M(\omega_1)] [S_{in}(\omega_1)] \quad (4.5.78a)$$

$$[S_{out}(\omega_2)] = [M(\omega_2)] [S_{in}(\omega_2)] \quad (4.5.78b)$$

Note that the input polarization vector does not change with the change of the frequency, $[S_{in}(\omega_1)] = [S_{in}(\omega_2)] = [S_{in}]$. Since the Mueller Matrix is normalized, $[M(\omega_1)][M(\omega_1)]^T = I$, where I is the unit matrix, Equations 4.5.78a and 4.5.78b can be combined to eliminate the input $[S_{in}]$,

$$[S_{out}(\omega_2)] = [M_\Delta] [S_{out}(\omega_1)] \quad (4.5.79)$$

where

$$[M_\Delta] = [M(\omega_2)] [M(\omega_1)]^T \quad (4.5.80)$$

The rotational matrix, $[M_\Delta]$, can be determined through the measurement using the setup shown in Figure 4.5.13. However, it is important to understand how to calculate the fiber DGD from the measured rotational matrix.

In fact, the rotational matrix can be defined as a spatial vector, which is described by a rotation angle ϕ around an axis $\vec{r}$ as shown in Figure 4.5.14 [35, 36]. Assume that $S_0 = S_{out}(\omega_0)$ is the polarization vector measured at frequency ω_0 (from the origin to point B as shown in Figure 4.5.14), $S_1 = S_{out}(\omega_1)$ is the polarization vector measured at frequency ω_1 (from the origin to point A), and the rotation is around an axis $\vec{r}$ with the rotation angle ϕ . In the following, we shall find the relationship between the vector rotation from S_0 to S_1 and the corresponding angular rotation ϕ , which is directly related to the fiber DGD.

According to the rule of vectoral addition, S_1 can be expressed as

$$S_1 = \vec{OA} = \vec{OO'} + \vec{O'B} = \vec{OO'} + \vec{O'C} + \vec{CA} \quad (4.5.81)$$

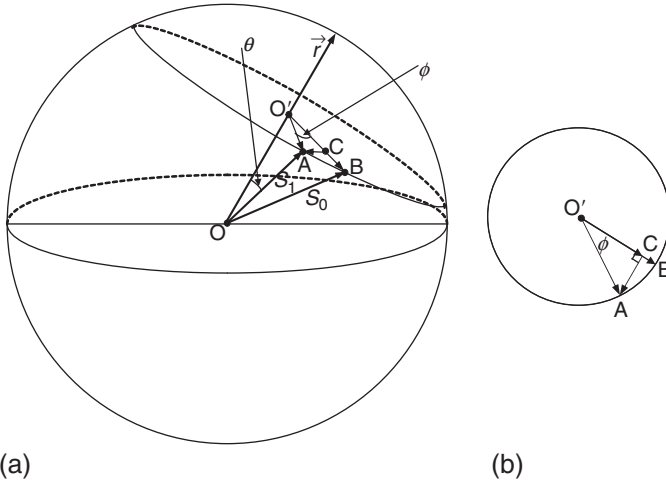


Figure 4.5.14 (a) Geometric expression of rotation matrix; (b) view of a plane perpendicular to the rotation axis.

Without loss of generality, we assume the sphere in Figure 4.5.14(a) has unit radius, and therefore $\vec{r}$, $\vec{S}_0$, and $\vec{S}_1$ all have the unity length. Then based on the geometric structure shown in Figure 4.5.14, we can find the following relations:

$$|\vec{OO'}| = |\vec{S}_0| \cos \theta = \vec{r} \cdot \vec{S}_0 \quad (4.5.82)$$

$$|\vec{O'A}| = |\vec{O'B}| \quad (4.5.83)$$

$$|\vec{O'B}| = |\vec{S}_0| \sin \theta = |\vec{S}_0 \times \vec{r}| \quad (4.5.84)$$

$$|\vec{CA}| = |\vec{O'A}| \sin \phi = |\vec{O'B}| \sin \phi \quad (4.5.85)$$

$$|\vec{O'C}| = |\vec{O'A}| \cos \phi = |\vec{O'B}| \cos \phi \quad (4.5.86)$$

Since $\vec{OO'}$ is in the same direction as $\vec{r}$,

$$\vec{OO'} = \vec{r} (\vec{r} \cdot \vec{S}_0) \quad (4.5.87)$$

Then the vector $\overrightarrow{O'B}$ can be expressed as

$$\overrightarrow{O'B} = \overrightarrow{OB} - \overrightarrow{OO'} = S_0 - \vec{r}(\vec{r} \cdot S_0) \quad (4.5.88)$$

Based on Equation 4.5.86 and take into account that $\overrightarrow{O'C}$ is in the same direction as $\overrightarrow{O'B}$, we can find

$$\overrightarrow{O'C} = [S_0 - \vec{r}(\vec{r} \cdot S_0)] \cos \phi \quad (4.5.89)$$

Using Equations 4.5.84 and 4.5.85 and noting that $\overrightarrow{CA}$ is in the same direction as $(S_0 \times \vec{r})$, we have,

$$\overrightarrow{CA} = \sin \phi (S_0 \times \vec{r}) \quad (4.5.90)$$

Substituting Equations 4.5.87, 4.5.89, and 4.5.90 into Equation 4.5.81, we obtain the following relation:

$$\begin{aligned} S_1 &= \vec{r}(\vec{r} \cdot S_0) + [S_0 - \vec{r}(\vec{r} \cdot S_0)] \cos \phi + \sin \phi (S_0 \times \vec{r}) \\ &= S_0 \cos \phi + (1 - \cos \phi) \vec{r}(\vec{r} \cdot S_0) + \sin \phi (S_0 \times \vec{r}) \end{aligned} \quad (4.5.91)$$

Now it is important to review a few matrix concepts and identities. First, a cross-product between two vectors $\vec{a} = [a_1, a_2, a_3]^T$ and $\vec{b} = [b_1, b_2, b_3]^T$ can be expressed in terms of matrix multiplication, which is the product between a skew-symmetric matrix of the first vector and the second vector,

$$\vec{a} \times \vec{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \vec{b} \quad (4.5.92)$$

The cross-product operator of $\vec{a}$ can be written as

$$\vec{a} \times = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \quad (4.5.93)$$

In addition, a dyadic product of the two vectors $\vec{a}$ and $\vec{b}$ is

$$\vec{a} \vec{b} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{bmatrix} \quad (4.5.94)$$

Another very useful identity involving matrix cross-product is

$$\vec{a} \times \vec{b} \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}) \quad (4.5.95)$$

where $\vec{c} = [c_1, c_2, c_3]^T$.

Based on Equation 4.5.93, $(\vec{r} \times)(\vec{r} \times)$ can be written as

$$\begin{aligned}
 (\vec{r} \times)(\vec{r} \times) &= \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} -r_3^2 - r_2^2 & r_1 r_2 & r_1 r_3 \\ r_2 r_1 & -r_3^2 - r_1^2 & r_2 r_3 \\ r_3 r_1 & r_3 r_2 & -r_1^2 - r_2^2 \end{bmatrix}
 \end{aligned} \tag{4.5.96}$$

Remember, $\vec{r}$ is a unit vector such that $r_1^2 + r_2^2 + r_3^2 = 1$. Based on the relation given in Equation 4.5.94, Equation 4.5.96 can be simplified as

$$\begin{aligned}
 (\vec{r} \times)(\vec{r} \times) &= \begin{bmatrix} -1 + r_1^2 & r_1 r_2 & r_1 r_3 \\ r_2 r_1 & -1 + r_2^2 & r_2 r_3 \\ r_3 r_1 & r_3 r_2 & -1 + r_3^2 \end{bmatrix} \\
 &= \begin{bmatrix} r_1^2 & r_1 r_2 & r_1 r_3 \\ r_2 r_1 & r_2^2 & r_2 r_3 \\ r_3 r_1 & r_3 r_2 & r_3^2 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \vec{r} \vec{r} - I
 \end{aligned} \tag{4.5.97}$$

Then use Equation 4.5.95 and 4.5.96, the term $\vec{r}(\vec{r} \cdot S_0)$ can also be simplified as

$$\vec{r}(\vec{r} \cdot S_0) = \vec{r} \times \vec{r} \times S_0 + S_0(\vec{r} \cdot \vec{r}) = [\vec{r} \vec{r} - I] \cdot S_0 + S_0 = \vec{r} \vec{r} \cdot S_0 \tag{4.5.98}$$

Substituting Equation 4.5.98 to Equation 4.5.91, we find

$$\begin{aligned}
 S_1 &= S_0 \cos \phi + (1 - \cos \phi) \vec{r} \vec{r} \cdot S_0 + \sin \phi (S_0 \times \vec{r}) \\
 &= S_0 \cos \phi + (1 - \cos \phi) \vec{r} \vec{r} \cdot S_0 - \sin \phi (\vec{r} \times S_0) \\
 &= [I \cos \phi + (1 - \cos \phi) \vec{r} \vec{r} - \sin \phi (\vec{r} \times)] S_0
 \end{aligned} \tag{4.5.99}$$

This provides the relationship between the polarization vectors S_0 and S_1 measured at two different frequencies. According to the rotation vector defined in Equations 4.5.79 and 4.5.80, the relationship can be simplified as

$$[M_\Delta] = I \cos \phi + (1 - \cos \phi) \vec{r} \vec{r} - \sin \phi (\vec{r} \times) \tag{4.5.100}$$

To obtain further explicit relationships between entries of matrix $[M_\Delta]$, the rotation angles and the rotation axis, we can expand Equation 4.5.100 into a matrix form,

$$\begin{aligned}
 [M_\Delta] &= \begin{bmatrix} m_{\Delta 11} & m_{\Delta 12} & m_{\Delta 13} \\ m_{\Delta 21} & m_{\Delta 22} & m_{\Delta 23} \\ m_{\Delta 31} & m_{\Delta 32} & m_{\Delta 33} \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cos \phi + (1 - \cos \phi) \begin{bmatrix} r_1^2 & r_1 r_2 & r_1 r_3 \\ r_2 r_1 & r_2^2 & r_2 r_3 \\ r_3 r_1 & r_3 r_2 & r_3^2 \end{bmatrix} - \sin \phi \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} \cos \phi + (1 - \cos \phi)r_1^2 & (1 - \cos \phi)r_1 r_2 + \sin \phi r_3 & (1 - \cos \phi)r_1 r_3 - \sin \phi r_2 \\ (1 - \cos \phi)r_2 r_1 - \sin \phi r_3 & \cos \phi + (1 - \cos \phi)r_2^2 & (1 - \cos \phi)r_2 r_3 + \sin \phi r_1 \\ (1 - \cos \phi)r_3 r_1 + \sin \phi r_2 & (1 - \cos \phi)r_3 r_2 - \sin \phi r_1 & \cos \phi + (1 - \cos \phi)r_3^2 \end{bmatrix} \quad (4.5.101)
 \end{aligned}$$

From Equation 4.5.101, it is straightforward to derive the following relations:

$$Tr([M_\Delta]) = 2\cos \phi + 1 \quad (4.5.102)$$

$$m_{\Delta 12} - m_{\Delta 21} = 2\sin \phi r_3 \quad (4.5.103)$$

$$m_{\Delta 31} - m_{\Delta 13} = 2\sin \phi r_2 \quad (4.5.104)$$

$$m_{\Delta 23} - m_{\Delta 32} = 2\sin \phi r_1 \quad (4.5.105)$$

where $Tr([M_\Delta])$ is the trace of the matrix $[M_\Delta]$. Thus the explicit forms relating $[M_\Delta]$ with the rotation axis $\vec{r}$ and rotation angle ϕ are

$$\cos \phi = \frac{1}{2}(Tr([M_\Delta]) - 1) \quad (4.5.106)$$

$$r_1 = \frac{m_{\Delta 23} - m_{\Delta 32}}{2\sin \phi} \quad (4.5.107)$$

$$r_2 = \frac{m_{\Delta 31} - m_{\Delta 13}}{2\sin \phi} \quad (4.5.108)$$

$$r_3 = \frac{m_{\Delta 12} - m_{\Delta 21}}{2\sin \phi} \quad (4.5.109)$$

The procedure of the Mueller Matrix technique can be summarized as follows:

1. At a certain signal optical frequency ω_1 , launch three independent optical signal SOPs into the fiber with their normalized polarization vectors $[1 \ 0 \ 0]$, $[0 \ 1 \ 0]$, and $[0 \ 0 \ 1]$ and measure the corresponding output SOPs

to complete the matrix using Equation 4.5.73 so that the nine elements of a Mueller matrix $[M(\omega_1)]$ can be obtained. (Theoretically, only two independent optical signal SOPs are sufficient to obtain a Mueller Matrix based on the discussion leading to Equations 4.5.75 and 4.5.76 with proper matrix manipulations.)

2. Repeat step 1 but at a different optical frequency $\omega_2 = \omega_1 + \Delta\omega$ and obtain a Mueller Matrix $[M(\omega_2)]$.
3. Derive the rotation matrix $[M_\Delta]$ using Equation 4.5.80, which indicates the Mueller Matrix change due to the change in optical signal frequency.
4. Use the rotation matrix $[M_\Delta]$ to obtain the rotation angle ϕ and the orientation of the PMD vector $\vec{r}$ based on Equations 4.5.106–4.5.109.
5. Calculate DGD based on $\Delta\tau = \phi/\Delta\omega$, which was used in the Poincare arc measurement as indicated in Equation 4.5.16.

An important feature of the Mueller Matrix technique is that it measures the frequency-dependent orientation of the PMD vector, $\vec{r}$, which is not generally provided by the Jones Matrix technique.

4.6 DETERMINATION OF POLARIZATION-DEPENDENT LOSS

In an optical device, the power transfer function may depend on the SOP of the optical signal. An extreme example is a linear polarizer, in which the transmission loss is nearly zero for a specific state of polarization while the transmission loss is very high for the SOP orthogonal to it. For most of the optical devices such as mirrors, beam splitters, and lenses, it is desirable to have low polarization dependency in the transmission loss. In a fiber-optic system, the polarization-dependent loss (PDL) of the optical fiber itself is usually very small, whereas the overall PDL in the system is affected by PDLs of all the individual optical components. From the measurement point of view, even if the PDL of each component may be well defined with a deterministic value, the PDL of the entire system may randomly vary with time. The overall system PDL cannot be obtained by simply adding up the PDL of each component used in the system; this is due to the random variations of the relative orientations between various PDL elements. In addition, the presence of polarization mode dispersion in the fiber further enhances this uncertainty.

A block diagram of PDL measurement of a fiber-optic system is shown in Figure 4.6.1, where the optical system is treated as a two-terminal device under test (DUT); we should have access to both terminals of the device. In this setup, the SOP of the optical signal entering the DUT is adjusted through a polarization controller, and a photodetector is used to measure the signal optical power

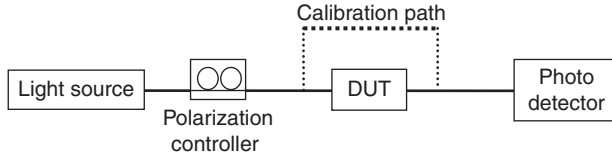


Figure 4.6.1 Measurement setup for polarization-dependent loss.

that passes through the DUT. With the scanning of the signal SOP across the Poincare sphere, a maximum and a minimum value can be detected by the photodetector, and their difference indicates the PDL of the DUT.

In fact, the general definition of PDL is

$$PDL = \frac{T_{\max} - T_{\min}}{T_{\max} + T_{\min}} \quad (4.6.1)$$

where $T_{\max}$ and $T_{\min}$ are the maximum and the minimum power transmission efficiencies of the DUT. If the signal optical power, P_{in} , entering the DUT is polarization-independent, Equation 4.6.1 can be written as

$$PDL = \frac{P_{\max}/P_{in} - P_{\min}/P_{in}}{P_{\max}/P_{in} + P_{\min}/P_{in}} = \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}} \quad (4.6.2)$$

where $P_{\max}$ and $P_{\min}$ are the maximum and the minimum optical powers received by the photodetector when the SOP of the optical signal scans to cover the entire Poincare sphere.

PDL in a fiber-optic system is more complicated than that of a simple optical device due to the random mode coupling and birefringence in the optical fiber. It is also impractical to scan the SOP of the optical signal randomly to cover the entire Poincare sphere. A more systematic measurement procedure is necessary, one in which the PDL can be obtained and that requires only a few individual SOP settings of the optical signal. The most commonly used method to characterize PDL in a fiber-optic system is based on the Mueller Matrix method we introduced in the last section. However, when PDL is considered, the 3×3 matrix is no longer sufficient to describe the system, and a 4×4 matrix needs to be used:

$$\begin{bmatrix} S_{out0} \\ S_{out1} \\ S_{out2} \\ S_{out3} \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix} \begin{bmatrix} S_{in0} \\ S_{in1} \\ S_{in2} \\ S_{in3} \end{bmatrix} \quad (4.6.3)$$

where $S_{in} = [s_{in0} \ s_{in1} \ s_{in2} \ s_{in3}]^T$ is the Stokes parameter of the input optical signal, and for a polarized optical signal, $s_{in0} = \sqrt{s_{in1}^2 + s_{in2}^2 + s_{in3}^2}$ represents the input optical power. Similarly, $S_{out} = [s_{out0} \ s_{out1} \ s_{out2} \ s_{out3}]^T$ is the Stokes parameter of the output optical signal and $s_{out0} = \sqrt{s_{out1}^2 + s_{out2}^2 + s_{out3}^2}$ is the output optical power if the signal is polarized.

Although a 4×4 Mueller Matrix may be used to characterize both the PMD and PDL, if we are only interested in measuring the PDL, the first row of the Mueller Matrix is sufficient, that is,

$$s_{out0} = m_{11}s_{in0} + m_{12}s_{in1} + m_{13}s_{in2} + m_{14}s_{in3} \quad (4.6.4)$$

The power transmission efficiency through the optical system is defined as the output power versus the input power,

$$T = \frac{s_{out0}}{s_{in0}} = \frac{m_{11}s_{in0} + m_{12}s_{in1} + m_{13}s_{in2} + m_{14}s_{in3}}{s_{in0}} \quad (4.6.5)$$

Since the vector $[m_{12} \ m_{13} \ m_{14}]$ describes the optical system property that is independent of the SOP of the input signal $[s_{in1} \ s_{in2} \ s_{in3}]$, the following constraint should be valid:

$$-\sqrt{m_{12}^2 + m_{13}^2 + m_{14}^2} \leq \frac{m_{12}s_{in1} + m_{13}s_{in2} + m_{14}s_{in3}}{\sqrt{s_{in1}^2 + s_{in2}^2 + s_{in3}^2}} \leq \sqrt{m_{12}^2 + m_{13}^2 + m_{14}^2} \quad (4.6.6)$$

Thus, the maximum and the minimum transmission through the fiber should be, respectively,

$$T_{\max} = m_{11} + \sqrt{m_{12}^2 + m_{13}^2 + m_{14}^2} \quad (4.6.7a)$$

$$T_{\min} = m_{11} - \sqrt{m_{12}^2 + m_{13}^2 + m_{14}^2} \quad (4.6.7b)$$

Then, by the definition of PDL as indicated in [Equation 4.6.1](#), we can relate the system PDL with the Mueller Matrix elements:

$$PDL = \frac{\sqrt{m_{12}^2 + m_{13}^2 + m_{14}^2}}{m_{11}} \quad (4.6.8)$$

Now the question is how to measure the Mueller Matrix elements m_{11} , m_{12} , m_{13} , and m_{14} . This can be accomplished using the experimental setup shown in [Figure 4.6.1](#) in which four well-defined SOPs of the input optical signal need to be used. [Table 4.6.1](#) summarizes the input signal SOP and the corresponding output optical power measured by the photodetector [37].

Table 4.6.1
Input Signal SOP Setup for PDL Measurement

Input Signal SOP	Input Stokes Vector	Measured Output Power
Linear horizontal (0°)	$S_{in,a} = (P_a \quad P_a \quad 0 \quad 0)$	$P_1 = m_{11}P_a + m_{12}P_a$
Linear vertical (90°)	$S_{in,b} = (P_b \quad -P_b \quad 0 \quad 0)$	$P_2 = m_{11}P_b - m_{12}P_b$
Linear diagonal (45°)	$S_{in,c} = (P_c \quad 0 \quad P_c \quad 0)$	$P_3 = m_{11}P_c + m_{13}P_c$
Right-hand circular	$S_{in,d} = (P_d \quad 0 \quad 0 \quad P_d)$	$P_4 = m_{11}P_d + m_{14}P_d$

In principle, an ideal polarization controller would not introduce additional signal optical power change when adjusting the signal SOP. However, in practice, there might be internal PDL created by the polarization controller and therefore [Table 4.6.1](#) assumes a slightly different optical power, P_a , P_b , P_c , and P_d , for each of the four input SOP of the optical signal. The output optical power, P_1 , P_2 , P_3 , and P_4 , measured corresponding to each input SOP relates to the Mueller Matrix elements through [Equation 4.6.3](#). Solving the four equations shown in the third column of [Table 4.6.1](#), the following expressions can be obtained:

$$m_{11} = \frac{1}{2} \left(\frac{P_1}{P_a} + \frac{P_2}{P_b} \right) \quad (4.6.9a)$$

$$m_{12} = \frac{1}{2} \left(\frac{P_1}{P_a} - \frac{P_2}{P_b} \right) \quad (4.6.9b)$$

$$m_{13} = \frac{P_3}{P_c} - \frac{1}{2} \left(\frac{P_1}{P_a} - \frac{P_2}{P_b} \right) \quad (4.6.9c)$$

$$m_{14} = \frac{P_4}{P_d} - \frac{1}{2} \left(\frac{P_1}{P_a} - \frac{P_2}{P_b} \right) \quad (4.6.9d)$$

The PDL value can finally be obtained by substituting [Equation 4.6.9](#) into [4.6.8](#). In practical measurements, to create the four well-defined SOPs of the input optical signal, a programmable polarization controller is required with a high level of repeatability. Otherwise, a polarimeter needs to be used to monitor the SOPs of the input optical signal as well.

It is worthwhile to note that both PMD and PDL are system properties that should be independent of the optical signal that is launched into the system. However, the impact of PMD and PDL on the performance of an optical system may depend on the polarization state of the input optical signal. The measurements of PMD and PDL in optical systems that carry live traffic will be discussed in Chapter 5.

4.7 PMD SOURCES AND EMULATORS

Polarization mode dispersion is an important source of performance degradation in fiber-optic transmission systems, especially at high data rates. In system characterization and verification tests, the effect of PMD has to be taking into account to evaluate system margin. It is important to make sure that the transmission equipment can tolerate the PMD level of the fiber system over which the equipment will be deployed. This requires a source to generate a known amount of PMD, either static or statistical, to simulate the PMD characteristic of a fiber system.

In general, a PMD source is referred to as a source that generates a fixed amount of DGD, whereas a PMD emulator is a device that generates a time-varying DGD to emulate the practical PMD characteristics in fiber systems. PMD sources are usually used to test the deterministic PMD impact on systems or subsystems because the added PMD level from the source is precisely known. On the other hand, a PMD emulator is often used to test the statistical nature of the system such as PMD-induced BER outage and the effectiveness of the PMD compensator in the receiver.

The basic configuration of the simplest PMD source is shown in [Figure 4.7.1](#), where two polarization beam splitters (PBS) are used. The input optical signal is first split by PBS1 into horizontal and vertical polarization components. Then one of them passes through an adjustable delay line before the two orthogonally polarized light beams combine in PBS2. The differential group delay (DGD) of the PMD source can be set by the time delay of the adjustable delay line.

To use this type of PMD source correctly, the input optical signal has to be linearly polarized. The SOP of the input signal has to be adjusted midway between the two principle axes of the PBS so that the powers of the horizontal and vertical polarization components are equal. In fact, the impact of the PMD in an optical receiver depends on the ratio of the signal optical power carried by the two PSPs in the fiber system [25],

$$P_{\text{penalty}}(\text{dB}) = \eta \left(\frac{\tau_{\text{DGD}}}{T_B} \right)^2 \gamma(1 - \gamma) \quad (4.7.1)$$

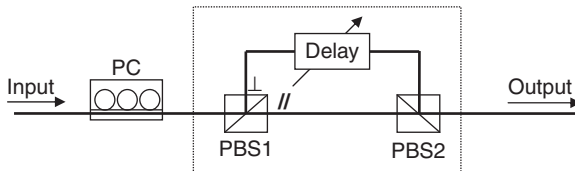


Figure 4.7.1 Block diagram of a simple PMD source.

where P_{penalty} is the power penalty caused by the PMD in the receiver, η is a proportionality parameter that depends on the modulation formats and receiver bandwidth, τ_{DGD} is the first-order DGD of fiber system, and T_B is the bit interval. $0 \leq \gamma \leq 1$ is the power-splitting ratio of the two PSPs, which is defined as $\gamma = P_{\text{fast}}/P_{\text{total}}$, where P_{fast} is the power carried by the fast axis and $P_{\text{total}} = P_{\text{fast}} + P_{\text{slow}}$ is the total signal power. Obviously for either $\gamma = 0$ or $\gamma = 1$, the signal optical power is carried by one of the two PSPs of the fiber and the PMD-induced penalty should be zero. The principle axes of the PBS in the PMD source are equivalent to the PSPs of a birefringence fiber in a transmission system. When the SOP of the input optical signal is midway between the two PSPs, $\gamma = 0.5$ and $\gamma(1 - \gamma) = 0.25$. If we define an angle θ between the SOP of the input optical signal and the PSP of the fiber, Equation 4.7.1 will become

$$P_{\text{penalty}}(\text{dB}) = \eta \left(\frac{\tau_{\text{DGD}} \sin(2\theta)}{2T_B} \right)^2 \quad (4.7.2)$$

where $0 \leq \theta \leq \pi/2$ corresponds to $0 \leq \gamma \leq 1$. For the PMD source using the configuration shown in Figure 4.7.1, if the SOP of the input optical signal is not equal to $\theta = \pi/4$, the equivalent DGD will be $|\tau_{\text{DGD}} \sin(2\theta)| \leq \tau_{\text{DGD}}$. Therefore, to provide the specified DGD value, we must make sure that the input SOP is correctly aligned at $\theta = \pi/4$. An easy procedure for this polarization alignment is to block one of the two arms and observe a 3 dB drop in the output optical power. This ensures that the optical powers in the two arms are equal.

The PMD source shown in Figure 4.7.1 can be constructed with polarization-maintaining optical fibers between the two PBSs. However, a disadvantage of a fiber-based PMD source is that the output signal SOP always drifts randomly and is hard to stabilize. In fact, if the optical field amplitude of the two arms is equal, $|E_{//}| = |E_{\perp}| = |E_1|$, the output optical field should be $E_0 = |E_1| [\vec{a}_x + \vec{a}_y e^{j\omega\Delta\tau} e^{j\varphi(t)}]$, where $\vec{a}_x$ and $\vec{a}_y$ are unit vectors, $\Delta\tau$ is the differential time delay, and $\varphi(t)$ is the relative optical phase difference between the optical signals passing through the two arms. The SOP of the combined output optical field E_0 is determined directly by the optical phase $\varphi(t)$. For a 10 cm long fiber delay line, for example, if the temperature sensitivity of the refractive index in silica is $\sim 10^{-5}/^\circ\text{C}$ and the signal wavelength is $1.5 \mu\text{m}$, there will be approximately $2\pi/3$ rotation of the output SOP for each degree change in temperature. Temperature-induced mechanical stress will also contribute to phase instability.

A compact PMD source can be made using multiple sections of birefringence crystals, as illustrated in Figure 4.7.2 [38]. An electrically controlled binary polarization switch cell is inserted between adjacent birefringence crystals, which provides a 90° polarization switch when a current is applied. Suppose the SOP of the linearly polarized input optical signal is $+45^\circ$ with respect to the extraordinary axis of the birefringence crystal. The first switch determines

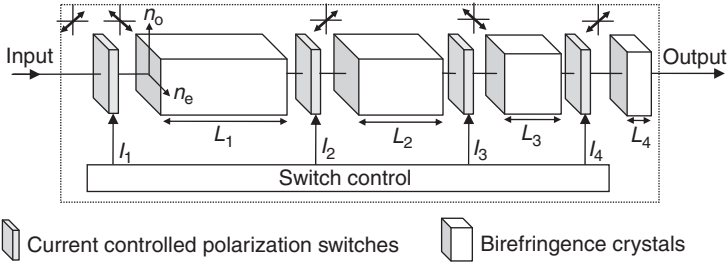


Figure 4.7.2 PMD source using concatenated birefringence crystals and polarization switches.

whether the SOP of the optical signal remains at $+45^\circ$ or it is switched to -45° when it reaches the first crystal section of length L_1 . Then, if the second switch cell makes a 90° switch, the (o, e) beams in the first crystal will become the (e, o) beams in the second crystal, otherwise, the signal SOP will remain unchanged. As a consequence, the total DGD for the optical signal will be $(n_o - n_e)(L_1 \pm L_2)/c$, depending on the status of the polarization switches. In general, for a PMD source with N birefringence crystal sections, the total DGD can be expressed as

$$\tau_{DGD} = \left| \sum_{i=1}^N (\pm \Delta\tau_i) \right| \quad (4.7.3)$$

where $\Delta\tau_i = (n_o - n_e)L_i/c$ is the DGD of the i -th crystal section and the $\pm$ sign is determined by the binary status (0 or 90°) of the polarization switch in front of each birefringence crystal. In this implementation, the instability problem for the output signal SOP can be eliminated because the optical signal does not have to be split when passing through the device. The maximum DGD this PMD source can provide depends on the total length of the birefringence crystals, $\tau_{DGD, \max} = (n_o - n_e) \sum_{i=1}^N L_i/c$. As an example, a YVO_4 crystal has the birefringence value of approximately $(n_o - n_e) \approx 0.2$ in the 1550 nm wavelength window. To make a PMD source with the maximum DGD of 50 ps, the aggregated crystal length has to be 75 mm.

Because binary polarization switches are used in this PMD source, it is able to specifically provide first-order DGD at the specified value while maintaining a negligible level of second-order DGD. By dynamically controlling the status of the polarization switches, it is possible to generate time-dependent DGD with any statistical distribution. The correlation time is determined by the speed of the polarization switches. With this dynamic control, the PMD source actually operates as a PMD emulator.

In high-speed optical systems, high orders of PMD may become important, affecting system performance. To create a high-order PMD effect, a large number of birefringence sections can be concatenated with random length and axes

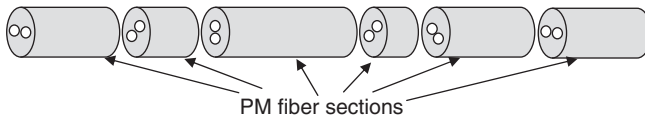


Figure 4.7.3 All-order PMD emulator created by splicing together a large number of PM fiber sections.

alignments. One simple way to make this type of all-order PMD emulator is to splice many pieces of polarization-maintaining (PM) fibers together with random lengths and orientations, as illustrated in [Figure 4.7.3](#). This closely simulates the propagation effect in a practical fiber system. However, since these fiber sections are permanently spliced together, the PMD emulator is no longer adjustable, except for the controlling through environmental perturbations such as temperature and mechanical stress or by sweeping the wavelength of the optical signal. Over a long time in a noncontrolled environment or by sweeping the wavelength across a wide window, a Maxwellian distribution of PMD can be obtained.

However, from system measurement point of view, for a transmission channel with a fixed signal wavelength, it is desirable to accelerate the PMD performance test using an emulator with adequate dynamic control such that the DGD converges rapidly to the expected Maxwellian distribution. This can be achieved by adding a polarization controller or a phase shifter between each pair of birefringence sections, as illustrated in [Figure 4.7.4](#). In this configuration, the birefringence section can be made by either PM fibers or YVO_4 crystals and their birefringence axes can be randomly aligned. The dynamic control of SOP rotation or signal optical phase retardation between birefringence sections accelerates the convergence of PMD statistics toward Maxwellian distribution.

PMD sources and emulators are commercially available with various optical configurations depending on the required maximum DGD level, PMD statistics, and dynamics. Different control mechanisms and algorithms, such as the control of SOP or optical phase retardation between birefringence sections or the

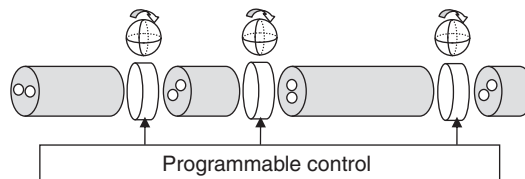


Figure 4.7.4 All-order PMD emulator created by using a polarization controller or a phase shifter between birefringence elements.

control of the DGD provided by each section, may exhibit different PMD statistics. Interested readers on the statistics of DGD generated by different types of PMD emulators can find details at [38, 39, 40, 41].

4.8 MEASUREMENT OF FIBER NONLINEARITY

Nonlinearity in an optical fiber refers to the transmission properties which depend on the optical power carried in the fiber. There are two basic types of nonlinear effects in optical fibers. The first type is due to the interaction of optical signal with phonons, which generates nonlinear scattering, such as stimulated Raman scattering (SRS) and stimulated Brillouin scattering (SBS). As a result, some of the input signal photons are converted into Stokes photons at a longer wavelength. This usually results in the loss of signal optical power, whereas it can also be utilized to perform optical amplification for signals at longer wavelengths.

The second type of nonlinearity is due to the Kerr effect, which produces the intensity-dependent refractive index in the fiber. Several well-known effects such as self-phase modulation (SPM), cross-phase modulation (XPM), four-wave mixing (FWM), and modulation instability are caused by the Kerr effect nonlinearity. In principle, instead of optical power, the power density is the most important quantity that determines the strength of the nonlinear effects in the fiber. Although a single-mode fiber usually carries signal optical power in the level of milliwatts, because the cross-section area of an optical fiber is very small, the power density can be very high. One important parameter to specify the power-handling capability of an optical fiber is its effective cross-section area A_{eff} , which is generally different from the geometric core cross-section area of the fiber.

As illustrated in Figure 4.8.1, the optical power distribution in the fiber core is nonuniform; therefore the power density in the fiber is, strictly speaking, non-uniform. The concept of effective area assumes that a uniform power

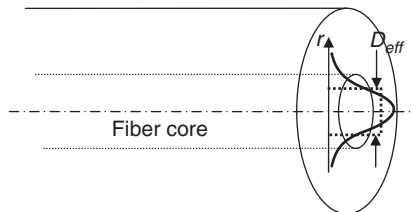


Figure 4.8.1 Illustration of fiber diameter D_{eff} .

distribution in the effective area A_{eff} would produce the correct amount of nonlinear effect, and therefore the definition of fiber effective area is

$$A_{eff} = \frac{\pi}{4} D_{eff}^2 = \frac{2\pi \left[\int_0^\infty P(r) r dr \right]^2}{\int_0^\infty P(r)^2 dr} \quad (4.8.1)$$

where D_{eff} is the effective cross-section diameter and $P(r)$ is the actual power distribution in the fiber. As an example, the effective cross-section area of a standard single-mode fiber is approximately $80 \mu\text{m}^2$. If the fiber carries an optical power of 10 mW, the power density is 12.5 kW/cm^2 .

In addition, optical signals propagate in the optical fiber for a very long distance, especially for optically amplified multispans fiber systems; the long interaction length between the optical power and the fiber material further magnifies the effect of nonlinearity. For the nonlinear scattering processes such as SBS and SRS, nonlinear gain coefficient and gain spectrum are important parameters, whereas the nonlinear Kerr effect is mainly determined by a nonlinear parameter γ , which is proportional to the nonlinear refractive index of the material and inversely proportional to the fiber core cross-section. All these parameters differ for different types of fibers, and they depend both on the composition of the glass material and on the mode-field distribution of the waveguide. In this section, we discuss measurement techniques of these nonlinear parameters.

4.8.1 Measurement of Stimulated Brillouin Scattering Coefficient

When an intense optical wave, commonly referred to as a *pump wave*, at a frequency f_p propagates in a fiber, it interacts with the mechanical structure of the material and creates an acoustic wave at a frequency f_a , which is about 11.1 GHz for fused silica-based optical fibers. Then a Stokes wave at a frequency of f_B is generated due to the interaction between the original pump wave and the acoustic wave. In this a three-wave mixing process. To satisfy the rule of energy conservation, the frequency of the Stokes wave has to be $f_B = f_p - f_a$. In addition, momentum conservation $\beta_B = \beta_p - \beta_a$ is important for the efficiency of SBS process, where $\beta_{B,p,a} = 2\pi f_{B,p,a} / v_{B,p,a}$ are the wave numbers of the Stokes wave, the pump wave, and the acoustic wave, respectively. Since the propagation speed of the acoustic wave is much slower than both the pump

wave and the Stokes wave, $\beta_a \gg \beta_p \approx \beta_B$, SBS in the forward direction does not exist, whereas the backscattering of SBS has the highest efficiency.

When a pump wave with the optical power $P_p(z)$ propagates along an optical fiber, the power of the Stokes wave generated along the fiber $P_B(f, z)$ can be described by the following propagation equation:

$$\frac{dP_B(f, z)}{dz} = -\frac{g_B(f, z)P_p(z)}{A_{eff}}P_B(f, z) + \alpha P_B(f, z) \quad (4.8.2)$$

where the negative sign indicates the backward propagation of the Stokes wave. α is the fiber attenuation coefficient and z is the position in the fiber longitudinal direction.

$$g_B(f, z) = \frac{g_0}{1 + \left[(f - f_B(z)) / (\delta f_B / 2) \right]^2} \quad (4.8.3)$$

is the backscattering Brillouin gain coefficient, which has a Lorentzian spectral shape with g_0 the peak gain. In Equations 4.8.2 and 4.8.3 we have assumed that in general the Brillouin gain coefficient may not be uniform along the fiber, and therefore it is a function of z . The Brillouin gain also has a certain spectral bandwidth δf_B , which is about 35 MHz in the 1550 nm wavelength window for fused silica fibers.

In most practical applications, the power of Brillouin scattering is much smaller than that of the pump and thus the pump depletion is usually negligible. Under this condition, the pump power distribution along the fiber can be simply described by $P_p(z) = P_p(0)e^{-\alpha z}$, and Equation 4.8.2 can be solved analytically so that the backward propagated SBS power at the fiber input can be calculated as

$$P_B(f, 0) = P_B(f, L) \exp \left(-\alpha L + \frac{P_p(0)}{A_{eff}} G(f) \right) \quad (4.8.4)$$

where L is the fiber length and $P_s(f, L)$ is the initial backward optical power at the far end of the fiber, which is usually created by spontaneous emission. The nonlinear optical gain introduced by SBS process in the optical fiber is

$$G(f) = \int_0^L \frac{g_0 \exp(-\alpha z)}{1 + \left[(f - f_B(z)) / (\delta f_B / 2) \right]^2} dz \quad (4.8.5)$$

The peak gain coefficient g_0 is mainly determined by the fiber material and can be expressed as [42]

$$g_0 = \frac{2\pi n^7 p_{12} K}{c \lambda^2 \rho v_a \delta f_B} \quad (4.8.6)$$

where n is the refractive index of the material, ρ is the material density, p_{12} is elasto-optic coefficient, λ is the average wavelength, v_a is the acoustic wave velocity, and c is the speed of light in vacuum. K is a polarization mismatch factor, which accounts for the fact that the three-wave mixing process requires the matched polarization states between the pump and the Stokes waves. If the birefringence along the fiber is random, the K value ranges between 1/3 and 2/3.

Because of the Lorentzian line shape of Brillouin gain, as shown in Equation 4.8.5, the FWHM linewidth of the scattered power $P_B(f, 0)$ due to SBS will be

$$\Delta f_{eff} = \frac{\delta f_B \sqrt{\pi}}{2} \sqrt{\frac{\alpha A_{eff}}{g_0 P_p(0)}} \quad (4.8.7)$$

The total backward-propagated Stokes power can be calculated by integrating all the contributions of spontaneous emission along the fiber, which is amplified by the SBS gain of the nonlinear process. It has been shown that for pure spontaneous scattering, this integration is approximately equivalent to the injection of a single Stokes photon at the point along the fiber where the nonlinear gain exactly equals to the attenuation of the fiber. Under this approximation, the effective optical gain of a spontaneous photon at the peak-gain frequency is [43]

$$G_{eff} = \frac{\exp\{P_p(0)g_0/(A_{eff}\alpha)\}}{P_p(0)g_0/(A_{eff}\alpha)} \quad (4.8.8)$$

For the Brillouin scattering process, the phonons are thermally activated and the effective starting Stokes power must be multiplied by $(kT/hf_a + 1) \approx kT/hf_a$, where, f_a is the frequency of the acoustic phonon, k is Boltzmann's constant, T is the absolute temperature, and h is the Planck's constant. Then the total backward-propagated Stokes power that reaches the terminal where the pump is injected can be found simply by

$$P_B(0) = G_{eff} \frac{kT}{hf_a} (hf_B) \Delta f_{eff} \quad (4.8.9)$$

Figure 4.8.2 shows the calculated power of SBS backscattering at the fiber input side using Equation (4.8.9) with the following parameters: $\alpha = 0.25 \text{ dB/km}$, $g_0 = 1 \times 10^{-11} \text{ m/W}$, $A_{eff} = 8 \times 10^{-11} \text{ m}^2$, $f_a = 11 \text{ GHz}$, and $f_B = 200 \text{ THz}$. Obviously, the backscattered optical power increases sharply when the input pump power reaches a certain level. A parameter that can be used to specify SBS performance in an optical fiber is the *SBS threshold*. A widely adopted definition of the term is the pump power level P_{th} at which the backscattered SBS power equals the launched pump power [43]. This threshold definition is somewhat arbitrary, and in fact from the energy conservation point of view it is even physically impossible for the backscattered Stokes power to be equal to the input pump power. However,

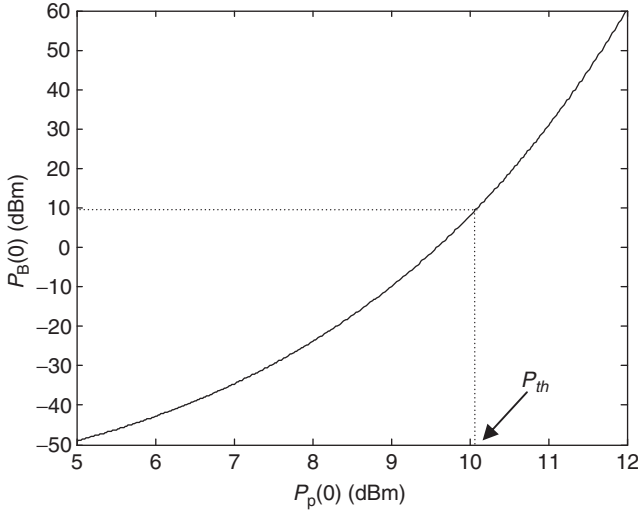


Figure 4.8.2 Backscattered optical power $P_B(0)$ as a function of the input pump power $P_p(0)$ due to SBS calculated with Equation 4.8.9.

since the backscattered power is increased very rapidly in the threshold region, it would not make a significant difference if this threshold was defined at which the backscattered power was equal to, for example, half the input pump power. A good approximation of the SBS threshold is

$$P_{th} \approx \chi_B \frac{A_{eff} \alpha}{g_0} \quad (4.8.10)$$

where the coefficient χ_B ranges from 19 to 21, depending on the characteristics of the fiber [44].

It is worthwhile to point out that the result shown in Figure 4.8.2 was calculated using Equation 4.8.9, where pump depletion was neglected; therefore the backscattered power due to SBS can be even higher than the input pump power in this figure. In this example, the threshold pump power is approximately 10 dBm; according to Equation 4.8.10, this threshold value corresponds to the coefficient of $\chi_B \approx 21$. This χ_B coefficient has a smaller value of 19 for older fibers in which the attenuation is much higher. Although Equation 4.8.9 helps explain and define the SBS threshold, it is valid only when the pump power is much less than the SBS threshold. Practically, when the input pump power increases to a certain value close to the SBS threshold, there will be a rapid increase in the backscattered Stokes power. As a result, the output pump power starts to saturate with further increasing the input pump, and thus pump depletion becomes significant.

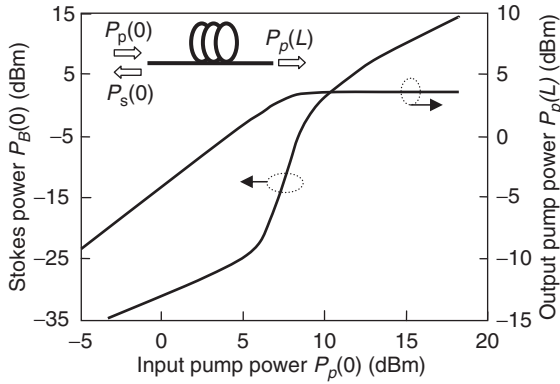


Figure 4.8.3 Illustration of the output optical power saturation in an optical fiber due to the increase of backscattered Stokes power at high input power levels.

Figure 4.8.3 shows the measured backscattering Stokes power at the input side of the fiber together with the pump power at the output side of the fiber as a function of the input pump power in a dispersion-shifted fiber [42]. The fiber parameters are: $\alpha = 0.31 \text{ dB/km}$, $A_{\text{eff}} = 35.8 \times 10^{-12} \text{ m}^2$ and the fiber length is 13 km. The measurement setup is shown in Figure 4.8.4, where a single-frequency laser is used as the light source whose linewidth is much narrower than the SBS bandwidth of 35 MHz. An optical isolator placed in front of the laser source prevents the optical reflection back to the laser. A 2×2 fiber directional coupler of 10 percent splitting ratio is used to deliver 90 percent of the optical power from the source into the fiber under test, and 10 percent of the power is sent to power meter 2 to monitor the source power. The backscattered optical power from the fiber due to SBS is directed into power meter 1 through the directional coupler. In this configuration the optical powers measured by the three power meters P_1 , P_2 , and P_3 are related to the parameters in Figure 4.8.3 as $P_1 = 0.1P_s(0)$, $P_2 = 0.1P_p(0)$, and $P_3 = P_p(L)$.

For SBS threshold determination, since the backscattered Stokes power can never be as high as the injected pump power, the SBS threshold can be

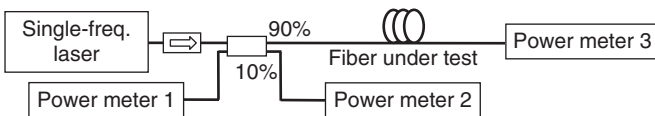


Figure 4.8.4 Experimental setup to measure SBS coefficient in a fiber. Power meters 1, 2, and 3 measure the backscattered power, the source power, and the transmitted power, respectively.

practically defined as the pump power level at which the backscattered power caused by SBS is equal to that due to Rayleigh backscattering. Again, since the power of SBS backscattering increases rapidly near the threshold, this modified definition would provide a very similar SBS threshold value as that defined by Equation 4.8.10.

Another important parameter in the SBS process is the frequency shift, which is in the neighborhood of 11 GHz, depending on the material composition of the fiber. For example, for the fiber doped with F and GeO_2 , an empirical equation for SBS frequency shift is

$$f_a(\text{GHz}) = 11.045 - 0.277[F] + 0.045[\text{GeO}_2] \quad (4.8.11)$$

where $[F]$ and $[\text{GeO}_2]$ are the weight percentage of F and GeO_2 concentrations in the glass material.

The measurements of the SBS frequency shift f_a and its bandwidth δf_B are important in fiber selection for optical transmission as well as fiber sensors. The precision of these measurements depends on the spectral resolution of the measurement setup. In general, a tunable laser and an optical spectrum analyzer cannot provide sufficient resolution for these measurements, while an RF spectrum analyzer may be better suited for this purpose.

Figure 4.8.5 shows the experimental setup to measure SBS frequency shift and the SBS bandwidth in a fiber [45]. This setup is based on coherent self-homodyne detection. A single-frequency optical signal from a DFB laser diode operating in CW is amplified by an EDFA and split into two parts through the first fiber coupler. One part is used as the local oscillator for coherent detection; the other part is amplified by another EDFA to boost the optical power

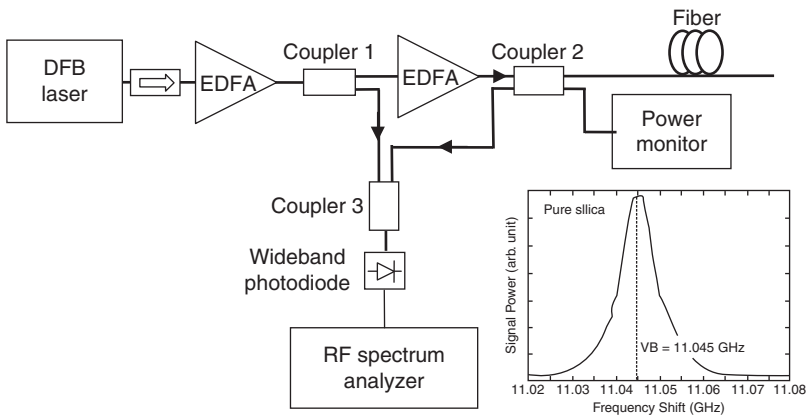


Figure 4.8.5 Experimental setup to measure SBS frequency shift and bandwidth in a fiber.

for SBS measurement. The amplified optical signal is injected into the fiber under test, and the second fiber coupler is used to redirect the backscattered SBS signal into coupler 3 to mix with the original optical signal, which acts as the local oscillator. A photodiode with bandwidth wider than the SBS frequency shift f_a is used for coherent homodyne detection, which brings the SBS optical spectrum into the RF domain. An RF spectrum analyzer is then used to display the SBS spectrum. The inset in Figure 4.8.5 is an example of the measured backscattering power spectral density of a pure silica fiber, where the central frequency shift is 11.045 GHz [45].

4.8.2 Measurement of the Stimulated Raman Scattering Coefficient

In contrast to SBS, in which the frequency shift is approximately 11 GHz and the bandwidth is narrow, SRS has a much higher frequency shift, on the order of 1.3 THz, and much wider bandwidth. In addition to the fact that they have different applications and their impacts on optical systems are very different, the measurement techniques for these two effects are also different.

Stimulated Raman scattering in an optical fiber is caused by the interaction between the optical signal photons and the energy states of the silica molecules. In this scattering process, a photon spends part of its energy to excite a molecule from a lower to a higher vibrational or rotational state. The frequency of the scattered photon f_s , commonly referred to as a *Stokes wave*, is

$$f_s = f_p - \frac{\Delta E_e}{h} \quad (4.8.12)$$

where ΔE_e is the molecular energy level difference of the material, f_p is the input photon frequency, and h is the Planck's constant. If the material structure has a regular crystal lattice, the vibrational or rotational energy levels of the molecules are specific; therefore the spectrum of SRS would be relatively narrow. For amorphous materials, such as fused silica, the SRS spectrum is wide because the electron energy levels are not very well regulated.

Figure 4.8.6 shows the normalized SRS spectrum as a function of the frequency shift $f = f_p - f_s$ in fused silica with $f_p = 1 \mu\text{m}$ [46]. The peak frequency shift is about 13 THz and the FWHM width of the spectrum is approximately 7 THz. Obviously, the spectral shape of SRS is not Lorentzian. In addition to the much higher frequency shift and the wider spectral bandwidth, SRS differs from SBS by the fact that it propagates in both forward and backward directions in an optical fiber.

When a pump wave with the optical power of $P_p(z)$ propagates along an optical fiber, the power of the Stokes wave $P_s(f, z)$ generated along the fiber can be described by the following propagation equations:

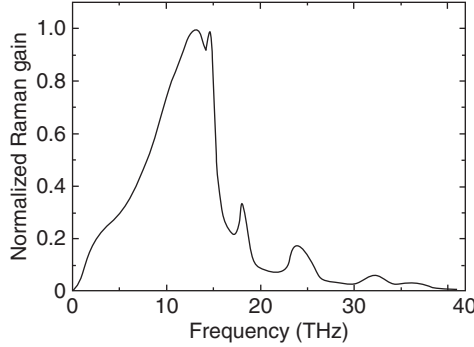


Figure 4.8.6 Raman-gain spectrum of fused silica as a function of the frequency shift. Pump wavelength is $1\mu\text{m}$.

$$\left[\frac{d}{dz} \pm \alpha \right] P_s(f, z) = \pm \frac{g_R(f, z) P_p(z)}{A_{eff}} P_s(f, z) \pm \frac{\sigma_{sp} P_p(z)}{A_{eff}} \quad (4.8.13)$$

where the $\pm$ sign represents forward (+) and backward (−) propagating Stokes waves, and $g_R(f, z)$ is the SRS gain coefficient, which is generally frequency dependent, as shown in Figure 4.8.6. We have assumed that the SRS gain coefficient may be nonuniform along the fiber and therefore is a function of z . σ_{sp} is the effective spontaneous Raman scattering coefficient. If we neglect pump depletion and assume that the power of the SRS scattering is much smaller than the pump, the pump power along the fiber can be simply described by $P_p(z) = P_p(0)e^{-\alpha z}$ and Equation 4.8.13 can be rewritten as

$$\left[\frac{d}{dz} \pm \alpha \right] P_s(f, z) = \pm \frac{g_R(f, z) P_s(f, z)}{A_{eff}} P_p(0)e^{-\alpha z} \quad (4.8.14)$$

where the spontaneous Raman scattering effect is neglected for simplicity. As a practical application, Raman gain in optical fiber can be used to make optical amplifiers. At the same time, Raman gain also introduces wideband amplified spontaneous emission. If an optical signal $P_s(f, 0)$ in the Stokes wave frequency is injected at the input side of the fiber, it can be amplified by the Raman effect and the output signal power will be

$$P_s(f, L) = P_s(f, 0) \exp \left[\frac{g_R(f)}{A_{eff}} P_p(0) \left(\frac{1 - e^{-\alpha L}}{\alpha} \right) - \alpha L \right] \quad (4.8.15)$$

where L is the fiber length; for simplicity we have assumed that the Raman gain coefficient $g_R(f)$ is uniform along the fiber. Because pump depletion is

neglected, the total optical gain will be the same for both the forward- and the backward-propagated optical signals at the Stokes frequency. The Raman gain coefficient is a material property that is proportional to the resonant contribution associated with molecular vibrations:

$$g_R(f) = \text{Im} \left[\chi_{\text{res}}^{(3)}(f) \right] \frac{8\pi^2 f_p}{cn_0} \quad (4.8.16)$$

where $\text{Im} \left[\chi_{\text{res}}^{(3)}(f) \right]$ is the imaginary part of the third-order susceptibility and n_0 is the linear refractive index of the material. The peak Raman gain coefficient of $g_R = 1.9 \times 10^{13} \text{ m/W}$ has been measured in fused silica at 526 nm wavelength. This value is about two orders of magnitude smaller than the SBS gain coefficient.

Similar to the SBS, the SRS Stokes power can also be calculated by integrating all the contributions of spontaneous emission along the fiber, which is amplified by the SRS gain process. If we define an *SRS threshold* P_{th} , at which the total SRS power equals the launched pump power [43], this threshold power can still be evaluated by an approximation similar to [Equation 4.8.10](#):

$$P_{\text{th}} \approx \chi_R \frac{A_{\text{eff}} \alpha}{g_R} \quad (4.8.17)$$

where the value of the coefficient χ_R is approximately 16, which is smaller than that for the SBS threshold. In addition, since the SRS gain coefficient g_R is much smaller than the SBS gain coefficient, the SRS-limited pump threshold can be much higher than that due to SBS. However, it is worthwhile to mention that since SBS is a narrowband process, the SBS threshold can be significantly increased if the optical bandwidth of the pump is much wider than the SBS bandwidth of 35 MHz. In fact, in practical optical fiber communication systems, frequency dithering is usually applied in optical transmitters to widen their spectral linewidths, therefore reducing the nonlinear SBS effect. For SRS, on the other hand, the bandwidth is several terahertz wide, so there is no simple technique for increasing SRS threshold. In WDM optical systems, the SRS effect converts energy from short wavelength channels to long wavelength channels, not only introducing inter-channel crosstalk but also causing power-level tilt over the wavelength window.

An important application of SRS is performing optical amplification. Fiber-based Raman amplifiers are commercially available and can be designed to operate at different wavelength windows as long as the suitable pump lasers are available for these wavelengths. Distributed Raman amplification can also be used in optical fiber transmission systems, where an intense pump at the wavelength shorter (by $\sim 13 \text{ THz}$) than the optical signal is injected into the optical fiber. In this case the optical signal is amplified due to the Raman gain provided by the pump, and the transmission fiber itself is used as the gain medium.

Raman gain, gain bandwidth, and frequency shift are important parameters for the design of Raman amplifiers. Since these parameters are determined by fiber types, precise measurement and characterization are critically important for amplifier design and system performance evaluation.

Figure 4.8.7 shows the schematic of an experimental setup to measure Raman gain in a fiber. A high power pump laser is used at wavelength λ_p and a low-power wideband light source is used as the probe, which can be a lamp or a superluminescence LED. To minimize the pump SBS effect, the linewidth of the pump laser needs to be controlled, either by frequency dithering through direct current injection or by a phase modulation externally. The pump and the probe are combined in a fiber directional coupler and delivered into the fiber under test. A power meter is used to monitor the pump and the probe optical powers that are actually launched into the fiber. An optical spectrum analyzer is used at the output of the fiber to measure the power spectrum of the Stokes wave.

Based on Equation 4.8.15, the optical gain at the Stokes frequency is

$$G_s(f) = \frac{P_s(f, L)}{P_s(f, 0)} = \exp \left\{ \frac{g_R(f)}{A_{eff}} P_p(0) L_{eff} \right\} e^{-\alpha L} \quad (4.8.18)$$

where $L_{eff} = (1 - e^{-\alpha L})/\alpha$ is the nonlinear length of the fiber, $P_s(f, 0)$ is the power spectral density of the probe that is injected at the fiber input, and $P_s(f, L)$ is that measured by the optical spectrum analyzer at the fiber output. There are a number of effects that may affect the accuracy of this Raman gain measurement. First, the contribution of spontaneous emission is included in the output optical power spectrum; second, the efficiency of power coupling into the fiber might not be easy to calibrate. However, at each Stokes frequency, if $G_s(f)$ is plotted in logarithm scale, $\log(G_s)$ should be linearly proportional to both the Raman gain coefficient and the injected pump power. In fact, from Equation 4.8.18 we can find

$$\frac{g_R(f)}{A_{eff}} L_{eff} = 0.23 \frac{dG_{s,dB}(f)}{dP_p(0)} \quad (4.8.19)$$

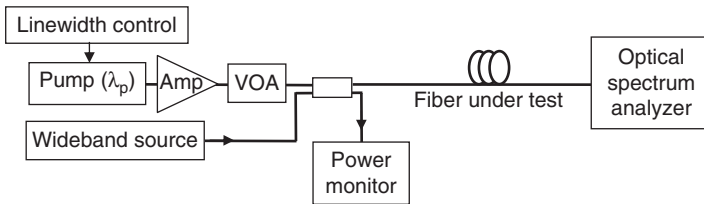


Figure 4.8.7 Experimental setup to measure SRS gain in a fiber. *Amp*: optical amplifier, *VOA*: variable optical attenuator.

where $G_{s,dB}(f) = 10\log G_s(f)$ is the Stokes gain expressed in decibels. By a programmable increase of the pump power through adjusting the variable optical attenuator (VOA), $G_{s,dB}(f)$ as a function of pump power can be measured at each wavelength. Then the composite Raman gain coefficient $g_R L_{eff}/A_{eff}$ can be found through a slope measurement based on Equation 4.8.19.

In the measurement setup shown in Figure 4.8.7, a wideband light source is used as the probe and the spontaneous Raman scattering is neglected because we assumed that the amplified probe is much stronger than the amplified spontaneous emission. In principle, amplified spontaneous Raman scattering can also be used to evaluate the Raman gain coefficient. However, the effective spontaneous Raman scattering coefficient σ_{sp} itself depends on many other effects such as temperature and the efficiency of coupling spontaneous scattering into the guided mode in the fiber.

Figure 4.8.8 shows the block diagram of another measurement technique that uses a square-wave pump for the measurement of fiber Raman gain. In this technique a CW pump generated by a DFB laser diode is phase modulated to minimize the SBS effect in the fiber. The pump wave is then intensity modulated by a square pulse train with duty cycle η and pulse duration τ_0 . An optical amplifier is used to boost the pump power, and a bandpass optical filter is applied at the pump wavelength to remove the ASE noise generated by the source and the optical amplifier. Due to the intense optical pump at wavelength λ_p , a Stokes wave is generated in the fiber at wavelength λ_s through the SRS process. Because of chromatic dispersion, the pump and the Stokes waves propagate in different speeds in the fiber; therefore the overlap between the pump and the Stokes waves depends on the pulse width of the pump and their relative delay. Considering the time-varying nature of the pump and the relative walk-off between the pump and the Stokes waves, the propagation Equation 4.8.13 of the forward Stokes power can be written as

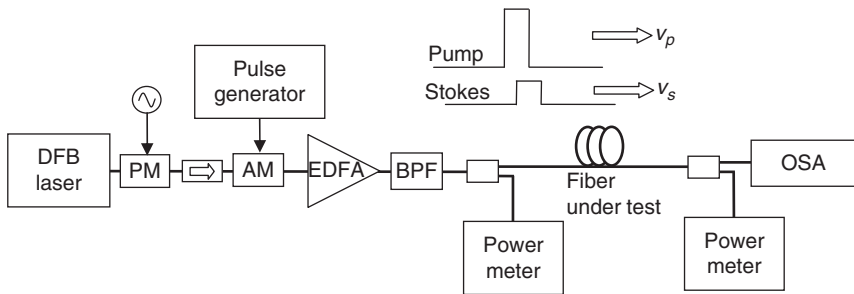


Figure 4.8.8 Experimental setup of SRS measurement using pulsed pump. *PM*: phase modulator, *AM*: amplitude modulator, *BPF*: optical bandpass filter.

$$\frac{\partial P_s(z, t)}{\partial z} = \frac{g_R(z, t)P_p(z, t - z/\Delta v)}{A_{eff}} P_s(z, t) + \frac{\sigma_{sp}P_p(z, t - z/\Delta v)}{A_{eff}} \quad (4.8.20)$$

where $\Delta v = v_p - v_s$ is the group velocity difference between the pump and the forward-propagating Stokes pulses, which is related to the chromatic dispersion parameter D of the fiber by $\Delta v = 1/[D(\lambda_p - \lambda_s)]$. Fiber attenuation has been neglected in Equation 4.8.20 for simplicity.

The effective interaction length between the pump and the Stokes pulses is $L_w = \tau_0 \cdot \Delta v$. For short optical pulses such that $L > L_w$, where L is the total fiber length, the average power density of the Stokes signal at a certain frequency f_s can be found by integrating Equation 4.8.20 as

$$\langle P_s(f_s) \rangle_{L > L_w} = P_{s0}\eta \left\{ (2 + P_0 g_s L) \frac{[\exp(P_0 g_s L_w) - 1]}{P_0 g_s L_w} - [\exp(P_0 g_s L_w) + 1] \right\} \quad (4.8.21)$$

Where $P_{s0} \equiv \sigma_{sp}/g_s(f_s)$, $g_s(f_s) \equiv g_R(f_s)/A_{eff}$ and P_0 is the peak power of the square pump waveform $P_p(0)$. For long optical pulses, on the other hand, if the walk-off length is longer than the length of the fiber, the average power density of the Stokes signal is

$$\langle P_s(f_s) \rangle_{L < L_w} = P_{s0}\eta \left\{ (2 + P_0 g_s L_w) \frac{[\exp(P_0 g_s L_w) - 1]}{P_0 g_s L_w} - \frac{L}{L_w} [\exp(P_0 g_s L) + 1] \right\} \quad (4.8.22)$$

The effect of fiber loss can be approximately counted for by replacing L and L_w with $[1 - \exp(-\alpha L)]/\alpha$ and $[1 - \exp(-\alpha L_w)]/\alpha$, respectively.

Figure 4.8.9(a) shows the power spectral density of the Stokes signal measured using the setup shown in Figure 4.8.8, where a 10.1 km standard single-mode fiber was used [47]. Different traces in the figure correspond to the pulse widths of 98, 33, 22, 18, 15, 13, 11, 8.3, 6.2, 4, and 1 ns. The Stokes power decreases monotonically with the decrease of pulse width. Figure 4.8.9(b) shows the power spectral density at the peak Stokes gain wavelength of 1680 nm as a function of the pulse width. For the fiber under test, the dispersion varies from $D_1 = 17$ ps/nm/km at 1550 nm to $D_2 = 25$ ps/nm/km near 1680 nm; therefore the pulse width at which the walk-off length equals the fiber length is approximately $\tau_0 \approx 0.5L(D_1 + D_2)\Delta\lambda \approx 26$ ns. Figure 4.8.9(b) clearly shows the transition between the two regimes described by Equations 4.8.21 and 4.8.22. The growth rate of the Stokes power as the function of pulse width increase gives the Raman gain coefficient. More precisely, Equation 4.8.20, with the consideration of fiber loss, can be numerically solved to fit the measured results shown in Figure 4.8.9(b) with the Raman gain coefficient as the free parameter for best fitting.

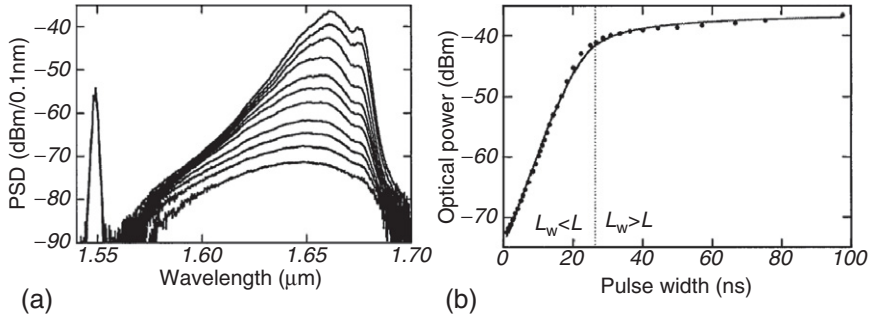


Figure 4.8.9 (a) SRS signal measured from a 10.1 km standard single-mode fiber for 11 different pump pulse widths (from top to bottom): 98, 33, 22, 18, 15, 13, 11, 8.3, 6.2, 4.0, and 1.0 ns, and (b) average Stokes power at the peak of the Raman gain spectrum as a function of pump pulse width [47]. Used with permission.

It is important to note that the Raman gain coefficient is polarization-dependent. The Raman gain for the Stokes wave parallel to the polarization of the pump $g_{R//}$ is much higher than that perpendicular to the pump polarization $g_{R\perp}$. Because of the PMD effect in the fiber, the polarization states of the pump and the Stokes waves walk off from each other rapidly because of their wide wavelength separation. Therefore the Raman gain coefficients measured by both of the discussed techniques are the average effect: $g_R = (g_{R\perp} + g_{R//})/2$.

4.8.3 Measurement of Kerr effect nonlinearity

Kerr effect nonlinearity arises from the intensity dependence of the refractive index in the fiber, which is originated from the third-order susceptibility of the silica material. The refractive index in the fiber can be expressed as

$$n = n_0 + n_2 \left(\frac{P}{A_{eff}} \right) \quad (4.8.23)$$

where n_0 is the linear refractive index, which may be wavelength-dependent representing the effect of chromatic dispersion. n_2 is the coefficient of nonlinear refractive index, which represents the dependency of refractive index on the optical power density in the material. For a lightwave signal with power P and wavelength λ_0 , the nonlinear phase shift created in a fiber of length L is

$$\Phi_{NL}(P, L) = \left(\frac{2\pi}{\lambda_0} \right) \left(\frac{n_2}{A_{eff}} \right) PL = \gamma PL \quad (4.8.24)$$

where a nonlinear parameter γ is introduced as

$$\gamma = \left(\frac{2\pi}{\lambda_0} \right) \left(\frac{n_2}{A_{eff}} \right) = \frac{n_2 \omega_0}{c A_{eff}} \quad (4.8.25)$$

It represents the nonlinear phase change over a unit fiber length with a unit power change. The implication of a nonlinear parameter in the nonlinear Schrödinger equation was discussed in Chapter 1. Several well-studied effects such as self-phase modulation, cross-phase modulation, and four-wave mixing are all originated from the nonlinear refractive index, which are major sources of performance degradation in optical fiber communication networks. On the other hand, Kerr effect nonlinearity has been used to support optical solitons and various other nonlinear optical signal processing techniques such as supercontinuum generation [48] and wavelength conversion.

Kerr effect nonlinearity in an optical fiber depends on the glass composition, material doping, cross-section design, and the core size. In recent years, photonic crystal fibers (PCFs) have been introduced and may provide very high nonlinearity with small solid-core PCFs, or very low nonlinearity with hollow-core PCFs. Precise measurement of the fiber nonlinear parameter is important for both optical system performance evaluation and nonlinear optical signal processing in optical fibers. A number of techniques have been developed to measure nonlinear parameters in fibers based on different effects of the nonlinearity, including SPM, XPM, and modulation instability.

4.8.3.1 Nonlinear Index Measurement Using SPM

When an intense optical pulse $P(t)$ is injected into an optical fiber, a nonlinear phase shift is introduced due to the nonlinear refractive index. The nonlinear phase shift is proportional to the instantaneous power of the optical signal,

$$\Delta\varphi_{NL}(t) = \gamma L_{eff} |A(t)|^2 \quad (4.8.26)$$

where $L_{eff} = (1 - e^{-\alpha L})/\alpha$ is the effective nonlinear length of the fiber, $A(t)$ is the optical field, and $|A(t)|^2 = P(t)$ is the optical power. The time derivative of this nonlinear phase shift is equivalent to a nonlinear frequency shift,

$$\Delta\omega_{NL} = -\frac{\partial \Delta\varphi_{NL}(t)}{\partial t} = -\gamma L_{eff} \frac{\partial}{\partial t} |A(t)|^2 \quad (4.8.27)$$

At the leading edge of an optical pulse, the nonlinear frequency is downshifted from the original optical frequency, whereas at the trailing edge, the frequency is upshifted. Therefore, after propagating through the fiber, the spectrum of the optical pulse will be broadened and even possibly split. The

frequency spectrum of an optical pulse affected by self-phase modulation is given by the Fourier transformation of the pulse optical field $A(t)$,

$$\tilde{A}(\omega) = \int_{-\infty}^{\infty} A(t) \exp[i\varphi_{NL}(t)] e^{i\omega t} dt \quad (4.8.28)$$

The power spectral density $|\tilde{A}(\omega)|^2$ can be experimentally measured by an optical spectrum analyzer. Nonlinear parameter γ of the fiber can be evaluated by the best fitting between the calculated and the measured optical spectra [49].

Suppose that the pulse shape is Gaussian without frequency chirp at the input of the fiber,

$$A_{in}(t) = A_0 \exp\left\{-\frac{(t-t_0)^2}{2\tau^2}\right\} \quad (4.8.29)$$

where A_0 is the peak amplitude, τ is the FWHM, and t_0 is the center of the pulse. At the fiber output, a time-dependent chirp is introduced by the SPM effect and the optical field becomes

$$A_{out}(t) \propto \exp\left\{-\frac{(t-t_0)^2}{2\tau^2}\right\} \cdot \exp\left\{j\Delta\varphi_{NL,max} \exp\left[\frac{(t-t_0)^2}{\tau^2}\right]\right\} \quad (4.8.30)$$

where $\Delta\varphi_{NL,max} = \gamma L_{eff} |A_0|^2$ is the maximum SPM phase shift that happens at the center of the pulse. The chromatic dispersion effect is neglected for simplicity, which is valid when the fiber is short enough, and therefore the amplitude waveform will not change over the fiber. The power spectral density of $A_{out}(t)$ can then be easily obtained by a fast-Fourier transform (FFT) as shown in [Figure 4.8.10](#) for several different $\Delta\varphi_{NL,max}$ values indicated in the figure.

The experimental setup for measuring fiber nonlinear parameters using the SPM effect is shown in [Figure 4.8.11](#). A mode-locked laser is used to generate short optical pulses with the width on the order of 50 ps. Two optical amplifiers are used to boost the optical power. Two bandpass optical filters eliminate the broadband ASE noise produced by the optical amplifiers. A variable optical attenuator (VOA) is used to adjust the optical power injected into the fiber under test. An OSA at the input side of the fiber monitors the optical spectrum without the SPM effect and another OSA at the fiber output measures the broadened optical spectrum caused by the SPM effect. High-speed photodiodes and oscilloscopes are also used at both the input and the output sides to monitor the pulse waveforms in the time domain for comparison.

With the increase of the pulse peak power injected into the fiber, the nonlinear phase shift due to SPM increases. [Figure 4.8.12](#) shows examples of measured optical spectra at two different pulse peak optical powers, corresponding to

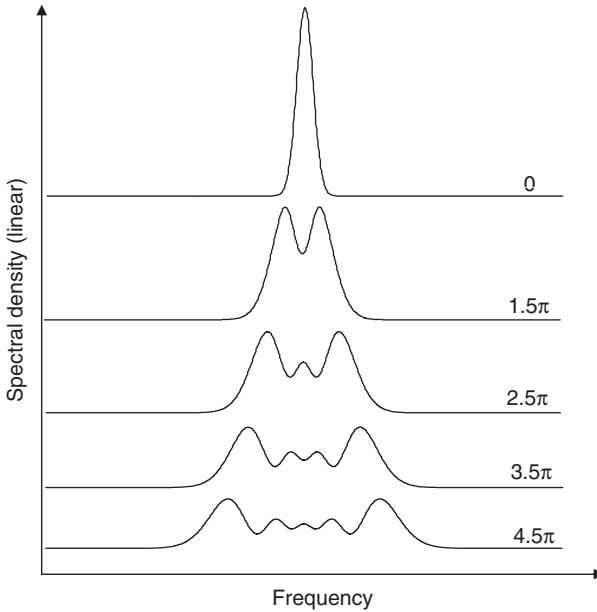


Figure 4.8.10 Spectrum of Gaussian pulses with different phase modulations.

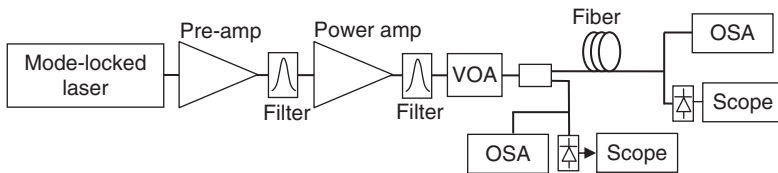


Figure 4.8.11 Block diagram of experimental setup for nonlinear parameter characterization using the SPM technique.

$\Delta\varphi_{NL,max} = 3.5\pi$ and $\Delta\varphi_{NL,max} = 7.5\pi$, respectively. If the effect of chromatic dispersion can be neglected, such as in dispersion-shifted fibers, the amplitude of the waveform $|A(t)|^2$ does not change significantly over the fiber and the output optical spectrum can be directly calculated by Equation 4.8.28. However, in general, the effect of chromatic dispersion and its interaction with the nonlinear phase modulation cannot be neglected and the intensity waveform may change dramatically along the fiber. In that case, a numerical simulation is necessary to solve the nonlinear Schrödinger Equation 1.3.96 using the split-step Fourier transform technique. If fiber parameters such as chromatic dispersion, attenuation and input pulse waveform and chirp are all known, the output waveform

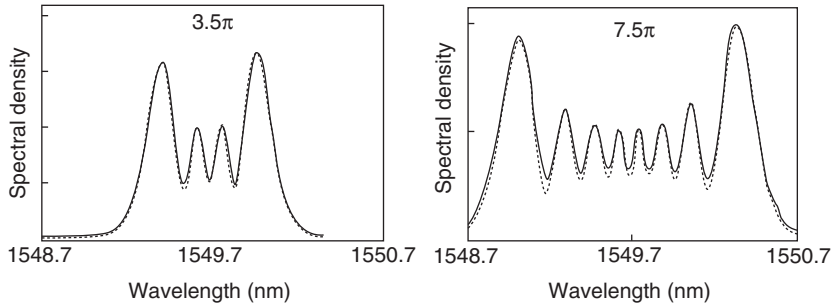


Figure 4.8.12 Measured (solid line) and calculated (dotted line) optical spectra corresponding to the peak phase shift of 3.5π (left) and 7.5π . [50]. Used with permission.

and its optical spectrum can be obtained accurately through numerical analysis. By fitting the calculated optical spectrum using the nonlinear coefficient (n_2/A_{eff}) as the free parameter with the experimentally measured spectrum, the value of n_2/A_{eff} can be precisely determined [50]. The dotted lines in Figure 4.8.12 show the calculated optical spectra that best fit the experimental results, where a 20 m long dispersion-shifted fiber was measured.

This measurement technique using the SPM effect can be very accurate. However, it requires detailed knowledge of the input pulse waveform as well as its frequency chirp. Chromatic dispersion and attenuation of the fiber are also important parameters for the numerical calculation and may change the output spectrum significantly. Asymmetry in the input pulse waveform and its frequency chirp may result in asymmetric broadening of the output optical spectrum. Adequate numerical modeling is also critical to ensure the accuracy of the nonlinear parameter evaluation.

In addition, the Kerr effect nonlinearity in an optical fiber also depends on the state of polarization of the optical signal. The nonlinear refractive index is the maximum for linearly polarized light and drops to 2/3 of its maximum value for circular polarized light. For long fibers, because the SOP of the optical signal varies randomly along the fiber between linear and circular polarization states, the average nonlinear refractive index should be approximately 8/9 of its maximum value [57].

4.8.3.2 Nonlinear Index Measurement Using FWM

When two continuous-wave optical carriers copropagate along an optical fiber, the beating between them through nonlinear Kerr effect modulates the refractive index of the fiber at their frequency difference. Meanwhile, each of these two waves is phase-modulated by this index modulation, creating two

extra modulation sidebands. This is commonly known as *degenerate FWM*, as discussed in Chapter 1.

In the simplest case, if we assume that the two carriers at frequencies of ω_1 and ω_2 are coupled into the fiber with identical amplitude A_0 , the composite optical field is

$$A_{in}(t) = A_0 [\exp(j\omega_1 t) + \exp(j\omega_2 t)], \quad (4.8.31)$$

and the corresponding optical power at the fiber input is

$$P(t) = |A_{in}(t)|^2 = 2P_{ave} [1 + \cos(\omega_2 - \omega_1)t] \quad (4.8.32)$$

where $P_{ave} = |A_0|^2$ is the average optical power of each CW carrier. Due to the SPM effect, the intensity modulation of the optical signal creates a phase modulation, and at the fiber output, the optical field is approximately

$$A(t) = 2A_0 e^{-\alpha L/2} \cos(\Delta\omega t) \cos(\omega_0 t) \exp[i\varphi_{NL, \max} \cos^2(\Delta\omega t)] \quad (4.8.33)$$

where $\omega_0 = (\omega_1 + \omega_2)/2$ is the average frequency of the two CW carriers and $\Delta\omega = (\omega_1 - \omega_2)$ is their frequency difference. $\Delta\varphi_{NL} = \gamma L_{eff} P_{ave}$ is the magnitude of the nonlinear phase shift, and α is the power attenuation coefficient of the fiber. This expression of optical field in Equation 4.8.33 can be expanded into a Bessel series using

$$\exp[ix \cos(\phi)] = \sum_{n=-\infty}^{\infty} i^n J_n(x) e^{in\phi} \quad (4.8.34)$$

where $J_n(x)$ is the Bessel function of the n -th order. Each term of the expanded series represents a discrete frequency component, and there will be a large number of frequency components generated by this degenerate FWM process. However, if we only consider the intensities of the 0th and the first-order harmonics, the ratio between them can be found as [51]

$$\frac{P_0}{P_1} = \frac{J_0^2(\varphi_{NL}/2) + J_1^2(\varphi_{NL}/2)}{J_1^2(\varphi_{NL}/2) + J_2^2(\varphi_{NL}/2)} \quad (4.8.35)$$

where the higher-order terms were neglected because the powers of these terms are much weaker than those at the fundamental frequency and at the first-order sidebands. By experimentally measuring the intensities of P_0 and P_1 , the nonlinear phase shift φ_{NL} can be found and the nonlinear parameter γ of the fiber can be determined.

Figure 4.8.13 shows a block diagram of the experimental setup, which measures the fiber nonlinear parameter utilizing the FWM effect. Two separate DFB laser diodes are used and their wavelengths can be adjusted thermally by

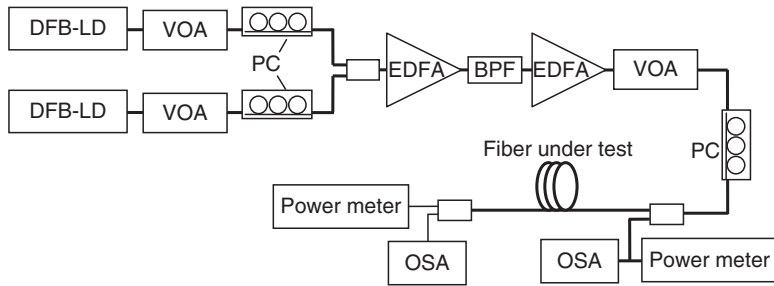


Figure 4.8.13 Block diagram of experimental setup for nonlinear parameter characterization using degenerate FWM technique.

varying the temperature control inside the laser diode package. A variable optical attenuator (VOA) adjusts the optical power of each laser diode, ensuring that the two channels have the same power level. Since FWM requires the polarization alignment between the two signal channels, a polarization controller (PC) is used in each branch to align their polarization states. Two optical amplifiers are used to obtain high optical power levels, and an optical bandpass filter is used between the two amplifiers to avoid the saturation of the second amplifier caused by the wideband ASE noise from the first amplifier. A third VOA at the optical amplifier output allows the control of the signal optical power coupled into the fiber under test. Both the optical power and the optical spectrum are monitored at the fiber input and output with a power meter and an optical spectrum analyzer. If the spacing between the two optical channels is too narrow, the spectral resolution of an OSA may not be sufficient to resolve these channels. In that case, coherent detection may be more appropriate for the spectral measurement.

Figure 4.8.14 shows an example of the measured optical spectrum, where two CW optical carriers separated by approximately 0.25 nm are injected into a fiber and several discrete sidebands are generated in the optical spectrum at the fiber output due to FWM. The ratio between P_0 and P_1 shown in Figure 4.8.14 can be used to calculate the nonlinear phase shift φ_{NL} based on Equation 4.8.35.

Since this technique uses two equal-powered CW optical pumps, the power of the composite optical signal at the fiber input shown in Equation 4.8.32 is equivalent to a single-channel optical signal that is intensity modulated at frequency $\Delta\omega = (\omega_1 - \omega_2)$ and the modulation index is $m = 1$. Therefore this technique is also known as the CW SPM method [51].

It is important to note that the optical field expression given in Equation 4.8.33 neglected chromatic dispersion in the fiber. When dispersion is considered, there is a relative walk-off between the two optical channels, and the FWM efficiency may be reduced, depending on the wavelength separation between

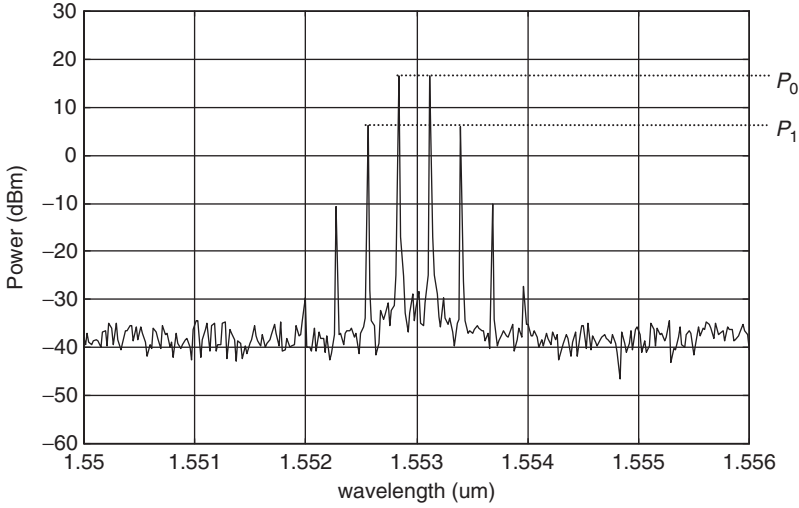


Figure 4.8.14 Example of the measured optical spectrum in FWM measurement.

the two channels, as described by Equation 1.3.119 in Chapter 1. [Figure 4.8.15](#) shows the calculated FWM efficiencies of standard single-mode fiber with dispersion parameter $D = 17$ ps/nm-km and non-zero-dispersion shifted fiber with $D = 5$ ps/nm-km operating in a 1550 nm wavelength window. The wavelength separation between the two channels is assumed to be 0.5 nm and the fiber attenuation is $\alpha = 0.25$ dB/km. Under these conditions, to ensure FWM efficiency of higher than 99 percent, the fiber should not be longer than 110 m for standard single-mode fiber and 360 m for nonzero-dispersion shifted fiber.

The FWM technique may also use two optical signals with very different power levels, also referred to as the *pump-probe technique*, as illustrated in [Figure 4.8.16](#). In this case, because of the large power unbalance between the pump and the probe, the simple relationship given by [Equation 4.8.35](#) does not hold, and the intensity of the generated FWM component at the fiber output is

$$P_{FWM}(L) = \eta_{FWM} \gamma^2 L_{eff} P_{pump}^2(0) P_{probe}(0) \quad (4.8.36)$$

where $P_{pump}(0)$ and $P_{probe}(0)$ are the powers of the pump and the probe at the input of the fiber and

$$\eta_{FWM} = \frac{\alpha^2}{\Delta\beta^2 + \alpha^2} \left[1 + \frac{4e^{-\alpha L} \sin^2(\Delta\beta \cdot L/2)}{(1 - e^{-\alpha L})^2} \right] \quad (4.8.37)$$

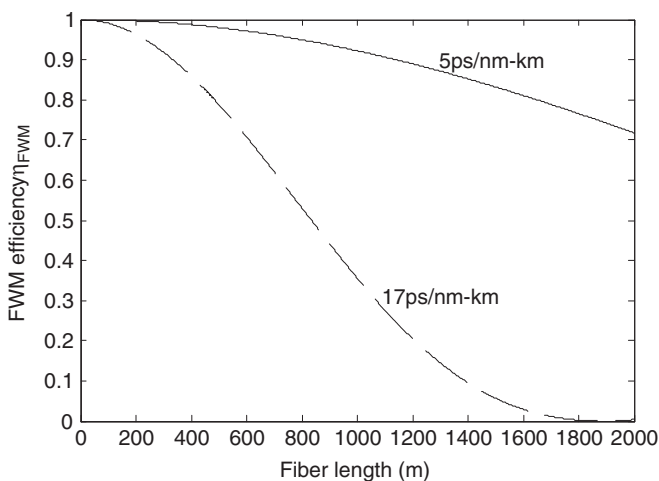


Figure 4.8.15 FWM efficiency for two wavelengths separated by 0.5 nm in a 1550 nm wavelength window. Fiber attenuation: $\alpha = 0.25$ dB/km, dispersion parameter: 17 ps/nm-km (dashed line) and 5 ps/nm-km (solid line).

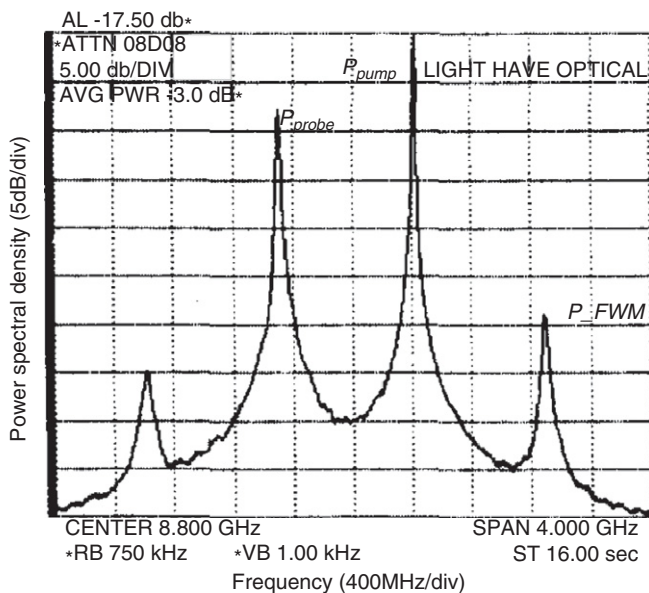


Figure 4.8.16 FWM spectrum with a strong pump and weak probe, measured with coherent detection.

is the efficiency of the degenerate FWM, which is a special case of Equation 1.3.119 for $\lambda_j = \lambda_k$. $\Delta\beta = 2\pi cD(\Delta\lambda/\lambda)^2$ is the phase walk-off and $\Delta\lambda$ is the wavelength separation between the pump and the probe.

In the measurement, the ratio between the output powers of the FWM component and the probe is proportional to the square of the output pump power,

$$R_p = \frac{P_{FWM}(L)}{P_{probe}(L)} = \eta_{FWM} \gamma^2 L_{eff}^2 P_{pump}^2(L) e^{-3\alpha L} \quad (4.8.38)$$

By varying the pump power, the slope of $\log(P_{FWM}(L)/P_{probe}(L))$ versus $\log(P_{pump}^2(L))$ gives the information $\eta_{FWM} \gamma^2 L_{eff}^2 e^{-3\alpha L}$. In this way the power-coupling efficiency of the pump and the probe at the fiber input does not have to be calibrated, and the relative change in the pump power can easily be controlled by a programmable VOA. In this particular example, as shown in Figure 4.8.16, the spectrum was measured using coherent heterodyne detection, which involves the use of a tunable laser as the local oscillator. In this case, the frequency spacing between the pump and the probe is only 900 MHz, which is approximately 0.008 nm in terms of the wavelength separation; a grating-based OSA would not be able to provide a good measurement.

4.8.3.3 Nonlinear Index Measurement Using Cross-Phase Modulation

Cross-phase modulation (XPM) is another effect originated from the nonlinear refractive index. The process of XPM can be explained briefly as follows: An intensity-modulated strong optical carrier propagating along an optical fiber modulates the refractive index of the fiber. As a consequence, another optical signal propagating in the same fiber will be phase modulated by this index modulation in the fiber. XPM is often a notorious source of performance degradation in dense WDM optical systems due to the interchannel crosstalk it introduces. On the other hand, the effect of XPM can also be utilized to measure the nonlinear parameter of the fiber.

Figure 4.8.17 shows a typical experimental setup for the measurement of a fiber nonlinear index using XPM effect. In this setup, two lasers are used with wavelengths of λ_1 and λ_2 , respectively. The output of the laser at wavelength λ_1 , commonly referred to as the pump, is intensity modulated by an external optical modulator and amplified by a high-power EDFA. The other laser known as the probe at wavelength λ_2 is used, which operates in continuous wave with a low power level. The pump and the probe are combined by a fiber coupler and sent to the fiber under test. At the output side of the fiber, an optical bandpass filter is used to reject the pump and only allows the probe to pass through. Due to the effect of XPM, the optical phase of the CW probe is modulated by the intensity waveform of the pump. This phase modulation can be

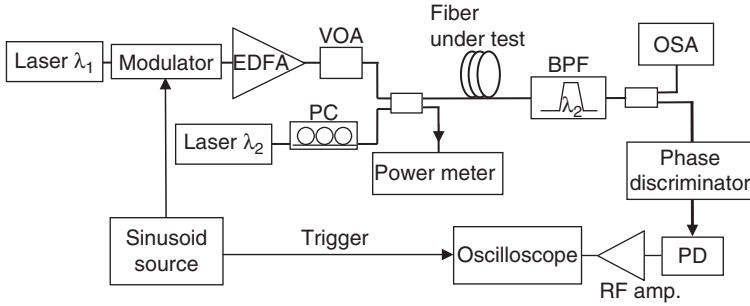


Figure 4.8.17 Block diagram of experimental setup for nonlinear parameter characterization using cross-phase modulation.

monitored either by an optical spectrum analyzer or by an optical phase discriminator that converts the phase modulation into an intensity modulation. The waveform of this converted intensity modulation is then detected by a photodiode and displayed on an oscilloscope.

Theoretically, at the fiber output the nonlinear phase shift of the probe channel at wavelength λ_2 can be expressed as

$$\Delta\varphi_{NL,\lambda_2}(t) = \gamma L_{eff} \left\{ \left| A_{\lambda_2}(t) \right|^2 + 2b \left| A_{\lambda_1}(t) \right|^2 \right\} \quad (4.8.39)$$

where $\left| A_{\lambda_1}(t) \right|^2 = P_{\lambda_1}(t)$ and $\left| A_{\lambda_2}(t) \right|^2 = P_{\lambda_2}(t)$ are the powers of the pump and the probe channels, and $1/3 < b < 1$ is a parameter representing the effect of polarization mismatch between the pump and the probe along the fiber. The first term represents the SPM effect of the probe channel itself; the second term represents the effect of XPM. The highest value of b is 1, which happens when the pump and the probe are both linearly polarized and parallel to each other; the minimum value $b = 1/3$ also happens when the pump and the probe are both linearly polarized but are perpendicular with each other. In practice, if the fiber is not polarization maintained and the SOP of the optical signal is randomly evolving over the fiber, an average value of $b = 2/3$ should be used [52].

To measure the effect of XPM, we can simply use a low-power probe such that the SPM effect in the probe is negligible. Assume that the intensity of the pump is modulated by a sinusoid, $P_{\lambda_1}(t) = P_{ave,\lambda_1}[1 + \cos\Omega t]$, where Ω is the modulation frequency and P_{ave,λ_1} is the average power of the pump at the fiber input. The phase modulation in the probe channel created by XPM is

$$\Delta\varphi_{XPM,\lambda_2}(t) = \Delta\varphi_{XPM,\max} \left(\frac{1 + \cos\Omega t}{2} \right) \quad (4.8.40)$$

where $\Delta\varphi_{XPM,\max} = 4b\gamma L_{\text{eff}} P_{\text{ave},\lambda_1}$ is the peak phase shift. Therefore at fiber output the optical field of the probe is

$$A_{\lambda_2}(t) = A_{0,\lambda_2} \exp(i\varphi_{XPM,\max}) \cdot \exp\left[i\frac{\varphi_{XPM,\max}}{2}\cos\Omega t\right] \quad (4.8.41)$$

where A_{0,λ_2} is the probe field amplitude at the fiber output. This expression of the phase-modulated probe can be expanded in a Bessel series based on [Equation 4.8.34](#),

$$A_{\lambda_2}(t) = A_{0,\lambda_2} \exp(i\varphi_{XPM,\max}) \cdot \sum_{n=-\infty}^{\infty} i^n J_n\left(\frac{\varphi_{XPM,\max}}{2}\right) e^{in\Omega t} \quad (4.8.42)$$

A Fourier transform of [Equation 4.8.41](#) yields the optical spectrum of the output probe. In addition to the central optical carrier, discrete sidebands exist with equal frequency separation by Ω between each other. The amplitude of each sideband is proportional to the n^{th} order Bessel function $J_n(\varphi_{XPM,\max}/2)$.

The nonlinear index of the fiber can be evaluated by measuring the relative amplitudes of the carrier and the sidebands of the probe optical spectrum as a function of the pump power. By fitting the measured amplitude with the theoretical prediction as shown in [Figure 4.8.18](#), the XPM-induced peak nonlinear phase shift $\varphi_{XPM,\max}/2$ can be precisely determined. From the practical instrumentation point of view, a conventional grating-based OSA usually has the frequency resolution in the order of 0.05–0.1 nm, or equivalently, 6.25–12.5 GHz. To clearly resolve the sidebands of different orders, the modulation frequency Ω has to be higher than 15 GHz; this imposes the requirement for high-speed electronics and wideband electro-optic modulators. Otherwise, a scanning FP interferometer or coherent detection may be used to provide higher spectral resolution.

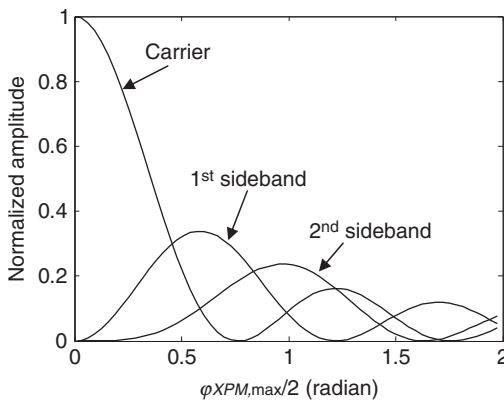


Figure 4.8.18 Relative amplitude of sidebands due to phase modulation.

The phase modulation in the probe can be characterized using either a phase discriminator or a frequency discriminator. Several different techniques can be used for this purpose, such as delayed self-heterodyne detection [53], a Michelson interferometer, or a Mach-zehnder interferometer [54]. The phase discriminator converts the XPM-induced phase modulation into intensity modulation, which can be easily detected by a photodiode, and the waveform can be displayed on an oscilloscope. In this case, the calibration of phase-to-intensity modulation efficiency is extremely important to ensure the accuracy of the measurement.

It is important to note that in the previous analysis, the effect of chromatic dispersion in the fiber was neglected. With the consideration of chromatic dispersion, the pump and the probe waves may propagate in different group velocities in the fiber, and the relative walk-off between them will reduce the efficiency of XPM. At the same time, chromatic dispersion may also convert phase modulation into intensity modulation in the probe channel [55]. This introduces additional complications in the measurement of the optical spectrum and its comparison to the theory. For the dispersion effect to be negligible, the dispersion-induced walk-off between the pump and the probe along the fiber has to be much smaller than the time period of the modulating signal, that is,

$$f < \frac{1}{100} \cdot \frac{1}{D \cdot L \cdot \Delta\lambda} \quad (4.8.43)$$

where $f = \Omega/2\pi$ is the RF modulation frequency, D is the dispersion parameter of the fiber, L is the fiber length and $\Delta\lambda$ is the pump-probe wavelength separation. For example, to measure a 10-km standard single mode fiber in a 1550 nm wavelength window with $D = 17$ ps/nm-km and $\Delta\lambda = 0.2$ nm, the modulation frequency f should not exceed 300 MHz.

In the XPM measurement, a strong pump and a weak probe copropagate along the same fiber. In the experimental setup shown in Figure 4.8.17, only the phase modulation induced in the probe channel is measured, whereas the pump channel is rejected at the fiber output by a bandpass optical filter. In fact, instead of rejecting the pump channel at the fiber output, if we use an OSA to measure the spectral characteristics of both the pump and the probe, the similar experimental setup shown in Figure 4.8.17 can be expanded to measure the effects of both XPM and SPM so that the precise relationship between XPM and SPM can be characterized. Figure 4.8.19 shows an experimental setup that is useful to measure both XPM and SPM in the same fiber, where three lasers are used. The two lasers at wavelengths λ_{11} and λ_{12} are combined in the first directional coupler and amplified by a high-power EDFA to provide the pump waves. The degenerate FWM between these two equal-powered pumps creates sidebands in the output optical spectrum, as already discussed in Equations

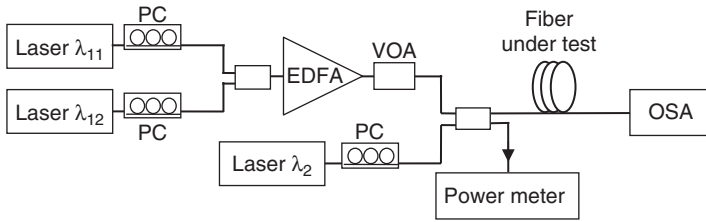


Figure 4.8.19 Experimental setup to measure SPM and XPM simultaneously.

4.8.31–4.8.35. This technique is also known as *CW self-phase modulation* because the combination of these two pump lasers is equivalent to a single laser that is intensity modulated at the difference frequency $\Delta\omega$ between them.

The third laser at wavelength λ_2 is used as the probe and is combined with the pump before coupled into the testing fiber through the second directional coupler. Due to XPM, the probe is phase modulated along the fiber by the intensity modulation created by the pumps. The OSA at the fiber output simultaneously measures the spectra of both the pump and the probe. Figure 4.8.20 shows an example of the measured optical spectrum where the pump and the probe are separated by approximately 2 nm [56]. The relative amplitudes of P_0 and P_1 in the pump output can be used to determine the nonlinear phase shift responsible for SPM with Equation 4.8.35. Meanwhile, by measuring the amplitudes of the carrier and the first-order sideband in the probe, the nonlinear phase shift due to XPM can also be estimated.

It has been observed that if the modulation frequency is lower than 1 GHz, the measurement using the XPM effect often observed a larger nonlinear

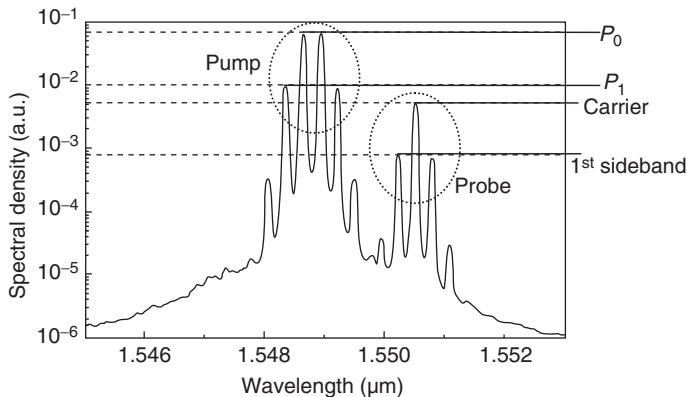


Figure 4.8.20 Experimental setup to measure SPM and XPM simultaneously [56]. Used with permission.

parameter compared to SPM-based techniques. An explanation for this discrepancy is related to electrostrictive contribution to the nonlinear refractive index that occurs at relatively low modulation frequencies [57]. By varying the frequency difference between the two pump lasers, the experimental setup shown in Figure 4.8.19 may be used to verify this discrepancy. However, for the modulation frequency to be lower than 1 GHz, it might be easier to use an intensity-modulated single-pump laser to replace the beating between the two pumps. For the spectrum measurement, a high-resolution optical spectrum analyzer such as coherent detection has to be used.

4.8.3.4 Nonlinear Index Measurement Using Modulation Instability

Modulation instability (MI) is another phenomenon originated from the nonlinear refractive index in an optical fiber. When an intensive optical pump propagates along an optical fiber with anomalous dispersion, optical gain is created in two spectral regions symmetrically located on each side of the pump wavelength. This nonlinear parametric gain process can also be explained as the FWM between the pump and the wideband ASE noise.

Based on the nonlinear Schrödinger Equation 1.3.96, assuming that the steady-state solution of the optical field envelope is $A = \sqrt{P_0}$, where P_0 is the signal optical power. Considering small perturbations in the amplitude and the phase, the general solution can be assumed to be

$$A = (\sqrt{P_0} + a)\exp(i\Phi_{NL}) \quad (4.8.44)$$

where a is the amplitude perturbation and ϕ_{NL} is the nonlinear phase shift as defined in Equation 1.3.106. Substituting Equation 4.8.44 back into the nonlinear Schrödinger equation, the amplitude perturbation can be found as [57]

$$a(z, t) = a_0 \exp[i(Kz - \Omega t + \varphi_0)] \quad (4.8.45)$$

where a_0 and φ_0 are the complex amplitude and the initial phase, and the wave number is

$$K = \pm \frac{1}{2} |\beta_2| \Omega \sqrt{\Omega^2 + \text{sgn}(\beta_2) \Omega_C^2} \quad (4.8.46)$$

where $\text{sgn}(\beta_2)$ indicates the sign of β_2 : $\text{sgn}(\beta_2) = 1$ for normal dispersion and $\text{sgn}(\beta_2) = -1$ for anomalous dispersion. Ω_C is related to the nonlinear parameter γ , the dispersion parameter β_2 , and the signal optical power P_0 by

$$\Omega_C^2 = \frac{4\gamma P_0}{|\beta_2|} \quad (4.8.47)$$

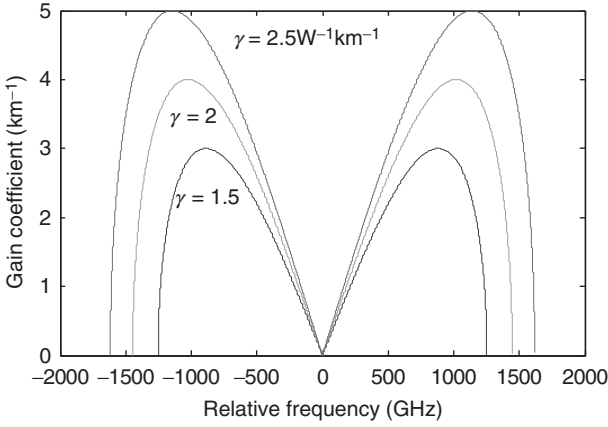


Figure 4.8.21 Parametric gain coefficient $g(\Omega)$ calculated using Equation 4.8.48 with $\beta_2 = 3.82$ ps²/km, $P_0 = 1$ W and $\gamma = 1.5, 2, 2.5$ W⁻¹km⁻¹.

For a fiber with normal dispersion, the wave number K is real and the amplitude of the perturbation does not increase over the fiber length z . On the other hand, if the fiber dispersion is anomalous, the wave number becomes imaginary for $\Omega < \Omega_C$; therefore the amplitude of the perturbation will grow exponentially along the fiber. This is the origin of modulation instability, which is also referred to as *parametric gain*. From Equations 4.4.46 and 4.4.47, it is evident that the power gain coefficient in the fiber caused by modulation instability is

$$g(\Omega) = 2\text{Im}(K) = |\beta_2|\Omega\sqrt{\frac{4\gamma P_0}{|\beta_2|} - \Omega^2} \quad (4.8.48)$$

The frequency of the maximum gain can be found where $dg(\Omega)/d\Omega = 0$, that is at, $\Omega_{\max} = \sqrt{2\gamma P_0/|\beta_2|}$. Therefore, if the signal optical power P_0 and the fiber dispersion parameter β_2 are known, the nonlinear parameter of the fiber can be determined by measuring the peak frequency of the parametric gain.

Figure 4.8.21 shows the calculated parametric gain coefficient using Equation 4.8.48 with the optical power level $P_0 = 1$ W at 1550 nm wavelength and the fiber chromatic dispersion parameter is $D = 1$ ps/nm/km (or equivalently, $\beta_2 = 3.82$ ps²/km). Three values of fiber nonlinear parameters are used $\gamma = 1.5$ W⁻¹km⁻¹, $\gamma = 2$ W⁻¹km⁻¹, and $\gamma = 2.5$ W⁻¹km⁻¹, respectively. Obviously, with a higher nonlinear parameter, the nonlinear gain is increased and the gain peak shifts to a higher frequency.

It is worthwhile to note that in our analysis of modulation instability so far, fiber attenuation was not included. Optical attenuation in the fiber will decrease the nonlinear effect because the optical power level is reduced along

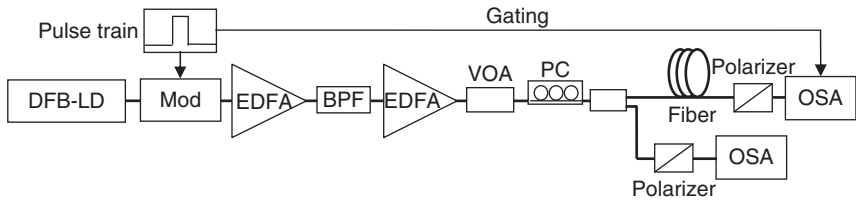


Figure 4.8.22 Nonlinear parameter characterization based on modulation instability. *BPF*: optical bandpass filter, *Mod*: electro-optic modulator, *PBS*: polarization beam splitter, *VOA*: variable optical attenuator, *PC*: polarization controller.

the fiber. However, with the consideration of fiber attenuation, it is not possible to obtain a simple analytic solution for the perturbation term; therefore a numerical simulation becomes necessary. Or a semi-analytic method can be used based on the transfer matrix analysis [58], which will be further discussed in Chapter 5.

Figure 4.8.22 shows the experimental setup for the nonlinear parameter characterization based on MI effect. The output of a laser diode is intensity modulated by a pulse train with a low duty cycle. Two EDFAs are used to produce high power in the optical signal. The purpose of the bandpass optical filter between the two EDFAs is to avoid the gain saturation of the second EDFA due to wideband ASE noise produced by the first EDFA. A variable optical attenuator allows the adjustment of optical power coupled into the fiber, and a polarization controller is used to adjust the SOP of the optical pump before injecting into the fiber. At the fiber output, an optical spectrum analyzer measures the wideband optical noise spectrum which is generated and amplified by the nonlinear parametric gain in the fiber due to MI. Since the MI effect only amplifies the ASE noise of the same polarization as the pump, a polarization beam splitter at the fiber output is adjusted to reject the perpendicular polarization component of the ASE noise. The purpose of using low duty cycle intensity modulation for the pump is to further increase the peak power, since the MI measurement especially requires a high power level for the pump. Generally, if the signal pulse width is shorter than the carrier lifetime of an EDFA, the output power of the EDFA is only limited by its average value. For example, suppose that an EDFA output is limited to a maximum of 10 dBm average power and the EDFA carrier lifetime is 10 ms; if the optical signal is a pulse train with 100 ns pulse width and 10 kHz repetition rate, the pulse peak power at the output of the EDFA could be as high as 40 dBm, or 10 W.

Due to the fact that the pump is pulsed, MI only exists during the peaks of the pulses and therefore the spectral measurement at the fiber output should only be limited to these time windows. Most of the commercial OSAs provide the gating function, in which the OSA operation is enabled by a gating pulse

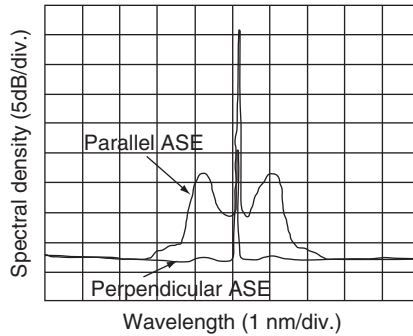


Figure 4.8.23 Measured ASE spectrum created due to modulation instability in the parallel and perpendicular polarization directions with respect to the pump [59]. Used with permission.

train. When the gating signal is at its low level, signals measured by the OSA will be discarded. By comparing the ASE noise spectra measured before and after the fiber, the nonlinear parametric gain spectrum caused by MI can be obtained and the nonlinear parameter of the fiber can be determined.

Figure 4.8.23 shows an example of the measured optical spectrum at the output of 10.1 km fiber. The zero-dispersion wavelength of the fiber is at 1539 nm and the optical signal wavelength is 1553 nm. For the pump optical pulse train, the pulse width is 25 ns, the pulse repetition period is 250 ns, and the pulse peak power is 180 mW [59]. With an intense optical signal propagating in the fiber, the parametric gain generated by MI amplifies the spontaneous emission noise and therefore a very pronounced side peak is generated on each side of the optical carrier, as predicted by Figure 4.8.21. The measurement of the amplitude and the peak frequencies of these amplified spontaneous emission noises shown in Figure 4.8.23 can be used to evaluate the fiber nonlinear parameter. Taking into account fiber chromatic dispersion, dispersion slope, attenuation, and Kerr effect nonlinearity, the measured optical spectra have to be compared with numerical simulations so that the actual fiber nonlinear parameter can be determined accurately.

Figure 4.8.23 shows the measured output ASE spectra at the same polarization as the pump and perpendicular to it. The two traces in the figure were obtained by adjusting the polarizer at the fiber output, as indicated in Figure 4.8.22, to maximize and minimize the pump. Since the modulation instability process only amplifies the ASE noise in the same polarization as the pump, the difference between the parallel and the perpendicular ASE shown in Figure 4.8.23 represents the actual parametric gain caused by the modulation instability.

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